

TSIX Course

English Edition

25 module

Kernel, VFS, Networking, GUI & Desktop, Development

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Platform: TSIX (educational OS-like runtime on Node.js/TypeScript)

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RFC-TSIX-000 | Mental map of the TSIX architecture — for contributors (human & AI), hobbyists, educators, and professionals.

This document is the **system's mental map**. It explains *what*, *where*, and *why* — not a per-file tutorial. Use it as an entry point before reading the code, and as a reference while exploring subsystems.

This is **Module 00** of the TSIX curriculum. For the full roadmap, see [toc.md](#).

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1. Philosophy & Big Picture

TSIX is a **simulated operating system built on Node.js + TypeScript**. It is not a VM that emulates a CPU — it builds **OS abstractions on top of the existing Node.js runtime**.



Five core principles:

1. **"Everything is a File"** — files and devices are both `IDevice`; `read/write` is polymorphic. Open a regular file → `FileSystemDevice`; open `/dev/tty1` → `TTYDevice`; even pipes, sockets, and the display are devices.
2. **"Distributed by Design"** — built-in IPC via the `SEND_MSG` syscall + identity-based messaging. Processes communicate through identities (UUIDs), not memory addresses.
3. **"Small, Sharp Tools"** — 80+ utilities that combine via pipes & redirection; one tool does one thing well, then tools are composed.

4. **"Security via Simplicity"** — a UID/GID permission model, process isolation (Worker Threads), and straightforward root privileges.
5. **"Unix fidelity first, pragmatic later"** — mirror Unix/Linux behavior and architecture as closely as practical; deviate only when the Node.js/V8 runtime truly cannot model it, and always document the deviation.

Key: the kernel never runs apps. All of userland (including PID 1 / init) runs in Worker Threads. The Ring 1/2 vs Ring 4 boundary is a *thread* + *IPC* boundary, reinforced by the uid/gid/mode-based `PermissionManager`.

2. Ring Model & Privilege Boundaries

Rings are a **concept** (documentation), not a hardware or V8 isolation mechanism:

Ring	Contents	Main files
0	Host: Linux + Node/V8	— (reserved)
1	Kernel core: Scheduler, Syscall, Permission	<code>src/kernel/*</code> , <code>src/common/*</code>
2	Driver & FS: HAL devices, VFS backends, MountManager	<code>src/kernel/devices/*</code> , <code>src/vfs/*</code>
3	Library framework: UserLib, Application	<code>src/mirror/lib/*</code>
4	Applications: <code>/bin/*</code> , <code>init</code> , <code>tsh</code> , <code>daemon</code>	<code>src/mirror/bin/*</code>

The **real privilege boundaries** live in two layers:

- **WorkerEntry sandbox** — apps may only require the `@tsix/*` / `@common/*` framework; `process.exit`/`process.kill` are sabotaged. Privileged apps (names containing `server`/`daemon`/`dome`/`tbuid`/`vfs`) get access to an allow-list of host modules (`http`, `ws`, `fs`, `crypto`, `esbuild`, etc.).
- **PermissionManager (kernel)** — `rx` checks: `root` (uid 0) `bypass` → `owner` → `group` → `others`. The `SetUID` bit is supported (e.g. `/bin/login 0o4755`).

 **Contributor note:** privileged status is based on **name substrings** — a fragile heuristic. This is an architectural gap worth fixing (ideally capability-based).

3. Boot Sequence

```
main.ts
Config.load()           → baca sysconfig.conf
new Kernel()            → Logger, PermissionManager, MountManager, GUIRegistry
kernel.boot()
  initializeSubsystems()
    mount BKFS("/")      → SQLite system.db (root filesystem)
    processFstab()       → /tmp (RamFS), /mnt/* (HostVFS), /mnt/sbak (BKFS)
    new Scheduler()
    new PermissionManager()
    new PortManager()
    new SyscallDispatcher() → scheduler.setSyscallHandler(...)
    rebuildVFSCache()    → pre-compile /lib ke memori
  new TTYManager(32) + TTYDevice tty1..32
  devices{}              → stdin, fb0, stdout, stderr, null
  init network interfaces → SimpleMQTNDriver per config
  loadAuxDevices()       → plugin driver (random, mysql, mcp23017)
  SerialDeviceManager    → auto-detect ttyUSB*
  applyDeviceConfigs()   → "udev": mode/uid/gid dari sysconfig
  pastikan /dev ada di VFS
  load RSA identity       → fingerprint visual di TTY
  ensureDefaultAuth()    → seed /etc/passwd + /etc/shadow (root)
  wire keyboard + TTY callbacks → Ctrl+C → SIGINT fg; Alt+F1-6 → switch
kernel.runInit()
  resolve /bin/init.js
```

```

createProcess("init", fds:[tty1x3]) → PID 1
setForegroundProcess(1, 1)
[init.ts di Worker]
  enforce setuid (passwd, sudo)
  generate/verify RSA identity
  exec /etc/rc.local.js + waitpid
  spawn login TTY2..6 (monitorProcess → respawn)
  spawn login TTY1 (foreground) → loop forever
keepAlive (100ms di main.ts)
jika PID 1 EXITED → process.exit(exitCode) (1 = reboot)

```

rc.local order (daemons): airtermd (remote access) → tpkgd (package server) → scp (SCP) → otad (ESP OTA) → iot-listener (MQTNL sensor) → dome (display server, must run after the listener) → Asteracea WM (1s delay, waits for DOME) → crond.

4. Kernel Core

4.1 Processes (PCB)

```

interface PCB in Scheduler.ts: pid, ppid, name, state, pc, owner/uid/gid/ruid/groups, cwd, worker?, fdTable,
env, exitCode?, ttyId?, uuid?.

```

Lifecycle: READY → RUNNING → BLOCKED → EXITED

- **Spawn:** createProcess() → allocate pid → PCB → spawnWorker() (Worker Thread).
- **Exit:** EXIT(code) → FD cleanup → worker.terminate(). The worker.on("exit") event → reparent orphans to PID 1 → resolve waitQueue → reap() (zombie if there is no waiter).
- **waitpid:** if EXITED → reap and return exitCode; if not yet → register in waitQueue.
- **daemonize:** DETACH → resolve waiters, detach the foreground TTY, clear ttyId.
- **REEXEC:** terminate the old worker, spawn a new one with the **same PID/PCB** (REEXECING state prevents the cleanup hook).

4.2 Syscall Dispatch

```

App → UserLib.dispatch(code, args)
→ postMessage({ requestId: uuid, pid, code, args })
→ Kernel: scheduler → syscallHandler.handleRequest(req)
  → validateArgs(code, args) // kontrak argumen
  → dispatch(pid, code, args) // switch-case ~65 syscall
  → PermissionManager.check() // satpam
  → MountManager.resolve() // path → backend
  → VFS / device / scheduler
→ postMessage({ requestId, success, data|error })
→ UserLib: cocokkan requestId di responseMap → resolve/reject

```

Request-response correlation via requestId (UUID) + responseMap — pure asynchronous RPC, no ordering guaranteed. Many syscalls can be in-flight at once.

Push events (kernel → worker, without requestId): { type, data }. Types: signal, ipc_message (SEND_MSG), gui_request (forwarding to queued), resize.

4.3 Signals

Signal	Effect
SIGKILL (9)	Hard kill: worker.terminate() immediately
SIGINT (2)	Graceful: push event, 100ms grace → terminate if not handled. Default exit 130
SIGTERM (15)	Graceful: 300ms grace. Default exit 143. Used by SHUTDOWN
SIGSTOP/SIGCONT	Soft: change pcb.state to BLOCKED/RUNNING + event
SIGSEGV	Sent to the PID that violates GUI window ownership

Signal	Effect
SIGHUP/USR1/USR2/WINCH	Generic push events

PID 1 is protected from kill — only SHUTDOWN/REBOOT (root) are valid. SHUTDOWN is staged: SIGTERM broadcast → 5s → SIGKILL survivors → flush network 1s → kill PID 1.

4.4 MountManager & PortManager

- **MountManager:** array of MountPoint {vfsPath, vfs, type, source, readOnly, uid, gid}. mount() sorts descending by path length; resolve() → the longest prefix wins → {vfs, relativePath, mountPoint}.
- **PortManager:** virtual ports 0-65535 for MQTNL; allocatePort, allocateRandomPort(10000-20000) for bind(0), releasePortsByPid() on process exit (safety net for leaked sockets).

5. VFS & Device Drivers (HAL)

5.1 VFS Layers

```
Aplikasi → syscall OPEN/READ/WRITE
→ MountManager.resolve(path) → backend IVFS
  "/"      → BKFS      (SQLite system.db, persisten)
  "/tmp"   → RamFS     (volatile, in-memory)
  "/mnt/*" → HostVFS   (bridge folder host, anti-escape)
→ path /dev/xxx → kernel.devices[xxx] (HAL, bypass MountManager)
```

- **IVFS** = a single contract (ls/mkdir/read/touch/stat/chmod/chown/.../readChunk/writeChunk/getSize) for all backends — allows "swapping backends" without changing syscalls.
- **BKFS:** a file = a row in the vnodes table (parent_id tree). Chunked I/O runs **inside SQL** (SUBSTR/CONCAT) without pulling the full content — for large files/OTA.
- **HostVFS:** toHostPath() + anti-escape checks (Security Violation) — read-only view of a host folder.

5.2 Device Model (HAL)

IDevice: read/write/ioctl + optional init/open/close + metadata uid/gid/mode.

/dev/	Class	Function
stdin	KeyboardDevice	Keyboard input (cooked/raw)
fb0/stdout/stderr	TTYDevice	Standard output → active TTY
tty1..tty32	TTYDevice	Virtual console
null	NullDevice	Black hole
smqtnl0/1	SimpleMQTNLDriver	MQTNL network interface
randomdevice	RandomDevice	Random numbers
mysql (experimental)	MySQLDevice	External DB connection — integration POC, see note below
mcp23017	MCP23017Device	I2C GPIO
ttyUSB*	SerialDevice	Auto-detect
(virtual)	PipeDevice / SocketDevice	Runtime instances (PIPE/SOCKET syscalls), not paths

The "everything is a file" key: the FD table holds FDEntry {device, context, flags} → all IDevice objects. Regular files are wrapped in FileSystemDevice. Pipe refcount via ioctl (INC_REF/DEC_REF), EOF when writeRefs==0.

Plugin drivers: the aux-devices/ folder is scanned at boot; new DeviceClass() + the static autoRegister(kernel) convention for hardware (MCP23017). applyDeviceConfigs() = "udev" from sysconfig.conf.

[!NOTE] **About MySQLDevice (/dev/mysql) — the first transport, not the only one** MySQLDevice is an external database integration through the device model — it is **not** a real hardware driver, but the **first transport** for DB access (an example of HAL extension). **DbLib is already implemented** (a UserLib sub-library, following the lib.fs/lib.net pattern) with **pluggable dual transport**:

- **Device** (/dev/mysql): DB_* syscalls (67-69) → kernel → device → mysql2
- **Service daemon** (mysql2): DB_* syscalls → kernel → Ring 4 daemon → mysql2

The kernel routes **dynamically**: if mysql2 is registered (syscall DB_SERVICE_REGISTER=70) → to the daemon; otherwise → to the device (fallback). The daemon replies via DB_SERVICE_REPLY=71. Apps only see db.connect()/query()/disconnect() — the medium is invisible. The sandbox is safe because mysql2 is only touched by the kernel or a privileged daemon.

6. Worker Thread & Sandboxing

6.1 Bootstrap Worker

```
Kernel.spawnWorker()
  workerData = { pid, appName, args, appPath, appContent, env, vfsCache }
  execArgv:
    *.js → JS-Direct (FAST)    : --enable-source-maps
    *.ts → TS-Transpile        : -r esbuild-register -r tsconfig-paths/register

WorkerEntry.ts (bootloader)
  1. realExit = process.exit (sebelum sandbox)
  2. hijack Module._load → resolve @tsix/* & @common/* dari vfsCache (DME)
  3. pasang unhandledRejection/uncaughtException → kirim error ke parent
  4. new UserLibClass(pid) → global._tsixLib
  5. muat app:
      DME : appContent di-transpile, __filename = BKFS path
      fisik: hostRequire(finalAppPath)
  6. cari export main/Main/default
  7. restrictHostAPI(appName) ← kunci pintu
  8. new AppClass() → await execute(lib, args)
  9. return string → std.print; lalu shell.exit(0)
```

6.2 Direct Memory Execution

The kernel pre-compiles /lib → vfsCache → sent via workerData → the worker _compiles from memory. Key trick: the framework module is named @tsix_Application.js so relative imports ./x can be rewritten to @tsix/x — a consistent alias cycle. Apps have **two filename identities**: physical (so require finds node_modules) vs BKFS (so stack traces are correct).

6.3 Sandbox Table

Layer	Restrictions
All apps	process.exit / process.kill → throw; non-framework require blocked
Non-privileged apps	http/fs/crypto etc. fully blocked
Privileged apps	allow-list: http, ws, path, fs, url, esbuild, crypto, os, bcryptjs
Kernel	PermissionManager + validateArgs + root-only SETUID

7. Networking MQTNL

MQTNL (MQTT Network Layer) = the TSIX networking protocol. Instead of IP / TCP/IP routing, it uses **MQTT pub/sub as the wire**, plus:

- **Address = node name** (string: "tsix", "esp32S3") — not an IP.
- **Port = application endpoint** (PortManager 0-65535).
- **Topic = mqtt@1.0/<dstAddress>** (JSON) / mqtt@1.1/<dstAddress> (Binary).
- **Packet:** 9-field header + payload; 32KB fragmentation; 30s reassembly TTL; RSA + ChaCha20-Poly1305 encryption.

```
App → socket() → bind(port) → driver.registerHandler(port)
→ sendto(addr, port, data) → driver.send → publish topic
→ MQTT broker → semua node subscribe 'mqtt@1.x/#'
→ filter dstAddress → reassembly → decrypt → socket.push()
→ recvfrom() → buffer.shift()
```

Dual protocol: MQTTNProtocolJSON (v1.0, legacy, readable) vs MQTTNProtocolBinary (v1.1, compact, no encryption — for byte-exact OTA). The driver detects from the **first magic byte** per srcAddress. src/common/protocols/ holds the IMQTNProtocol contract — userland **never touches this directly**.

Syscalls: SOCKET=30, BIND=31, SENDTO=32, RECVFROM=33, NETSTAT=34. There is no listen/accept — UserLib emulates them: listen() = socket+bind, accept() = polling recv().

8. TTY & Virtual Console

- **TTYManager** manages 1-32 virtual consoles; switching only 1-12, login on 1-6.
- **TTY** (src/kernel/tty/TTY.ts): char[x][y] buffer, cursor, ANSI subset parser, full re-render.
- **TTYDevice** (/dev/ttyN): ioctl clear(1), raw(10), inject input(0x2001), read output(0x2002), window size(4) — lets remote apps (pixelterm) control a TTY.
- **TTY allocation to processes:** via ttyId in the EXEC syscall — the kernel overrides FD 0/1/2 to tty{n}.
- **Keyboard:** cooked/raw mode, Ctrl+C → SIGINT to fg, Alt+F1-6 → TTY switch, resize → SIGWINCH broadcast.

9. Userland

9.1 Init (PID 1)

No /etc/inittab — the logic is hardcoded in init.ts: enforce setuid → RSA identity → exec rc.local → spawn login per TTY (via monitorProcess respawn, anti-crash-loop) → loop.

9.2 Login → Shell

login.ts: verify /etc/passwd + /etc/shadow (bcrypt) → setgroups → setgid → setuid (POSIX order) → exec \$SHELL → tsh.ts.

9.3 Two Application Styles

1. **Class IProgram** (majority legacy): export class main implements IProgram { async execute({fs, shell, std}: OSContext, args) {...} } — ls, cat, ps, kill, tsh.
2. **Program() wrapper + proxy singletons** (new): import { Program, std, fs, shell, net } from "@tsix/Application" — esp-send, dome, iot-dashboard.

9.4 Error = "Window"

std.error() does 4 things: write syslog → broadcast to parent → print red on TTY → send GUI_WINDOW_ERROR to the WM (with wid, pid, fileHint from the stack trace).

9.5 DbLib — Database Sub-Library

DbLib (@tsix/DbLib, lib.db) is the fifth UserLib sub-library (after std/fs/shell/net). It wraps the DB_* syscalls (67-71) into the db.connect()/query()/disconnect() API. **Pluggable transport:** the kernel routes to /dev/mysql (device) or

mysqld (Ring 4 service daemon) dynamically. Apps don't know the medium — exactly like the `lib.fs` pattern, which doesn't care about its VFS backend.

```
App → db.query(sql) → dispatch(DB_QUERY=68)
  → Kernel: mysqld terdaftar?
    YA → sendEvent(daemon, "db_request") → mysql2 → dbServiceReply → resolve
    TIDAK → /dev/mysql device → mysql2 → resolve
  → App: rows
```

10. PixelSpace & TDE

Full details in `PIXELSPACE_DEVELOPER_GUIDE.md`. Architectural summary:

```
Worker (app) → GUI_REQ (61) → Kernel (GUIRegistry auth pid↔wid)
  → DOME daemon (WS broadcast) → Browser (DOM)
Browser → event → DOME → Kernel (SEND_MSG) → Worker (callback)
```

- **Contract:** `GUITypes.ts` (`IDOMNode`, `IGUIPayload`, `GUIAction`, `IBrowserEvent`, `IGUIEventIPC`) — the "constitution", don't change it carelessly.
- **GUIRegistry (kernel)** = the single authority over window ownership: `CREATE_WINDOW` = register `wid↔pid`; accessing someone else's window → `SIGSEGV`; corrupted payload → `SIGKILL`; process death → window auto-destroy.
- **DOME** = display server (Ring 4 daemon, port 8080): WS relay + DOM primitive producer + **compositor** (titlebar, drag, resize, focus, replay). Monolithic due to drag/resize latency considerations.
- **Emerald** = widget toolkit (`@tsix/emerald`): `Screen`, `Window`, factory functions, connected widgets.
- **Cashew** = component framework (`@tsix/cashew`): OOP/Delphi-style `TForm`/`TButton`/`TEdit`, auto-bind lifecycle, `TDialogs` & `TTimer` — a layer above Emerald.
- **Asteracea** = window manager (fullscreen frameless app): taskbar, launcher, login, wallpaper; listens to `GUI_WINDOW_*` lifecycle events via `/etc/asteracea/wm-pid`.

11. Development Flow

Dev cycle (host ↔ VFS):

```
scripts/vfs-bootstrap.ts  host → system.db (transpile TS→JS, chmod, setuid)
scripts/sync-vfs.ts       sync satu file host ↔ VFS
scripts/vfs-pull.ts       system.db → src/root (pull balik)
SYNC_TO_HOST (syscall)    DB → host (app dalam VFS menulis /lib → src/.tsix_sdk)
userlib-update.ts         sinkronkan /lib → src/.tsix_sdk/lib (agar Node.js host bisa require)
```

Configuration: `src/sysconfig.conf` (database path, `workerEntryPath`, `bootEntry`, network interfaces). `tsconfig.json` maps `@tsix/*` → `src/.tsix_sdk/lib/*`, `src/root/lib/*`, `src/mirror/lib/*`.

12. Design Insights & Gotchas

Insights (why it was designed this way):

1. **Syscalls as a mini ABI** — `SyscallCode` (1-66) + `validateArgs` + `requestId` correlation. One mechanism for everything: files, processes, IPC, GUI, networking.
2. **Pure asynchronous RPC** — request/response via UUID; different from Linux synchronous traps.
3. **Staged kill** — `SIGINT`/`SIGTERM` = event + grace period, then terminate. Replicates the Unix default (unhandled signal → exit).
4. **Faithful zombies & reparenting** — orphans → PID 1; zombies wait for `waitpid` + `reap`.

5. **UUID identity** — `SET_IDENTITY` allows `SEND_MSG` to a stable name even when the PID changes (well-known services).
6. **GUI auth in the kernel** — `payload.pid` is overridden by the kernel (don't trust userland).
7. **MQTNL = free no-IP** — the MQTT broker is the backbone; by design for IoT (ESP32) without a public IP.
8. **TTY = full PTY emulation** — master-side I/O via `ioctl` enables remote terminals.

Gotchas (be careful when contributing):

⚠ **Stale compile artifacts:** `src/mirror/lib/*.js` and `src/userland/WorkerEntry.js` can differ from their `.ts` sources. The runtime uses the `.ts` recompiled by the kernel, so committed `.js` files are prone to drift (example: the privileged list in `WorkerEntry.js` doesn't include `dome`).

⚠ **Docs vs code:** `rc.local.ts` (source) holds the full daemon list, but `rc.local.js` (the executed one) only has 3 daemons. The wiki `Networking-MQTNL.md` mentions the topic `tsix/net/*` but the code uses `mqtnl@1.0/`. **Code is the truth.**

⚠ **Dead code:** `ExecutableRegistry` is defined but not wired into the Kernel — binary resolution goes through `MountManager.resolve()` + `vfs.read()`.

⚠ **No inittab** — `init` hardcodes the TTY logins.

⚠ **Dual NetworkLib** — `UserLib.ts` (inline, `lib.net`) vs `src/mirror/lib/NetworkLib.ts` (legacy, `OSContext`-based) both target the same syscalls.

⚠ **Name-based privilege** — app name substring heuristic (fragile; ideally capability-based).

13. Code Reading Map

Start here (required)

File	Role
<code>src/main.ts</code>	Host entry point + keep-alive
<code>src/kernel/Kernel.ts</code>	Boot orchestrator for all subsystems
<code>src/common/IPCTypes.ts</code>	IPC contract (request/response/event)
<code>src/common/SyscallCode.ts</code>	Syscall ABI (1-66)
<code>src/common/GUITypes.ts</code>	PixelSpace contract

Kernel Core

File	Role
<code>src/kernel/Scheduler.ts</code>	PCB, processes, signals, reexec
<code>src/kernel/Syscalls.ts</code>	~65 syscall dispatcher + permission
<code>src/kernel/PermissionManager.ts</code>	rwX checks
<code>src/kernel/MountManager.ts</code>	Path → backend routing
<code>src/kernel/PortManager.ts</code>	Network ports
<code>src/kernel/GUIRegistry.ts</code>	Window authority

VFS & Devices

File	Role
<code>src/vfs/IVFS.ts</code>	Filesystem contract
<code>src/vfs/BKFS.ts</code>	Root FS (SQLite)

File	Role
src/vfs/VFS.ts / RamFS.ts / HostVFS.ts	Other backends
src/kernel/devices/IDevice.ts	HAL contract
src/kernel/devices/FileSystemDevice.ts	File ↔ IVFS bridge
src/kernel/devices/*.ts	Drivers: TTY, Keyboard, Pipe, Socket, MQTNL
src/kernel/devices/aux-devices/	Plugin drivers

Worker & Userland

File	Role
src/userland/WorkerEntry.ts	Worker bootloader + sandbox + DME
src/mirror/lib/UserLib.ts	Worker-side "libc"
src/mirror/lib/Application.ts	Program() v2.1 framework
src/mirror/lib/emerald.ts	Widget toolkit
src/mirror/bin/init.ts	PID 1

Networking & Protocols

File	Role
src/kernel/devices/SimpleMQTNLDriver.ts	MQTNL driver
src/common/protocols/IMQTNLProtocol.ts	Protocol contract
src/common/protocols/MQTNLProtocolJSON.ts	v1.0
src/common/protocols/MQTNLProtocolBinary.ts	v1.1

GUI/TDE

File	Role
src/mirror/bin/dome.ts	DOME server
src/mirror/bin/dome-client.html	DOME browser
src/mirror/bin/asteracea.ts	WM
src/mirror/lib/theme.ts	Theme

Appendix: Key data flow summary

Read a file:

```
fs.readFile("/etc/passwd")
→ OPEN: resolve → MountManager → BKFS → FileSystemDevice → fd
→ READ: fdTable[fd].device.read() → vfs.read → SQL SELECT
→ CLOSE: ioctl DEC_REF → fdTable[fd] = null
```

Spawn an app:

```
shell.exec("/bin/ls")
→ EXEC: resolve → vfs.read(appContent) → permission EXECUTE
→ createProcess → spawnWorker → WorkerEntry → DME app → execute()
```

Send a message between apps:

```
shell.send(pidOrUuid, {type, ...})  
→ SEND_MSG → kernel resolve pid (atau uuidMap) → sendEvent(pid, "ipc_message", data)
```

Open a window:

```
Screen → CREATE_WINDOW → kernel (GUIRegistry.auth) → DOME → WS → browser
```

TSIX Architecture Guide v1.0 — for contributors & developers. This document is alive — update it as the code changes.
"Code is the truth."

RFC-TSIX-EDU-002 | The first module of the TSIX curriculum. Builds a mental map: what TSIX is, why it was designed this way, and the five core principles that form the foundation of the entire system.

Before writing code, understand the **philosophy** first. TSIX is not just an "OS emulator" — it is an **OS abstraction built on top of the existing Node.js runtime**. This module explains *why* it was designed this way, and the five principles that are consistent across all subsystems.

Learning Objectives

- ☐ Explain what TSIX is and how it differs from a VM/emulator
- ☐ Name the five core principles of the TSIX architecture
- ☐ Explain why the kernel, drivers, and FS are combined into a single thread
- ☐ Recognize the Ring 0-4 layers and their contents in broad outline
- ☐ Explain why the kernel never executes applications

Core Concepts

What is TSIX?

TSIX is a **simulated operating system based on Node.js + TypeScript**. It is not a VM that emulates a CPU. It builds an **OS abstraction on top of the existing Node.js runtime** — using `Worker Thread` as the process boundary, `postMessage` as IPC, and `SQLite` as the filesystem.



With this approach, TSIX adopts **UNIX architecture concepts** — process isolation, UID/GID permissions, POSIX filesystem, signal handling, device abstraction (HAL), syscall communication — expressed in TypeScript on top of Node.js.

Why not a microkernel?

One of the most fundamental architectural decisions: **the kernel, drivers, and filesystem are combined into a single main thread**, not split into separate threads like a pure microkernel (Minix, QNX, seL4).

Reason	Explanation
IPC overhead	A pure microkernel = 3-4x <code>postMessage</code> per operation (App → Kernel → FS → Kernel → App). A single thread = 1x (App → Kernel). Each <code>postMessage</code> performs a structured clone (serialize → deserialize).
Worker limitation	Node.js Worker Threads cannot use shared memory . Each transfer = new allocation + data copy. The larger the data, the larger the overhead.

Reason	Explanation
Embedded target	TSIX is for IoT with limited CPU and small RAM. Each additional Worker = ~4-8MB heap.
Isolation already exists	Applications (Ring 4) are already in a separate worker → 90% of the isolation benefit at 10% of the cost.

The result: the most frequent crashes are user applications — and they are already isolated. Kernel + drivers + FS rarely crash, and even if they do, the system dies (but that is rare).

Five Core Principles

1. "Everything is a File"

Files and devices are both `IDevice`; `read/write` is polymorphic.

- Opening a regular file → `FileSystemDevice`
- Opening `/dev/tty1` → `TTYDevice`
- Opening `/dev/smqtnl0` → `SimpleMQTNLDriver`

Even pipes, sockets, and the display are devices. One contract, many implementations.

2. "Distributed by Design"

Built-in IPC via the `SEND_MSG` syscall + identity-based messaging. Processes communicate through identities (UUIDs), not memory addresses — fitting the platform's distributed nature.

3. "Small, Sharp Tools"

80+ utilities that combine via pipes & redirection — the Unix tradition: one tool does one thing well, then tools are composed.

4. "Security via Simplicity"

A UID/GID permission model, process isolation (Worker Threads), and straightforward root privileges — security comes from a clear design, not from complexity.

5. "Unix fidelity first, pragmatic later"

TSIX mirrors Unix/Linux behavior and architecture as closely as practical — syscall semantics, UID/GID permissions, credentials (saved UID), file formats (`/etc/shadow`, `passwd`), filesystem hierarchy — as the design **north star**. Not because it "must match exactly", but because **observable** behavior must be consistent: non-root cannot read `/etc/shadow`, a non-root process cannot `setuid` arbitrarily, and so on.

Deviating from Unix is **allowed**, but only when the Node.js/V8 runtime truly cannot model it — never as a "it's easier this way" shortcut. Every deviation **must be documented** (a changelog entry and/or a code comment) with a clear technical reason, so it is not mistaken for a bug.

[!NOTE] **Semantics > Mechanisms** — there is no need to emulate bare-metal (hardware interrupts, MMU, etc.). What matters is that behavior observed from userland matches Unix. Example: `setuid` is simulated with a saved UID (`pcb.suid`) in the kernel, not with a CPU register — but the effect is the same.

Source Code

File	Role
<code>src/main.ts</code>	Host entry point + keep-alive

File	Role
src/kernel/Kernel.ts	Boot orchestrator for all subsystems
src/kernel/Syscalls.ts	Syscall dispatcher
src/kernel/Scheduler.ts	Process management
src/userland/WorkerEntry.ts	Worker bootloader + sandbox
src/common/IPCTypes.ts	IPC contract

Exercises / Practice

1. Read `src/main.ts` — find where the 100ms keep-alive is. What happens if PID 1 exits with code 1?
2. Read `src/kernel/Kernel.ts` — list the order of `initializeSubsystems()`. Compare it with the boot diagram in Module 03.
3. Run TSIX (`npm start` / per `README.md`) and observe the boot log.

References

- `wiki/Arsitektur-Sistem.md` — layer diagram and microkernel analysis
- `wiki/course/00-overview.en.md §1` — global mental map
- `src/main.ts`, `src/kernel/Kernel.ts`

Module 01 — complete. Continue to [Module 02 — Ring Model & Privilege Boundaries](#).

RFC-TSIX-EDU-002 | The second module of the TSIX curriculum. Understand the "Ring" concept as a privilege boundary, and where the *actual* security boundary sits in TSIX.

Important: **A Ring in TSIX is a concept (documentation), not a hardware isolation mechanism.** TSIX runs on top of V8, not on bare metal. The real boundaries live in two layers: the **WorkerEntry sandbox** and the **PermissionManager (kernel)**.

Learning Objectives

- ☐ Explain the contents of each Ring 0-4 and its main files
- ☐ Distinguish a Ring as a concept vs. a real isolation mechanism
- ☐ Explain the two real privilege boundary layers: the WorkerEntry sandbox & the PermissionManager
- ☐ Explain what the setuid bit is and an example of its use
- ☐ Know the weakness of privilege based on app name

Core Concepts

Ring Map

Ring	Contents	Main files
0	Host: Linux + Node/V8	— (reserved)
1	Kernel core: Scheduler, Syscall, Permission	src/kernel/*, src/common/*
2	Drivers & FS: HAL devices, VFS backends, MountManager	src/kernel/devices/*, src/vfs/*
3	Library framework: UserLib, Application	src/mirror/lib/*
4	Apps: /bin/*, init, tsh, daemon	src/mirror/bin/*

 Ring 1 & 2 — kernel internals (dispatcher, scheduler, FS/Net/HAL stacks) Source: [wiki/diagram/Arsitektur-System-2.mmd](#)

A Ring is a **map of responsibility** — not isolation enforced by V8. Two files from different rings may call each other; what keeps the system secure are the rules below.

Comparison with Linux

Feature	Linux Equivalent	TSIX Ring
Core OS Logic	Ring 0 (Kernel Mode)	Ring 1
Drivers & FS	Ring 0 (Kernel Mode)	Ring 2
Standard Library	Ring 3 (glibc)	Ring 3
User Apps	Ring 3 (User Mode)	Ring 4

[!NOTE] **Ring 0 is the host domain.** Because TSIX is not bare metal, "Ring 0" means Linux/Windows + V8. TSIX starts numbering from Ring 1.

The Two Real Privilege Boundary Layers

Layer 1 — WorkerEntry Sandbox (worker side)

`src/userland/WorkerEntry.ts` is the **bootloader** of every worker. It locks the door before the app runs:

- An app may only require the `@tsix/*` / `@common/*` frameworks — everything else is blocked.
- `process.exit` / `process.kill` are sabotaged (thrown).
- **Privileged** apps (name contains `server`, `daemon`, `dome`, `tbuild`, `vfs`) get an **allow-list** of host modules: `http`, `ws`, `path`, `fs`, `url`, `esbuild`, `crypto`, `os`, `bcryptjs`.

Layer 2 — PermissionManager (kernel side)

`src/kernel/PermissionManager.ts` is the "security guard" in the kernel. It performs layered rwx checks:

```
check(pid, path, mode):
  root (uid 0) → bypass
  owner       → check owner bit
  group       → check group bit
  others      → check others bit
```

The **SetUID bit** (`004000`) is supported: a process runs with the file's ownership. Example: `/bin/login` has mode `004755` — whoever runs `login`, the process runs as root (to perform `setuid/setgid` to the target user).

Source Code

File	Role
<code>src/userland/WorkerEntry.ts</code>	Worker-side sandbox (restrictHostAPI)
<code>src/kernel/PermissionManager.ts</code>	rwx checks + setuid
<code>src/kernel/Syscalls.ts</code>	Call the permission check before an action
<code>src/mirror/bin/login.ts</code>	Example of setuid usage (<code>004755</code>)

Snippet (code level)

```
// src/kernel/PermissionManager.ts — core of the permission check (condensed)
check(pid: number, path: string, mode: number): boolean {
  const { uid, gid } = this.scheduler.getProcess(pid);
  const stat = this.getStat(path);
  if (uid === 0) return true; // root bypass
  if (uid === stat.uid) return !(stat.mode & mode); // owner
  if (gid === stat.gid) return !(stat.mode & (mode >> 3)); // group
  return !(stat.mode & (mode >> 6)); // others
}
```

[!WARNING] **Known weakness.** Privileged status is based on **app name substring** — a fragile heuristic. Anyone who names their app `my-daemon` automatically becomes privileged. Ideally this should be replaced with *capability-based*.

Exercises / Practice

1. Read `src/userland/WorkerEntry.ts` — find the allow-list of host modules.
2. Read `src/kernel/PermissionManager.ts` — test the rwx check logic with several uid/gid/mode combinations.
3. Run a non-privileged app that tries to `require("fs")` — observe the error that appears.

References

- [wiki/ARCHITECTURE_RINGS.md](#) — official ring definitions
 - [wiki/Keamanan-dan-Sandboxing.md](#) — sandbox details
 - [wiki/course/00-overview.md](#) §2
-

Module 02 — done. Continue to [Module 03 — Boot Sequence](#).

RFC-TSIX-EDU-002 | Third module of the TSIX curriculum. Traces the boot sequence from `main.ts` → `kernel.boot()` → PID 1 (`init`) → login, up to the 100ms keep-alive on the host.

TSIX boot mimics a UNIX/Linux boot: from host bootstrap → kernel subsystem init → `init` (PID 1) → `rc.local` → `getty/login` per TTY. What decides whether the system is "alive or dead" is the **100ms keep-alive** in `main.ts`.

Learning Objectives

- ☐ Explain the role of `main.ts` and the 100ms keep-alive
- ☐ List the order of `initializeSubsystems()` in the kernel
- ☐ Explain the content and role of `init` (PID 1)
- ☐ Explain the order of daemons in `rc.local`
- ☐ Explain what exit code 1 means when PID 1 dies (reboot)

Flow / How It Works

TSIX boot follows the classic UNIX/Linux pattern: host bootstrap → kernel subsystem init → `init` (PID 1) in a Worker → `rc.local` → `getty/login` per TTY → host keep-alive. The diagram below summarizes the entire sequence, from the first second the host starts until the system is "alive".

```
sequenceDiagram
    participant H as Host (Node.js)
    participant M as main.ts
    participant K as Kernel.ts
    participant V as VFS (BKFS/SQLite)
    participant W as Worker Thread (Ring 4)
    participant I as init (PID 1)
    participant R as rc.local
    participant L as login (TTY1-6)

    H->>M: node src/main.ts
    M->>M: Config.load() → sysconfig.conf
    M->>K: new Kernel() (Logger, PermissionManager, MountManager, GUIRegistry)
    M->>K: await kernel.boot()

    activate K
    K->>K: initializeSubsystems()
    K->>V: mount BKFS "/" (system.db)
    K->>K: processFstab() → /tmp (RamFS), /mnt/shared, /mnt/sbak
    K->>K: new Scheduler() + new PermissionManager() + new PortManager()
    K->>K: new SyscallDispatcher() → scheduler.setSyscallHandler(...)
    K->>K: rebuildVFSCache() → pre-compile /lib (DME)
    K->>K: TTYManager(32) + /dev devices (stdin, fb0, stdout, stderr, null, tty1-32)
    K->>K: network interfaces (SimpleMQTNLDriver) + loadAuxDevices() + SerialDeviceManager
    K->>K: applyDeviceConfigs() (udev) + pastikan /dev di VFS
    K->>K: load RSA identity → fingerprint visual
    K->>K: ensureDefaultAuth() + ensureDefaultGroups()
    K-->>M: boot() selesai

    M->>K: kernel.runInit()
    K->>V: resolve /bin/init.js (cfg.scheduler.bootEntry)
    K->>W: createProcess("init", fds:[tty1x3]) → PID 1
    K->>K: setForegroundProcess(1, 1)
    deactivate K

    activate W
    activate I
    I->>I: enforce setuid (/bin/passwd.js, /bin/sudo.js)
    I->>I: RSA identity: generate/verify keypair + fingerprint
```

```

I->>R: exec /etc/rc.local.js + waitpid
activate R
R->>R: airtermd → tpkgd → scpd → otad → iot-listener
R->>R: tunggu /var/run/dome.ready → dome → asteracea
R->>R: crond → mysqld
R->>I: exit 0
deactivate R

I->>L: spawnLogin TTY2-6 (monitorProcess → respawn)
I->>L: spawnLogin TTY1 (foreground)
I->>I: while(true) { wait 10s } ← PID 1 tak pernah exit
deactivate I
deactivate W

M->>M: keepAlive: setInterval(100ms)
M->>M: scheduler.getProcess(1) → EXITED?
M->>M: exitCode === 1 → reboot | else → power off

```

[!NOTE] This mermaid version is more complete than `wiki/boot_sequence.md` and follows the current code (`Kernel.ts`, `init.ts`, `rc.local.ts`).

Concise flow (ASCII)

```

main.ts
Config.load()           → baca sysconfig.conf
new Kernel()            → Logger, PermissionManager, MountManager, GUIRegistry
kernel.boot()
  initializeSubsystems()
    mount BKFS("/")      → SQLite system.db (root filesystem)
    processFstab()       → /tmp (RamFS), /mnt/shared (HostVFS), /mnt/sbak (BKFS)
    new Scheduler()
    new PermissionManager()
    new PortManager()
    new SyscallDispatcher() → scheduler.setSyscallHandler(...)
    rebuildVFSCache()    → pre-compile /lib ke memori (DME)
  new TTYManager(32) + TTYDevice tty1..32
  devices{}              → stdin, fb0, stdout, stderr, null
  init network interfaces → SimpleMQTNLDriver per config
  loadAuxDevices()       → plugin driver (random, mysql, mcp23017)
  SerialDeviceManager    → auto-detect ttyUSB*
  applyDeviceConfigs()   → "udev": mode/uid/gid dari sysconfig
  pastikan /dev ada di VFS
  load RSA identity      → fingerprint visual di TTY
  ensureDefaultAuth()    → seed /etc/passwd + /etc/shadow (root)
  ensureDefaultGroups()  → seed users (100) + sudo (27)
  wire keyboard + TTY callbacks → Ctrl+C → SIGINT fg; Alt+F1-6 → switch
kernel.runInit()
  resolve /bin/init.js
  createProcess("init", fds:[tty1×3]) → PID 1
  setForegroundProcess(1, 1)
[init.ts di Worker]
  enforce setuid (passwd, sudo)
  generate/verify RSA identity
  exec /etc/rc.local.js + waitpid
  spawn login TTY2..6 (monitorProcess → respawn)
  spawn login TTY1 (foreground) → loop forever
  keepAlive (100ms di main.ts)
  jika PID 1 EXITED → process.exit(exitCode) (1 = reboot)

```

initializeSubsystems() order (from Kernel.ts)

```

this.bootLogStart("VFS: Mounting root filesystem (BKFS/SQLite)");
this.bkfs = new BKFS(cfg.kernel.database);
this.mountManager.mount("/", this.bkfs, "bkfs", cfg.kernel.database, false);
this.bootLogEnd(true);

await this.processFstab();           // /tmp (RamFS), /mnt/* (HostVFS/BKFS)

```

```

this.scheduler = new Scheduler();           // Core: Process Scheduler
this.satpam = new PermissionManager();      // Security: Permission Manager
this.portManager = new PortManager();      // Network: Port Manager
this.syscall = new SyscallDispatcher(
  this.bkfs, this.mountManager, this.scheduler, this, this.satpam,
);
this.scheduler.setSyscallHandler(async (req) => {
  return await this.syscall!.handleRequest(req);
});
this.rebuildVFSCache();                    // VFS: pre-compile /lib (DME)
this.scheduler.setVFSCacheProvider(() => this.vfsCache);

```

rc.local order (daemons, from rc.local.ts)

1. **SetUID** /bin/login.js → 0o4755, chown root:root.
2. airtermd (/sbin/airtermd.js) — remote access.
3. tpkgd (/sbin/tpkgd.js) — package server (TPKG).
4. scpd (/sbin/scpd.js) — SCP file transfer.
5. otad (/sbin/otad.js) — ESP firmware OTA.
6. iot-listener (/sbin/iot-listener.js) — MQTNL sensor.
7. Clean old marker /var/run/dome.ready, then **wait for DOME ready** (polling 200ms, timeout 10s).
8. dome (/opt/dome/dome.js) — display server (PixelSpace).
9. asteracea (/opt/asteracea/asteracea.js) — window manager (after DOME is ready).
10. crond (/sbin/crond.js) — cron scheduler.
11. mysqld (/opt/mysqld/mysqld.js) — database service.

[!WARNING] **Documentation vs code.** The rc.local.ts source contains the 11 steps above. However, the compiled rc.local.js file (the one actually executed) only contains 3 daemons: airtermd, tpkgd, scpd. **Code is truth** — at runtime, only the first 3 daemons actually run from rc.local.

Source Code

File	Role
src/main.ts	Host entry point + keep-alive
src/kernel/Kernel.ts	boot() + initializeSubsystems() + runInit()
src/mirror/bin/init.ts	PID 1: setuid, identity, rc.local, spawn login
src/mirror/etc/rc.local.ts	Startup daemon list
src/common/Config.ts	Reads sysconfig.conf

Snippet (code level)

Keep-alive in main.ts (the host must not die)

```

const keepAlive = setInterval(() => {
  const scheduler = kernel.getScheduler();
  const pl = scheduler?.getProcess(1);
  if (!pl || pl.state === "EXITED") {
    clearInterval(keepAlive);
    if (process.stdin.isTTY) {
      (process.stdin as any).setRawMode(false);
    }
  }

  const exitCode = (kernel as any).wantedExitCode ?? (process.exitCode ?? 0);
  if (exitCode === 1) {

```

```

        console.log("\n[Kernel] System is rebooting...");
    } else {
        console.log("\n[Kernel] System halted. Powering off...");
    }
    process.exit(exitCode);
}
}, 100); // 100ms biar satset responnya

```

Key point: **as long as PID 1 is alive, the host does not stop**. When PID 1 exits with **code 1** → reboot; otherwise → power off. Before exiting, the terminal raw mode is restored so the host terminal is not left "broken".

Boot log: `bootLogStart()` + `bootLogEnd()`

`Kernel.ts` prints boot progress with a `[ ]` bracket that updates in place (`\r` + ANSI `\x1b[K`). Usage pattern:

```

this.bootLogStart("VFS: Mounting root filesystem (BKFS/SQLite)");
this.bkfs = new BKFS(cfg.kernel.database);
this.mountManager.mount("/", this.bkfs, "bkfs", cfg.kernel.database, false);
this.bootLogEnd(true); // → [ OK ] VFS: Mounting root...

```

`bootLogEnd(false, "pesan")` → `[ FAIL ]`. Both are no-ops if `cfg.kernel.verbose` is false.

runInit: spawning PID 1

```

const initPath = "/bin/" + cfg.scheduler.bootEntry; // bootEntry = "init.js"
const res = this.mountManager.resolve(initPath);
const raw = res.vfs.read(res.relativePath); // kontrak IVFS, bukan stat()

const initPcb = this.scheduler?.createProcess("init", {
  fds: [tty1, tty1, tty1],
  appName: "init",
  appContent: initContent,
  env: {
    PATH: cfg.scheduler.defaultPath,
    HOME: "/root",
    HOSTNAME: cfg.shell.defaultHostname,
    PROMPT_FORMAT: cfg.shell.promptFormat,
    LINES: (process.stdout.rows || cfg.shell.defaultRows).toString(),
    COLUMNS: (process.stdout.columns || cfg.shell.defaultColumns).toString(),
  },
  cwd: cfg.scheduler.defaultCwd,
  ttyId: 1,
});
if (initPcb) this.scheduler?.setForegroundProcess(initPcb.pid, 1);

```

init: enforce setuid

```

// Runtime mengeksekusi sidcar .js (bukan source .ts), jadi chmod harus di .js
await lib.fs.chmod("/bin/passwd.js", 2541); // 2541 = 0o4755 (setuid root)
await lib.std.print(`${ok} [INIT] SetUID bit applied to: /bin/passwd.js\n`);
await lib.fs.chmod("/bin/sudo.js", 2541);
await lib.std.print(`${ok} [INIT] SetUID bit applied to: /bin/sudo.js\n`);
await lib.std.print(`${ok} [INIT] System binary permissions enforced.\n`);

```

[!NOTE] 2541 decimal = 0o4755 octal (setuid + rwxr-xr-x). This SetUID lets `passwd` and `sudo` elevate privileges to root even when run by a regular user.

init: exec `rc.local` + `waitpid`

```

const result = await lib.shell.exec("/etc/rc.local.js", [], undefined, undefined, undefined);
if (result) {
  const exitCode = await lib.shell.waitpid(result.pid);
  if (exitCode === 0) {

```

```

    await lib.std.print(`${ok} [INIT] Startup scripts completed successfully.\n`);
  } else {
    await lib.std.print(`Init: Warning - rc.local exited with code ${exitCode}\n`);
  }
}

```

init: spawnLogin + monitorProcess (respawn to avoid crash-loop)

```

const spawnLogin = async (ttyId: number) => {
  try {
    if (ttyId > 1)
      await lib.std.print(`${ok} Initializing session on TTY${ttyId}...\n`);
    const result = await lib.shell.exec("/bin/login.js", [], undefined, undefined, ttyId);
    if (result && result.pid) {
      terminals.set(ttyId, result.pid);
      await lib.std.log(`Login service spawned on TTY${ttyId} (PID ${result.pid})`, "init");
      // Kita nungguin di background (thread terpisah di Worker)
      this.monitorProcess(lib, ttyId, result.pid, spawnLogin);
    }
  } catch (e: any) {
    await lib.std.print(`Init: Error spawning login on TTY${ttyId}: ${e.message}\n`);
  }
};

for (let i = 2; i <= 6; i++) await spawnLogin(i); // TTY2-6 di background
await spawnLogin(1); // TTY1 foreground

```

`monitorProcess` waits on `waitpid`; if login dies, it waits 1 second then `respawns` — preventing crash-loop spam:

```

private async monitorProcess(
  lib: UserLib, ttyId: number, pid: number, respawn: (tty: number) => Promise<void>,
) {
  const exitCode = await lib.shell.waitpid(pid);
  await lib.std.print(
    `Init: Process on TTY${ttyId} (PID ${pid}) exited with code ${exitCode}. Respawning...\n`,
  );
  await new Promise(r => setTimeout(r, 1000)); // delay anti crash-loop
  await respawn(ttyId);
}

```

init: eternal loop (PID 1 must never exit)

```

// Init process must never exit
while (true) {
  await new Promise(r => setTimeout(r, 10000));
}

```

rc.local: waiting for DOME ready (marker, not a fixed sleep)

```

const DOME_READY_MARKER = "/var/run/dome.ready";
const DOME_WAIT_MS = 10000;
const DOME_POLL_MS = 200;
let domeReady = false;
const waitStart = Date.now();
while (Date.now() - waitStart < DOME_WAIT_MS) {
  try {
    if (await lib.fs.stat(DOME_READY_MARKER)) { domeReady = true; break; }
  } catch (_) { /* marker belum ada */ }
  await new Promise(r => setTimeout(r, DOME_POLL_MS));
}

```

Asteracea is only started after the marker appears — otherwise, `CREATE_WINDOW` is rejected by the kernel ("GUI_REQ: DOME engine is not running") and the screen stays blank.

ensureDefaultAuth / ensureDefaultGroups (identity seeding)

At the end of `boot()`, the kernel ensures the identity files exist before userland runs:

```
// ensureDefaultAuth() - seed bila belum ada
const rootPasswd = "root:x:0:0:root:/root:/bin/tsh.ts\n";      // /etc/passwd
const rootShadow = "root:$2b$10$...KupG:19750:0:99999:7:::\n";  // /etc/shadow (bcrypt "root")

// ensureDefaultGroups() - group wajib
missing.push("users:x:100:");  // GID 100
missing.push("sudo:x:27:");    // GID 27 (gaya Ubuntu)
```

Exercises / Practice

1. Run `npm start` — note the boot log order (`[ OK ]` / `[ FAIL ]`) and compare it with the mermaid diagram above.
 2. Kill the `init` process from inside the system (`kill 1` as root) — what happens to the host? (100ms keep-alive → exit code → reboot/halt).
 3. Read `src/mirror/bin/init.ts` — where are `enforce setuid` and RSA identity called? How many login TTYs are spawned?
 4. Read `src/mirror/etc/rc.local.ts` — match the 11 daemon steps above with the code. Compare with `rc.local.js` (only 3 daemons) — why are they different?
 5. Read `src/kernel/Kernel.ts` — find the order of `initializeSubsystems()` and `rebuildVFSCache()`; explain the role of DME (Direct Memory Execution).
-

References

- `wiki/boot_sequence.md` — mermaid sequence diagram
 - `wiki/course/00-overview.en.md §3` — Boot Sequence
 - `src/main.ts` — host entry point + keep-alive
 - `src/kernel/Kernel.ts` — boot, `initializeSubsystems`, `runInit`, `ensureDefaultAuth`
 - `src/mirror/bin/init.ts` — PID 1 (setuid, identity, rc.local, spawn login)
 - `src/mirror/etc/rc.local.ts` — startup daemons
 - `src/common/Config.ts` — reads `sysconfig.conf`
-

Module 03 — done. Continue to [Module 04 — Processes & Scheduler](#).

RFC-TSIX-EDU-002 | Fourth module of the TSIX curriculum. Understand the process lifecycle in TSIX: PCB, spawn → run → block → exit, zombie, orphan reparent, daemonize, and reexec.

One worker thread = one process. TSIX multitasking is **Node.js concurrency**, not preemption — the kernel never forcibly stops a process. What the kernel manages is **the lifecycle and signals**.

Learning Objectives

- ☐ State the main `PCB` fields and distinguish enum states vs transient states
- ☐ Explain the lifecycle: `READY` → `RUNNING` → `BLOCKED` → `EXITED` (including zombie, orphan, reexec)
- ☐ Explain `waitpid` + `reap` (zombie) and reparenting orphans to `PID 1`
- ☐ Explain `daemonize` (`DETACH`) and `REEXEC`
- ☐ Explain the signal model (`SIGKILL`, `SIGINT`, `SIGTERM`, `SIGSTOP/CONT`) and how it is delivered
- ☐ Trace the real flow: `spawn` → `waitpid` → `exit` → `reap`

Core Concepts

PCB (Process Control Block)

Every TSIX process is a single **worker thread**. The kernel (main thread) never runs application code — it only manages process metadata through the **PCB** (interface `PCB` in `src/kernel/Scheduler.ts`). One PCB = one process.

PCB Fields

Field	Type	Meaning
<code>pid</code>	<code>number</code>	Unique process ID; allocated from <code>nextPid++</code>
<code>ppid?</code>	<code>number</code>	Parent PID — forms the process tree
<code>name</code>	<code>string</code>	Process name (e.g. <code>"Shell"</code> , <code>"init"</code>)
<code>state</code>	<code>ProcessState</code>	<code>READY</code> / <code>RUNNING</code> / <code>BLOCKED</code> / <code>EXITED</code>
<code>pc</code>	<code>number</code>	Program counter — address of the next instruction
<code>owner</code>	<code>string</code>	Name of the owning user (e.g. <code>"root"</code>)
<code>uid</code>	<code>number</code>	User ID (effective)
<code>gid</code>	<code>number</code>	Group ID (effective)
<code>ruid</code>	<code>number</code>	Real user ID — who actually launched it
<code>groups</code>	<code>number[]</code>	Supplementary groups
<code>cwd</code>	<code>string</code>	Current working directory
<code>worker?</code>	<code>Worker</code>	Worker thread — the process body (exists after <code>spawnWorker</code>)
<code>fdTable</code>	<code>(FDEntry null)[]</code>	File descriptor table; 0=stdin, 1=stdout, 2=stderr
<code>env</code>	<code>Record<string, string></code>	Environment variables
<code>exitCode?</code>	<code>number</code>	Explicit exit code from the <code>EXIT</code> syscall
<code>ttyId?</code>	<code>number</code>	Virtual console (TTY) where the process runs
<code>uuid?</code>	<code>string</code>	Application identity — persistent across PIDs (for <code>SEND_MSG</code> by identity)

```

export interface PCB {
  pid: number;           // ID Unik Proses
  ppid?: number;         // Parent PID – buat process tree
  name: string;          // Nama Proses (misal: "Shell")
  state: ProcessState;   // Status Saat Ini
  pc: number;            // Program Counter (Alamat instruksi berikutnya)

  owner: string;         // User yang menjalankan (misal: "root")
  uid: number;           // User ID (Effective)
  gid: number;           // Group ID (Effective)
  ruid: number;          // Real User ID (Siapa yang aslinya nge-jalanin)
  groups: number[];      // Supplementary Groups
  cwd: string;           // Current Working Directory (Posisi sekarang)
  worker?: Worker;       // Raga asli proses di thread terpisah

  // FD TABLE sekarang berisi object FDEntry yang menunjuk ke Driver nyata.
  fdTable: (FDEntry | null)[];

  // Environment Variables
  env: Record<string, string>;
  exitCode?: number;     // Explicit exit code set by Syscall
  ttyId?: number;        // ID Virtual Console (TTY) tempat proses ini berjalan
  uuid?: string;         // Unique Application Identity (Persistent across PID changes)
}

```

[!NOTE] **State** in TSIX has only the four enum values of `ProcessState`: `READY`, `RUNNING`, `BLOCKED`, `EXITED`. There is no `NEW` state — the PCB is already fully formed when `createProcess()` returns it. `REEXECING` is **not** part of the enum; it is a transient state set via `(pcb as any).state = "REEXECING"` to prevent the cleanup hook during `reexec()`.

Each `fdTable` entry is an `FDEntry`:

```

export interface FDEntry {
  device: IDevice;       // Driver nyata (FileSystemDevice, TTYDevice, dll)
  context: any;          // Konteks driver (path, offset, ...)
  flags?: string;        // "r" | "w" | ...
}

```

Lifecycle

```

stateDiagram-v2
    [*] --> READY : createProcess()\nnpid = nextPid++ → PCB
    READY --> RUNNING : spawnWorker()\nWorker Thread dibuat
    RUNNING --> BLOCKED : SIGSTOP(19) / menunggu event / IPC
    BLOCKED --> RUNNING : SIGCONT(18) / event selesai
    RUNNING --> EXITED : worker.on("exit")
    EXITED --> ZOMBIE : tidak ada waiter waitpid
    ZOMBIE --> [*] : waitpid() → reap()
    RUNNING --> REEXECING : reexec()\nterminate worker lama
    REEXECING --> READY : PCB sama, worker baru (spawnWorker)
    note right of EXITED
        orphan: semua child (ppid === pid)
        di-reparent ke init (PID 1)
    end note

```

[!NOTE] `course-server.ts` does not render mermaid yet (it displays as a code block); text version: `READY` → `RUNNING` → `BLOCKED` → `EXITED`, where `EXITED` without a waiter becomes a **zombie** (it stays in the table until `waitpid` + `reap`).

Event	Mechanism
Spawn	<code>createProcess()</code> → <code>nextPid++</code> → PCB (<code>READY</code>) → <code>spawnWorker()</code> when <code>appName/appPath</code> exists → state <code>RUNNING</code>

Event	Mechanism
Exit	Syscall <code>EXIT(code)</code> → set <code>pcb.exitCode</code> → <code>cleanupProcess()</code> (close FDs, release ports) → <code>kill(pid)</code> → worker "exit" event → <code>EXITED</code>
Orphan	<code>worker.on("exit")</code> handler → all children (<code>ppid === pid</code> , not yet <code>EXITED</code>) are set <code>ppid = 1</code>
Zombie	Process <code>EXITED</code> without a waiter → stays in the table until <code>waitpid + reap()</code>
waitpid	Already <code>EXITED</code> → <code>reap()</code> + return <code>exitCode</code> ; not yet → registered in <code>waitQueue</code>
daemonize	<code>DETACH</code> → FDs 0/1/2 redirected to <code>/dev/null</code> , resolve waiters, detach foreground TTY, clear <code>ttyId</code>
REEXEC	Set (<code>pcb</code> as any). <code>state</code> = <code>"REEXECING"</code> → terminate old worker → same PCB → new <code>spawnWorker()</code>
REARENT (syscall)	<code>REARENT { pid, newPpid }</code> → <code>childPcb.ppid = newPpid</code> (manual; verify the parent exists or is PID 1)

Walkthrough: spawn → waitpid → exit → reap

Follow the real flow of a `sleep 2` process launched from the shell (`tsh`):

1. **Spawn.** `tsh` (e.g. PID 3) calls the `EXEC` syscall → the kernel calls `createProcess("sleep", { appPath: "/bin/sleep.js", ppid: 3, fds: [...], ttyId: 1 })`. The PCB is created in state `READY`, then `spawnWorker()` creates a Worker Thread and sets state `RUNNING`. The process gets `pid = 42`.
2. **waitpid.** `tsh` calls `WAITPID(42)` → `waitpid(42)`. The process is not finished → `setForegroundProcess(42, 1)` (so Ctrl+C goes to `sleep`), then the resolver is registered in `waitQueue[42]`. `waitpid` returns a Promise.
3. **Exit.** `sleep` finishes → it sends the `EXIT(0)` syscall → the kernel sets `pcb.exitCode = 0` → `cleanupProcess(42)` closes all FDs and releases ports → `kill(42)` (default `SIGKILL`) → `worker.terminate()`.
4. **Reap.** Node triggers the worker "exit" event → state `EXITED` → `finalCode = 0` → `onProcessExitCallback` (global cleanup) → `reparent orphans` (none) → `waitQueue[42]` has a waiter → `resolve(0)` (the shell receives exit code 0) → `reap(42)` removes the PCB from the table. No zombie. The foreground TTY is returned to `null`/the shell.
5. **Zombie (anti-pattern).** If the shell does **not** call `waitpid`, the exit handler logs `became a ZOMBIE` and the PCB stays in the table. The PCB is removed later when `waitpid(42)` is called (`reap`).

Signals

Signals are sent through `scheduler.kill(pid, signal)` — the only gateway. Two syscalls use it:

- `KILL(pid)` → always **SIGKILL (9)**.
- `SIGNAL({ pid, sig })` → any signal according to `sig`.

Both perform the same checks: the process exists, **PID 1 is protected**, and permission (root or process owner).

Signal	Value	Effect
SIGKILL	9	Hard kill: <code>worker.terminate()</code> immediately, no event
SIGINT	2	Graceful: push <code>SIGINT</code> event, 100ms grace → terminate if unhandled. Default exit 130
SIGTERM	15	Graceful: push event, 300ms grace → terminate. Default exit 143. Used by SHUTDOWN
SIGSTOP	19	Soft: <code>pcb.state = BLOCKED</code> + event
SIGCONT	18	Soft: <code>pcb.state = RUNNING</code> + event
SIGSEGV	—	Sent by the kernel when a process violates GUI window ownership
SIGHUP (1), SIGUSR1 (10), SIGUSR2 (12), etc.	—	Generic event push <code>sendEvent(pid, "signal", name)</code>

 Ctrl+C → `SIGINT` to the foreground process of the active TTY Source: [wiki/diagram/Kernel-dan-Scheduler-3.mmd](https://mmd.wiki/diagram/Kernel-dan-Scheduler-3.mmd)

[!IMPORTANT] **PID 1 (init) is protected** — both `KILL` and `SIGNAL` refuse to target PID 1, even for root. The only way to terminate the system is the `SHUTDOWN/REBOOT` syscall.

Staged SHUTDOWN (root only, `args = 0` shutdown / 1 reboot):

1. `broadcastEvent("signal", "SIGTERM")` to all processes (except itself and PID 1).
2. Wait a grace of up to 5 s (100 ms polling) — all processes `EXITED` → success.
3. Remaining processes that did not respond → `kill(pid, 9)` (`SIGKILL`).
4. Flush network 1 s (so the last packets like `!exit!` get sent).
5. `kill(1, 9, exitCode)` → PID 1 is terminated; `main.ts` `keepAlive` reads the exit code (1 = reboot).

Source Code

File	Role
<code>src/kernel/Scheduler.ts</code>	PCB, lifecycle, waitpid, detach, signals, reexec
<code>src/kernel/Syscalls.ts</code>	KILL, WAITPID, EXEC, EXIT, DETACH, REEXEC syscalls
<code>src/common/IPCTypes.ts</code>	Signal event contract

Snippets (code level)

All snippets below are copied from `src/kernel/Scheduler.ts` and `src/kernel/Syscalls.ts` (code is truth).

createProcess (spawn)

```
public createProcess(name: string, options: SpawnOptions = {}): PCB | null {
  const pcb: PCB = {
    pid: this.nextPid++,
    name: name,
    state: ProcessState.READY,
    pc: 0,
    owner: options.owner || "root",
    uid: options.uid ?? 0,
    gid: options.gid ?? 0,
    ruid: options.ruid ?? (options.uid ?? 0),
    groups: options.groups ?? [],
    cwd: options.cwd ?? "/",
    fdTable: [],
    env: options.env ?? {},
    ttyId: options.ttyId,
    ppid: options.ppid ?? 0,
  };

  // Initialize FDs (0=stdin, 1=stdout, 2=stderr)
  if (options.fds && options.fds.length >= 3) {
    pcb.fdTable = [
      { device: options.fds[0], context: "", flags: "r" },
      { device: options.fds[1], context: "", flags: "w" },
      { device: options.fds[2], context: "", flags: "w" }
    ];
  }

  this.processes.push(pcb);

  // Worker baru hanya dibuat bila ada appName / appPath
  if (options.appName || options.appPath) {
    this.spawnWorker(pcb, options);
  }
}
```

```
    return pcb;
}
```

[!NOTE] `createProcess` only creates the PCB (state `READY`) + (if an app exists) a Worker Thread. `pid = nextPid++`; standard FDs (0/1/2) are filled immediately when `options.fds ≥ 3`.

State transitions (spawnWorker, SIGSTOP/CONT, exit)

```
// spawnWorker() - worker dibuat → RUNNING
pcb.worker = new Worker(workerPath, { workerData, execArgv });
pcb.state = ProcessState.RUNNING;

// kill(pid, 19) - SIGSTOP → BLOCKED
pcb.state = ProcessState.BLOCKED;

// kill(pid, 18) - SIGCONT → RUNNING
pcb.state = ProcessState.RUNNING;

// Handler worker.on("exit") → EXITED (kecuali sedang REEXECING)
pcb.state = ProcessState.EXITED;
```

waitpid + reap (zombie)

```
public async waitpid(pid: number): Promise<number> {
    const pcb = this.getProcess(pid);

    // Jika proses sudah EXITED, langsung balikin 0 (atau simpan exit code di PCB)
    if (!pcb || pcb.state === ProcessState.EXITED) {
        const code = pcb?.exitCode ?? 0;
        // Jika proses sudah EXITED, kita reap sekalian biar gak jadi zombie
        this.reap(pid);
        return code;
    }

    // Proses yang ditunggu (foreground) berhak menerima interrupt Ctrl+C
    this.setForegroundProcess(pid, pcb.ttyId);

    return new Promise((resolve) => {
        if (!this.waitQueue.has(pid)) {
            this.waitQueue.set(pid, []);
        }
        this.waitQueue.get(pid)!.push(resolve);
    });
}
```

```
public reap(pid: number): void {
    const index = this.processes.findIndex(p => p.pid === pid);
    if (index !== -1) {
        const pcb = this.processes[index];
        if (pcb.state === ProcessState.EXITED) {
            this.logger.debug(`Reaping Process ${pcb.name} (PID ${pid}). Memory freed.`);

            // Remove from uuidMap if registered
            if (pcb.uuid && this.uuidMap.get(pcb.uuid) === pid) {
                this.uuidMap.delete(pcb.uuid);
                this.logger.debug(`UUID ${pcb.uuid} released from PID ${pid}`);
            }

            this.processes.splice(index, 1); // PCB dihapus dari tabel
        }
    }
}
```

[!IMPORTANT] `reap()` only removes PCBs in the `EXITED` state. As long as `waitpid` has not been called, the PCB stays in the table — that is the **zombie**.

Auto-reparent orphans + notify waiters (the `worker.on("exit")` handler)

```
pcb.worker.on("exit", (code) => {
  // If it was a reexec, we don't trigger the exit logic yet
  if (pcb.state === (ProcessState as any).REEXECING) return;

  pcb.state = ProcessState.EXITED;

  // Use explicit exitCode if set (from EXIT syscall), otherwise use worker exit code
  const finalCode = pcb.exitCode !== undefined ? pcb.exitCode : (code || 0);

  // Global Cleanup Hook (reset terminal, close FDs, etc)
  if (this.onProcessExitCallback) {
    this.onProcessExitCallback(pcb.pid);
  }

  // Auto-reparent orphans: semua child dari proses yang exit → init (PID 1)
  const allProcs = Array.from(this.processes.values());
  const orphans = allProcs.filter((p: PCB) => p.ppid === pcb.pid && p.state !== ProcessState.EXITED);
  for (const orphan of orphans) {
    orphan.ppid = 1;
    this.logger.info(`[REARENT] Auto-reparent orphan PID ${orphan.pid} (${orphan.name}) to init (PID 1)`);
  }

  // Notify Wait Queue
  if (this.waitQueue.has(pcb.pid)) {
    const waiters = this.waitQueue.get(pcb.pid)!;
    waiters.forEach(resolve => resolve(finalCode));
    this.waitQueue.delete(pcb.pid);
    // Jika ada yang nunggu (waitpid), kita bisa reap sekarang
    this.reap(pcb.pid);
  } else {
    this.logger.warn(`Process [${pcb.pid}] became a ZOMBIE (Needs waitpid)`);
  }

  // Jika proses foreground yang mati, kembalikan ke NULL
  if (this.getForegroundProcess(pcb.ttyId) === pcb.pid) {
    this.setForegroundProcess(null, pcb.ttyId);
  }
});
```

reexec (same PID, new body)

```
public async reexec(pid: number, appPath: string, args: string[]): Promise<boolean> {
  const pcb = this.getProcess(pid);
  if (!pcb || !pcb.worker) return false;

  // 1. Mark as Re-execing to prevent 'exit' hook cleanup
  (pcb as any).state = "REEXECING"; // Custom transient state

  // 2. Terminate current worker (Non-blocking to caller)
  pcb.worker.terminate().catch(() => { });
  pcb.worker = undefined;

  // 3. Update PCB info
  pcb.name = path.basename(appPath);
  pcb.state = ProcessState.READY;
  pcb.exitCode = undefined;

  // 4. Spawn new worker
  this.spawnWorker(pcb, { appPath, args });

  return true;
}
```

[!TIP] REEXECING is not a member of the `ProcessState` enum — it is a transient state set via `(pcb as any).state`. Its purpose: the `worker.on("exit")` handler skips the cleanup logic because of the `return` on the first line.

detach (daemonize)

```
public async detach(pid: number): Promise<boolean> {
  const pcb = this.getProcess(pid);
  if (!pcb) return false;

  // 1. Resolve Wait Queue (si Shell jadi gak nungguin lagi)
  if (this.waitQueue.has(pid)) {
    const waiters = this.waitQueue.get(pid)!;
    waiters.forEach(resolve => resolve(0)); // Kasih status sukses (0)
    this.waitQueue.delete(pid);
  }

  // 2. Lepas dari Foreground TTY (biar Ctrl+C gak masuk sini lagi)
  if (pcb.ttyId && this.ttyForegroundPids.get(pcb.ttyId) === pid) {
    this.ttyForegroundPids.delete(pcb.ttyId);
  }

  // 3. Clear TTY Association (Crucial for ps display)
  pcb.ttyId = undefined;

  return true;
}
```

[!NOTE] At the syscall level, `DETACH` also redirects FDs 0/1/2 to `/dev/null` before calling `scheduler.detach()` — so the daemon does not leak logs to the original TTY.

Signal delivery (scheduler.kill)

```
public async kill(pid: number, signal: number = 9, exitCode: number = 0): Promise<boolean> {
  const pcb = this.getProcess(pid);
  if (!pcb || !pcb.worker) return false;

  if (signal === 9) { // SIGKILL (Hard Kill)
    if (pcb.worker) {
      await pcb.worker.terminate().catch(() => { });
    }
    return true;
  }

  if (signal === 2) { // SIGINT (Ctrl+C)
    // Send event first to allow graceful handling
    const eventSent = this.sendEvent(pid, "signal", "SIGINT");
    // If no listener handled it, terminate after a brief delay
    // This preserves default Unix behavior: unhandled SIGINT = process exit
    setTimeout(async () => {
      const stillRunning = this.getProcess(pid);
      if (stillRunning && stillRunning.state !== ProcessState.EXITED && stillRunning.worker) {
        this.logger.debug(`PID ${pid} did not handle SIGINT, terminating...`);
        stillRunning.worker.terminate().catch(() => { });
      }
    }, 100); // 100ms grace period for handler to respond
    return eventSent;
  }

  if (signal === 19) { // SIGSTOP (Pause)
    pcb.state = ProcessState.BLOCKED;
    this.logger.info(`Process [${pid}] ${pcb.name} PAUSED (SIGSTOP)`);
    return this.sendEvent(pid, "signal", "SIGSTOP");
  }

  if (signal === 18) { // SIGCONT (Resume)
```

```

    pcb.state = ProcessState.RUNNING;
    this.logger.info(`Process [${pid}] ${pcb.name} RESUMED (SIGCONT)`);
    return this.sendEvent(pid, "signal", "SIGCONT");
}

if (signal === 15) { // SIGTERM (15)
    const eventSent = this.sendEvent(pid, "signal", "SIGTERM");
    setTimeout(async () => {
        const stillRunning = this.getProcess(pid);
        if (stillRunning && stillRunning.state !== ProcessState.EXITED && stillRunning.worker) {
            this.logger.debug(`PID ${pid} did not handle SIGTERM, terminating...`);
            stillRunning.worker.terminate().catch(() => { });
        }
    }, 300);
    return eventSent;
}

// Generic Signal Handling (HUP, USR1, etc)
const sigNames: Record<number, string> = { 1: "SIGHUP", 10: "SIGUSR1", 12: "SIGUSR2" };
const sigName = sigNames[signal] || `SIG${signal}`;
return this.sendEvent(pid, "signal", sigName);
}

```

Syscall gateway: KILL / SIGNAL / WAITPID

```

case SyscallCode.KILL: {
    const targetPid = args as number;
    const targetPcb = this.scheduler.getProcess(targetPid);
    if (!targetPcb)
        throw new Error(`kill: No such process (PID ${targetPid})`);
    // PID 1 (init) cannot be killed – not even by root
    if (targetPid === 1) {
        throw new Error(`kill: PID 1 (init) is protected – cannot be killed directly. Use 'shutdown' or 'reboot' instead.`);
    }
    // Permission check: root can kill anyone, non-root can only kill own processes
    if (!this.isRoot(pcb) && targetPcb.uid !== pcb.uid) {
        throw new Error(`kill: Permission denied – you do not own process ${targetPid} (owned by UID ${targetPcb.uid})`);
    }
    return await this.scheduler.kill(targetPid, 9); // SIGKILL
}

case SyscallCode.SIGNAL: {
    const { pid: targetPid, sig } = args as { pid: number; sig: number };
    // cek sama: proses ada → PID 1 diproteksi → permission →
    return await this.scheduler.kill(targetPid, sig);
}

case SyscallCode.WAITPID: {
    const targetPid = args as number;
    return await this.scheduler.waitpid(targetPid);
}

```

[!WARNING] The KILL syscall always maps to **SIGKILL (9)** — no other signal. For other signals (SIGTERM, SIGINT, SIGSTOP, ...) use the SIGNAL { pid, sig } syscall. Both refuse PID 1 and check process ownership.

Exercises / Practice

1. Run `ps` in the shell — identify the PID, PPID, state, and TTY of each process.
2. Run `sleep 30 &` then `ps` — find the process running in the background (state and ttyld empty).
3. Run an app, note its PID, then `kill <pid> (SIGKILL)` and `kill -15 <pid> (SIGTERM)` — observe the difference in behavior in the kernel logs.
4. Launch a foreground process (without `&`), press **Ctrl+C** — observe `SIGINT` being sent to the foreground process via `sendInterruptSignal()`.

5. Trace this module's walkthrough (spawn → waitpid → exit → reap) directly in the `Scheduler.ts` code: `createProcess()`, `waitpid()`, and the `worker.on("exit")` handler.
 6. Read `src/kernel/Syscalls.ts` — compare the `KILL` case (always `SIGKILL`) vs `SIGNAL` (free sig) and the `PID 1` protection.
-

References

- `wiki/Kernel-dan-Scheduler.md` — process & scheduler details
 - `wiki/course/00-overview.en.md` §4.1, §4.3
 - `src/kernel/Scheduler.ts` — PCB, lifecycle, waitpid, detach, signals, reexec
 - `src/kernel/Syscalls.ts` — `KILL`, `SIGNAL`, `WAITPID`, `EXIT`, `DETACH`, `REEXEC`, `SHUTDOWN` cases
 - `src/kernel/Scheduler.test.ts` — behavior confirmation (A2.19–A2.21 zombie/waitpid, A2.26 detach, reexec)
-

Module 04 — done. Continue to [Module 05 — Syscall & IPC](#).

RFC-TSIX-EDU-002 | Fifth module of the TSIX curriculum. Understand the TSIX mini ABI: how applications call the kernel via syscall, request-response correlation, and push events from the kernel to the worker.

TSIX syscalls are **purely asynchronous RPC**. No order is guaranteed — every request carries a `requestId` (UUID) and is answered with the same `requestId`. This differs from synchronous traps in Linux.

Learning Objectives

- ☐ Explain the complete flow of one syscall (from app to return)
- ☐ Explain the role of `requestId` + `responseMap`
- ☐ Distinguish request-response vs push events
- ☐ Name several important syscall codes (1-73)
- ☐ Explain the role of `validateArgs`

Core Concepts

TSIX uses asynchronous request-response IPC — not synchronous traps like Linux. The application and the kernel run in separate *threads* (Worker vs main thread). The only bridge is `postMessage` in both directions.

Concept	Explanation
<code>SyscallRequest</code>	Packet app → kernel: { <code>requestId</code> , <code>pid</code> , <code>code</code> , <code>args</code> }
<code>SyscallResponse</code>	Packet kernel → app: { <code>requestId</code> , <code>success</code> , <code>data?</code> , <code>error?</code> }
<code>requestId</code> (UUID)	Correlates request ↔ response. Created in <code>UserLib.dispatch</code> , reused by the kernel when replying.
<code>responseMap</code>	Map on the app side: <code>requestId</code> → callback <code>resolve/reject</code> . Replies are matched through this key.
Push event	Kernel → app notification <i>without</i> <code>requestId</code> : { <code>type</code> , <code>data</code> } (<code>signal</code> , <code>ipc_message</code> , <code>gui_request</code> , <code>db_request</code> , <code>resize</code>).
<code>validateArgs</code>	Argument contract in the kernel: checks which syscalls must have <code>args</code> , plus type checks per syscall.

[!IMPORTANT] Because RPC is purely asynchronous, **reply order is not guaranteed**. Two consecutive syscalls (A then B) can be answered with B first. Correlation always goes through `requestId`, never through order.

Flow / How It Works

Full syscall flow: PRINT

Simplest example: the app calls `std.print("hello")`. Here is the complete path (code is truth):

```
// src/mirror/lib/UserLib.ts - StdLib
public async print(text: string) {
  return await this.dispatch(SyscallCode.PRINT, text);
}
```

Step by step:

- `std.print("hello")` calls `UserLib.dispatch(PRINT, "hello")` on the worker side.
- `dispatch()` creates `requestId = uuidv4()`, stores the callback in `responseMap`, then `parentPort.postMessage({ requestId, pid, code: PRINT, args: "hello" })`.

3. The kernel receives the message in `Scheduler.spawnWorker()` → handler `pcb.worker.on("message", ...)`.
4. The handler calls `syscallHandler(request)` — wired in `Kernel.initializeSubsystems()` via `scheduler.setSyscallHandler(...)`.
5. `SyscallDispatcher.handleRequest(req)` → `validateArgs(PRINT, args)` → `dispatch(pid, PRINT, args)`.
6. The `PRINT` case takes `FD 1` from `pcb.fdTable`, then `entry.device.write("hello")` → output to screen.
7. The result (`0`) returns to the Scheduler handler; the Scheduler replies `postMessage({ requestId, success: true, data: 0 })`.
8. In the worker, `parentPort.on("message")` finds `requestId` in `responseMap`, calls the callback → `Promise resolve(0)` → `await print` finishes.

```
sequenceDiagram
    participant App as Aplikasi (Worker)
    participant Lib as UserLib (dispatch)
    participant Sch as Scheduler (worker.on message)
    participant Disp as SyscallDispatcher
    participant Dev as Device (fdTable[1])

    App->>Lib: std.print("hello")
    Lib->>Lib: requestId = uuidv4(); responseMap.set(requestId, cb)
    Lib->>Sch: postMessage({ requestId, pid, code: PRINT, args })
    Sch->>Disp: await syscallHandler(request)
    Disp->>Disp: validateArgs(PRINT, args)
    Disp->>Dev: dispatch(pid, PRINT, args) → fdTable[1].device.write()
    Dev-->>Disp: return 0
    Disp-->>Sch: return 0
    Sch->>Lib: postMessage({ requestId, success: true, data: 0 })
    Lib->>Lib: cocokkan requestId di responseMap → resolve(0)
    Lib-->>App: await print selesai
```

Request-response correlation via `requestId` (UUID) + `responseMap` — purely asynchronous RPC. Many syscalls can be in-flight at once.

Push events (kernel → worker, without requestId)

Unlike request-response, push events have no `requestId`. The kernel sends `{ type, data }` whenever needed:

```
{ type, data }
```

Types actually used in the code (`Scheduler.ts`, `Syscalls.ts`, `Kernel.ts`):

type	Usage in code	Purpose
signal	<code>sendEvent(pid, "signal", "SIGINT")</code> etc.	Signals to processes (SIGINT, SIGTERM, SIGSTOP, SIGCONT, SIGWINCH, SIGSEGV...)
ipc_message	<code>SEND_MSG</code> between apps; forward <code>NET_SNIFF</code> packets	Horizontal IPC / sniffer
gui_request	Window cleanup & forward to GUI daemon	GUI daemon (PixelSpace)
db_request	<code>forwardDbRequest</code> → DB service daemon	Alternative database transport
resize	<code>broadcastEvent("resize", { lines, columns })</code> when host terminal changes	All active processes

[!NOTE] On the worker side, `UserLib.parentPort.on("message")` distinguishes three things: (1) syscall reply — has a `requestId` matching in `responseMap`; (2) event — has `type` and `type !== "signal"`; (3) signal — `type === "signal"`, with default action `SIGINT` → `exit(130)` and `SIGTERM` → `exit(143)` when there is no listener.

Main Syscall List (mini ABI)

Kode	Nama	Kode	Nama
1	PRINT	29	PIPE
2	MKDIR	30–34	SOCKET/BIND/SENDTO/RECVFROM/NETSTAT
3	LS	35	UPTIME
4	EXIT	36	DETACH
5–8	OPEN/READ/WRITE/CLOSE	37	UNAME
9	SCREEN_INFO	38	SETGROUPS
10–12	PS/KILL/EXEC	40–41	GET_SYSPATH/GET_USAGE
13–14	CHDIR/GETCWD	50	SHUTDOWN
15–16	CHMOD/CHOWN	53–55	SYNC_TO_HOST/REEXEC/SYNC_FROM_HOST
17–19	WHOAMI/GETENV/SETENV	56–58	MOUNT/UMOUNT/GET_MOUNTS
20–23	STAT/UNLINK/RMDIR/IOCTL	60	SET_IDENTITY
24–28	SEND_MSG/WAITPID/SIGNAL/SETUID/SETGID	61–66	GUI_REQ/GET_PPID/READ_CHUNK/WRITE_CHUNK/GET_SIZE/REARENT
—	—	67–71	DB_CONNECT/DB_QUERY/DB_DISCONNECT/DB_SERVICE_REGISTER/DB_SERVICE_REPLY
—	—	72–73	NET_SNIFFER_REGISTER/NET_SNIFFER_UNREGISTER

`SyscallCode` is an enum in `src/common/SyscallCode.ts` — the ABI contract. Do not change existing numbers (breaking change).

Some key codes (number → name → purpose):

Number	Name	Purpose
1	PRINT	Prints text to the screen (via FD 1)
14	GETCWD	Gets the process working directory
24	SEND_MSG	Horizontal IPC (app-to-app)
25	WAITPID	Waits for a process to finish by PID
26	SIGNAL	Sends a signal to a process
61	GUI_REQ	PixelSpace Display Protocol (RFC-TSIX-002)
67	DB_CONNECT	Opens an external database connection

Source Code

File	Contents	Relevance
<code>src/common/SyscallCode.ts</code>	ABI enum 1–73	Syscall numbers
<code>src/common/IPCTypes.ts</code>	Packet shapes <code>SyscallRequest</code> , <code>SyscallResponse</code> , <code>IPCEvent</code>	Packet contract

File	Contents	Relevance
src/kernel/Syscalls.ts	SyscallDispatcher: handleRequest, validateArgs, dispatch	Dispatcher
src/kernel/Scheduler.ts	spawnWorker + worker.on("message") (reply) + sendEvent	Bridge & reply
src/kernel/Kernel.ts	Wiring syscallHandler(...)	Wiring
src/mirror/lib/UserLib.ts	dispatch(), responseMap, parentPort.on("message"), onEvent	Worker side

Snippets (code level)

All snippets below are copied from the source. "Code is truth."

1. UserLib dispatch (worker side) — src/mirror/lib/UserLib.ts

```
private async dispatch(code: SyscallCode, args: any): Promise<any> {
  return new Promise((resolve, reject) => {
    const requestId = uuidv4();
    const request: SyscallRequest = { requestId, pid: this.pid, code, args };

    this.responseMap.set(requestId, (response: SyscallResponse) => {
      if (response.success) {
        resolve(response.data);
      } else {
        reject(new Error(response.error || "Syscall Failed"));
      }
    });

    if (parentPort) {
      parentPort.postMessage(request);
    } else {
      reject(new Error("No parentPort found! Are you running in a Worker?"));
    }
  });
}
```

responseMap has type Map<string, (res: SyscallResponse) => void> — its values are **callbacks**, not { resolve, reject } objects. The Promise is resolved/rejected inside the callback when the reply arrives.

2. Receiver side (worker) — parentPort.on("message")

```
parentPort.on("message", (msg: any) => {
  // 1. Balasan Syscall
  if (msg.requestId && this.responseMap.has(msg.requestId)) {
    const resolve = this.responseMap.get(msg.requestId)!;
    resolve(msg as SyscallResponse);
    this.responseMap.delete(msg.requestId);
  }
  // 2. Event (Push Notification) – selain signal
  else if (
    msg.type &&
    msg.type !== "signal" &&
    this.eventListeners.has(msg.type)
  ) {
    const callbacks = this.eventListeners.get(msg.type)!;
    callbacks.forEach((cb) => cb(msg.data));
  }
  // 3. Signal (SIGINT/SIGTERM) dengan default action
  else if (msg.type === "signal") {
    const sig = msg.data;
    // ... default: exit(130) untuk SIGINT, exit(143) untuk SIGTERM
  }
});
```

```

    }
  });
}

```

3. Kernel handleRequest — src/kernel/Syscalls.ts

```

public async handleRequest(req: SyscallRequest): Promise<any> {
  const silentCodes = [
    SyscallCode.READ,
    SyscallCode.PS,
    SyscallCode.GETCWD,
    SyscallCode.WHOAMI,
  ];
  if (!silentCodes.includes(req.code)) {
    this.logger.debug(
      `[IPC] PID ${req.pid} Request: ${SyscallCode[req.code]} (${req.requestId})`,
    );
  }

  try {
    this.validateArgs(req.code, req.args);
    return await this.dispatch(req.pid, req.code, req.args);
  } catch (e: any) {
    this.logger.error(
      `Syscall Error [${SyscallCode[req.code]}]: ${e.message}`,
    );
    throw e;
  }
}

```

⚠ Important correction: `handleRequest` **does not reply by itself**. It only validates, executes, then returns the result (or throws an error). The `postMessage` reply is sent by the **Scheduler** in the `worker.on("message")` handler (see snippet 4).

4. Reply sent by Scheduler — src/kernel/Scheduler.ts (spawnWorker)

```

pcb.worker.on("message", async (request: SyscallRequest) => {
  if (this.syscallHandler) {
    try {
      const result = await this.syscallHandler(request);
      if (pcb.worker) {
        pcb.worker.postMessage({
          requestId: request.requestId,
          success: true,
          data: result
        });
      }
    } catch (error: any) {
      if (pcb.worker) {
        pcb.worker.postMessage({
          requestId: request.requestId,
          success: false,
          error: error.message
        });
      }
    }
  }
});

```

5. Real syscall handler example — src/kernel/Syscalls.ts

```

case SyscallCode.PRINT: {
  const entry = pcb.fdTable[1];
  if (!entry) return -1;
  entry.device.write(args as string);
  return 0;
}

```

```

}

case SyscallCode.GETCWD:
    return pcb.cwd;

```

`PRINT` does not write directly to the screen — it writes through **FD 1** (`stdout`) in `fdTable`. This embodies "everything is a file": output is just a device attached to `fdTable`. `GETCWD`, on the other hand, simply returns `pcb.cwd` (process state in the PCB).

6. Push events — `sendEvent` in `src/kernel/Scheduler.ts`

```

public sendEvent(pid: number, type: string, data: any): boolean {
    const pcb = this.getProcess(pid);
    if (pcb && pcb.worker && pcb.state !== ProcessState.EXITED) {
        pcb.worker.postMessage({ type, data });
        return true;
    }
    return false;
}

```

Example usage (forwarding sniffer packets in `Syscalls.ts`):

```

this.scheduler.sendEvent(pid, "ipc_message", { data: sniff });

```

Exercises / Practice

1. Read `src/common/SyscallCode.ts` — note all syscall numbers (1–73). Pay attention to the number gaps (39, 42–49, 51–52, 59) intentionally reserved.
2. Read `src/common/IPCTypes.ts` — recognize the shapes of `SyscallRequest`, `SyscallResponse`, `IPCEvent`.
3. Add a `log` in `handleRequest` for the `PRINT` syscall — then run `echo hello` in the shell. Observe the flow (same `requestId` in request & response).
4. Explain why `requestId` is needed — what happens without correlation?
5. Send two syscalls at once (e.g. `READ` large file + `PRINT`) and observe the reply order — prove that order is not guaranteed.
6. From the worker side, check the contents of `responseMap` right after `dispatch()` is called (before the reply arrives) — prove the values are callbacks.

References

- `src/common/SyscallCode.ts` — ABI enum (source of truth for syscall numbers)
- `src/common/IPCTypes.ts` — IPC packet shapes
- `src/kernel/Syscalls.ts` — `handleRequest`, `validateArgs`, `dispatch`
- `src/kernel/Scheduler.ts` — `worker.on("message")` (reply) + `sendEvent` (push)
- `src/kernel/Kernel.ts` — wiring `setSyscallHandler`
- `src/mirror/lib/UserLib.ts` — `dispatch()`, `responseMap`, `onEvent`
- `wiki/identity_guid_ipc_walkthrough.md` — IPC & identity walkthrough
- `wiki/course/00-overview.en.md` §4.2

Module 05 — complete. Continue to [Module 06 — Permission & Security](#).

RFC-TSIX-EDU-002 | The sixth module of the TSIX curriculum. Understand the rwx permission model, the setuid bit, root bypass, and known weaknesses in the TSIX permission system.

PermissionManager is the kernel's "security guard". Every file access goes through layered checks: root bypass → owner → group → others. Plus the setuid bit for privileged binaries like `login` and `sudo`.

Learning Objectives

- ☐ Explain the permission check order (root → owner → group → others)
- ☐ Explain the permission bits `r=4, w=2, x=1`
- ☐ Read the permission matrix and example modes (`600`, `644`, `755`, `4755`) with their decimal values
- ☐ Explain the setuid bit (`0o4000` = 2048) and its handling during `EXEC`
- ☐ Explain why `parseMode("4755")` is 2541 (decimal)
- ☐ Explain the `CHMOD` rule (owner or root) vs `CHOWN` (root only)
- ☐ Explain the app-name-based privilege weakness and its mitigation

Core Concepts

Layered rwx checks

```
check(pcb, node, requested):
  1. root (uid 0)          → true (God Mode)
  2. owner (uid == node.uid) → bit owner   ((mode >> 6) & 0x7)
  3. group (gid match)      → bit group    ((mode >> 3) & 0x7)
  4. others                 → bit others   (mode & 0x7)
```

 PermissionManager check flow (root → owner → group → others) Source: wiki/diagram/Keamanan-dan-Sandboxing-3.mmd

rwx permission matrix

One mode = **3 octal digits**, one for each class: owner, group, and others. `check()` shifts the bits according to the class:

Class	Calculation in <code>check()</code>	Bits	Example <code>644</code> (decimal <code>420</code>)
owner (user)	<code>(mode >> 6) & 0x7</code>	<code>r=4, w=2, x=1</code>	digit 6 → <code>rw-</code>
group	<code>(mode >> 3) & 0x7</code>	<code>r=4, w=2, x=1</code>	digit 4 → <code>r--</code>
others	<code>mode & 0x7</code>	<code>r=4, w=2, x=1</code>	digit 4 → <code>r--</code>

Encode permission

Bit	Value	Meaning
r (read)	4	Read content
w (write)	2	Write / modify
x (execute)	1	Execute

Bit combination per octal digit (used in the mode table below):

Digit	Binary	rwX	Meaning
0	000	---	no access
1	001	--x	execute
2	010	-w-	write
3	011	-wx	write + execute
4	100	r--	read
5	101	r-x	read + execute
6	110	rw-	read + write
7	111	rwX	read + write + execute

Common modes & decimal values

Modes are stored as **decimal in SQLite** (not an octal string). `parseMode("4755") = parseInt("4755", 8) = 2541` (decimal) — this is the value `init.ts` uses when it calls `chmod("/bin/passwd.js", 2541)`.

Mode	Octal	Decimal	owner	group	others	Example usage
600	0o600	384	rw-	---	---	private files: <code>/etc/shadow</code> , SSH keys
644	0o644	420	rw-	r--	r--	plain text files (default mode of <code>touch</code> in the <code>OPEN</code> syscall)
755	0o755	493	rwX	r-x	r-x	directories & binaries (default mode of <code>mkdir</code>)
4755	0o4755	2541	rwS	r-x	r-x	setuid binaries: <code>/bin/passwd.js</code> , <code>/bin/sudo.js</code>

The `s` in the owner position means the **setuid bit is active** (`0o4000 = 2048`), not a regular `x`. The decimal values are confirmed by test `A5.10` (`644 → 420`, `755 → 493`) and `A5.13` (`4755 & 0o4000`).

The SetUID bit (0o4000)

SetUID = execute as the **file owner**, not the caller. Example:

- `/bin/passwd.js → chmod 2541 (0o4755)`: a normal user runs `passwd`, and the process runs as root so it can modify `/etc/shadow`.
- `/bin/sudo.js → same`.

Handling during EXEC (in `Syscalls.ts`): the `0o4000` bit = **2048** (decimal). When executing a binary, the kernel checks `node.mode & 2048`. If active, `targetUid = node.uid` and `targetGid = node.gid`; the new process is born with the file owner's uid/gid, while `ruid` (real UID) stays with the caller. See the snippet below.

Note: the runtime executes the `.js` sidecar, so SetUID must be set on `/bin/*.js`, not the source `.ts` files. This is what `init.ts` does: `lib.fs.chmod("/bin/passwd.js", 2541)` and `lib.fs.chmod("/bin/sudo.js", 2541)`. `rc.local.ts` also sets `chmod("/bin/login.js", 0o4755)`.

The POSIX order during login: `setgroups → setgid → setuid` (this order is required so the group is correct). Visible in `login.ts`: `setgroups(supplementaryGids) → setgid(gid) → setuid(uid)`.

Snippet (code level)

PermissionManager.check()

```
public check(pcb: PCB, node: any, requested: Permission): boolean {
  if (pcb.uid === 0) return true; // 1. root bypass
```

```

const mode = node.mode; // decimal di SQLite

if (pcb.uid === node.uid) { // 2. owner
  const userMode = (mode >> 6) & 0x7;
  if ((userMode & requested) === requested) return true;
}

if (pcb.gid === node.gid || (pcb.groups && pcb.groups.includes(node.gid))) {
  const groupMode = (mode >> 3) & 0x7; // 3. group
  if ((groupMode & requested) === requested) return true;
}

const otherMode = mode & 0x7; // 4. others
if ((otherMode & requested) === requested) return true;

return false;
}

public static parseMode(octal: string | number): number {
  return parseInt(octal.toString(), 8); // "4755" → 2541
}

```

EXEC — permission enforcement & setuid (Syscalls.ts)

```

// 1. Cek x bit dulu – tanpa izin eksekusi, langsung ditolak
if (!this.satpam.check(pcb, node, Permission.EXECUTE)) {
  throw new Error(`Permission Denied: Cannot execute ${absoluteExecPath}`);
}

let targetUid = pcb.uid;
let targetGid = pcb.gid;
let targetOwner = pcb.owner;

// 2. SETUID BIT SUPPORT (0o4000 = 2048)
if (node && node.mode & 2048) {
  targetUid = node.uid; // eksekusi sebagai pemilik file
  targetGid = node.gid;
  if (targetUid === 0) targetOwner = "root";
  this.logger.info(
    `SetUID Execution detected for ${absoluteExecPath}: Running as UID ${targetUid}`,
  );
}

// 3. Proses baru lahir dengan uid/gid target; ruid (real UID) dipertahankan
const newPcb = this.scheduler.createProcess(binaryName, {
  // ...
  uid: targetUid,
  gid: targetGid,
  ruid: pcb.ruid, // Preserve Real UID
  owner: targetOwner,
  // ...
});

```

This path is the security key for `/bin/login.js`, `/bin/passwd.js`, and `/bin/sudo.js`: a normal user cannot setuid directly, but executing a setuid-root binary makes the process run as uid 0.

OPEN — read vs write (Syscalls.ts)

```

const requiredPerm =
  flags.includes("w") || flags.includes("+")
    ? Permission.WRITE
    : Permission.READ;

const node = vfs.stat(relativePath);

if (node) {

```

```
// File sudah ada → cek izin file itu sendiri
if (!this.satpam.check(pcb, node, requiredPerm)) {
  throw new Error(
    `Permission Denied: Cannot open ${absoluteOpenPath} for ${Permission[requiredPerm]}`,
  );
}
} else if (requiredPerm === Permission.WRITE) {
  // File belum ada → cek WRITE ke direktori parent, lalu buat mode 420 (644)
  // ...
  vfs.touch(relativePath, "", pcb.uid, pcb.gid, 420); // 644 equivalent
}
}
```

CHMOD — change mode (Syscalls.ts)

```
case SyscallCode.CHMOD: {
  const { path: argPath, mode } = args as { path: string; mode: number };
  const absolutePath = PathResolver.resolve(pcb.cwd, argPath);

  if (absolutePath.startsWith("/dev/")) {
    if (!this.isRoot(pcb)) return false; // chmod /dev/* → root only
    const devName = absolutePath.replace("/dev/", "");
    const device = this.kernel.devices[devName];
    if (!device) return false;
    device.mode = mode;
    return true;
  }

  const { vfs, relativePath } = this.mountManager.resolve(absolutePath);
  const node = vfs.stat(relativePath);
  if (!node) return false;
  if (pcb.uid !== 0 && pcb.uid !== node.uid) return false; // root ATAU owner
  return vfs.chmod(relativePath, mode);
}
```

CHMOD rule: a regular file may be changed by **its owner or root**. Verified by test A5.11.

CHOWN — change owner/group (Syscalls.ts)

```
case SyscallCode.CHOWN: {
  const {
    path: argPath,
    uid: targetUid,
    gid: targetGid,
  } = args as { path: string; uid: number; gid: number };
  const absolutePath = PathResolver.resolve(pcb.cwd, argPath);

  if (absolutePath.startsWith("/dev/")) {
    if (!this.isRoot(pcb)) return false; // chown /dev/* → root only
    // ...
  }

  if (!this.isRoot(pcb)) return false; // chown file → ROOT SAJA
  const { vfs, relativePath } = this.mountManager.resolve(absolutePath);
  return vfs.chown(relativePath, targetUid, targetGid);
}
```

CHOWN rule: **only root** may change a file's owner (test A5.15). Unlike CHMOD, which the owner may also do.

Known Weaknesses

[!WARNING] **App-name-based privilege (fragile).** In `WorkerEntry.ts`, `restrictHostAPI(appName)` marks an app as "privileged" if the **substring** of its name (after `toLowerCase()`) contains one of:

```
const isPrivileged = appName.toLowerCase().includes("server") ||
  appName.toLowerCase().includes("daemon") ||
  appName.toLowerCase().includes("dome") ||
  appName.toLowerCase().includes("tbuild") ||
  appName.toLowerCase().includes("vfs") ||
  appName.toLowerCase().includes("mysqld");
```

A privileged app gets an **allow-list of host modules**:

```
const allowedModules = ["http", "ws", "path", "fs", "url", "esbuild", "crypto", "os", "bcryptjs", "mysql2", "mysql2/promise"];
```

Exploitation scenario (example):

1. An attacker uploads an app named `evil-daemon` (the name contains `daemon`).
2. When executed, `isPrivileged = true` → `global.require = privilegedRequire`.
3. The app can now `require("fs")`, `require("crypto")`, `require("http")`, etc. — **direct host** Node.js modules, not TSIX syscalls.
4. With `fs`, the attacker can read files outside the TSIX VFS sandbox (e.g., the host Linux `/etc/shadow`) → **sandbox escape**.

[!IMPORTANT] Mitigation.

- **Long-term (architectural):** replace the substring heuristic with **capability-based** checks — an explicit list of `appName` → `capabilities` pairs (or a per-app manifest).
- **Now (defense-in-depth):** even if an app is "privileged" inside the sandbox, the kernel still has `PermissionManager + validateArgs + root-only SETUID`. The app name only opens the host module door — **not** full kernel access; every syscall is still checked per-invocation on the kernel side.

Exercises / Practice

1. Run `ls -l /bin/passwd.js` — observe the setuid bit (`s`) in the owner mode (mode `rwsr-xr-x`).
2. Run `stat /bin/passwd.js` — check the decimal mode value (should be `2541 = 0o4755`).
3. Run `chmod 755 /bin/passwd.js` then `ls -l` again — the `s` bit disappears; `init` will set it again (`init.ts`).
4. As a non-root user, try to read `/etc/shadow` — observe the permission error.
5. As a non-root user, run `chown root /tmp/foo` — observe the error (only root may chown); then `chmod 600 /tmp/foo` — must succeed because the owner may chmod.
6. Read `src/kernel/PermissionManager.ts` and `src/kernel/Syscalls.ts` — find all call sites of `satpam.check()`.
7. Read `src/userland/WorkerEntry.ts` → the `restrictHostAPI` function — test: run an app named `evil-daemon` that calls `require("fs")`, and compare it with a normal app named `hitung`.

References

- `wiki/Keamanan-dan-Sandboxing.md` — complete security model
- `wiki/course/00-overview.en.md §4.3` (ring model & privilege boundaries)
- `src/kernel/PermissionManager.ts` — implementation of `check()` & `parseMode()`
- `src/kernel/PermissionManager.test.ts` — tests A5.1–A5.30 (`rx`, `chmod`, `chown`, `umask`, `setuid`, `sudo`)
- `src/kernel/Syscalls.ts` — `OPEN/EXEC/CHMOD/CHOWN` permission enforcement + setuid in `EXEC`
- `src/userland/WorkerEntry.ts` — the `restrictHostAPI` sandbox (name-based privilege)
- `src/mirror/bin/init.ts`, `src/mirror/bin/rc.local.ts` — setuid bit setup
- `src/mirror/bin/login.ts`, `src/mirror/bin/sudo.ts` — the `setgroups/setgid/setuid` flow

Module 06 — complete. Continue to [Module 07 — Mount & Path Resolution](#).

RFC-TSIX-EDU-002 | Seventh module of the TSIX curriculum. Understand how VFS paths are routed to the correct filesystem backend — by the "longest prefix wins" principle.

MountManager is the "path router". When an app opens `/tmp/foo`, the kernel must know that it belongs to RamFS, not BKFS. The rule is simple: **the longest matching prefix wins**.

Learning Objectives

- ☐ Explain the `MountPoint` structure
- ☐ Explain the "longest prefix wins" principle
- ☐ Explain the contents of `fstab.conf` (default mounts)
- ☐ Distinguish resolution via MountManager vs device `/dev/*`
- ☐ Explain the role of `PathResolver`
- ☐ Walk through `PathResolver.resolve()` for relative, absolute, `.` and `..` paths
- ☐ Predict the result of `resolve()` by reading the mount sort order

Core Concepts

Two layers of resolution

Path resolution in TSIX runs in two layers:

1. `PathResolver.resolve()` — normalizes a path string. It only processes text (`///`, `.`, `..`), without knowing about filesystems.
2. `MountManager.resolve()` — routes the normalized path to the correct `IVFS` backend, using the "longest prefix wins" rule.

[!IMPORTANT] `PathResolver` works at the **string** level. `MountManager` works at the **filesystem backend** level. They are two different functions.

MountPoint

```
export interface MountPoint {
  vfsPath: string;    // titik kait di VFS, misal "/tmp"
  vfs: IVFS;          // backend filesystem
  type: string;       // "bkfs" | "ramfs" | "host"
  source: string;     // "system.db" | "shared" | "systembak.db" | ...
  readOnly: boolean;
  uid?: number;
  gid?: number;
}
```

`type` uses the real values from `fstab.conf`: `"bkfs"`, `"ramfs"`, and `"host"` (HostVFS).

PathResolver.resolve() — walkthrough

`PathResolver.resolve(cwd, targetPath)` takes two arguments: the working directory (`cwd`) and the target path. The result is always a normalized absolute path (without `///`, `.`, or `..`).

#	cwd	targetPath	Result	Explanation
1	/	/etc/passwd	/etc/passwd	Absolute path — cwd is ignored
2	/	etc/passwd	/etc/passwd	Relative path resolved at root
3	/bin	../etc/passwd	/etc/passwd	.. goes up one level out of /bin
4	/home/user	docs/../a.txt	/home/user/a.txt	.. cancels the docs segment
5	/	/tmp/./foo//bar	/tmp/foo/bar	// and . are cleaned up
6	/	/../x	/x	.. above root cannot go higher

[!NOTE] Examples 4 and 5 show that normalization happens **before** mount lookup. So `resolve("/tmp/./foo")` and `resolve("/tmp/foo")` are considered the same by `MountManager`.

Default mounts

Root `/` is **not** in `fstab.conf`. It is mounted directly in `initializeSubsystems()`:

```
this.mountManager.mount("/", this.bkfs, "bkfs", cfg.kernel.database, false);
```

`cfg.kernel.database` is `"system.db"` (see `src/sysconfig.conf`). The rest are loaded from `src/mirror/etc/fstab.conf` by `processFstab()`:

Path	type	Backend	Source	readOnly	uid/gid	mode	Nature
/	bkfs	BKFS (SQLite)	system.db	false	—	—	persistent, root fs
/tmp	ramfs	RamFS	in-memory (hostPath "RAM" is ignored)	false	0/100	0o1777	volatile, sticky
/mnt/shared	host	HostVFS	host folder shared	false	1000/100	0o775	host bridge, anti-escape
/mnt/sbak	bkfs	BKFS	systembak.db	false	1000/100	0o775	backup DB

[!NOTE] `fstab.conf` only contains `/tmp`, `/mnt/shared`, and `/mnt/sbak`. The `/tmp` entry uses `active = true`; the rest are active by default. The INI format spells permissions in explicit octal: `mode = 0o1777` (sticky, everyone can write — typical for `/tmp`), `mode = 0o775` for the others. (Decimal is still valid: `1023 = 0o1777`.)

Resolution: longest prefix wins

After all mounts are in place, the list is sorted **descending by path length**. The resulting sort order for the default mounts:

```
/mnt/shared (11) ← diperiksa dulu
/mnt/sbak   ( 9)
/tmp        ( 4)
/           ( 1) ← fallback terakhir
```

Example output of `MountManager.resolve()`:

Requested path	Selected MountPoint	relativePath	vfs
/tmp/a.txt	/tmp	/a.txt	RamFS
/tmp	/tmp (exact)	/	RamFS
/etc/passwd	/	/etc/passwd	BKFS

Requested path	Selected MountPoint	relativePath	vfs
/mnt/shared/readme.md	/mnt/shared	/readme.md	HostVFS
/mnt/shared	/mnt/shared (exact)	/	HostVFS
/mnt/sbak/backup.tar	/mnt/sbak	/backup.tar	BKFS
/mnt/sharedx/foo	/	/mnt/sharedx/foo	BKFS

[!WARNING] The last row is an important gotcha. `/mnt/sharedx/foo` does **not** match the prefix `/mnt/shared/` (because after `shared` there is `x`, not `/`). So it falls back to `/` → BKFS. This is why the prefix uses an extra slash (`m.vfsPath + "/"`), not a plain `startsWith(m.vfsPath)`.

📁 Path resolution: syscall → MountManager → VFS backend Source: [wiki/diagram/Home-2.mmd](#)

The `/dev/*` path does not pass through MountManager

`dev/xxx` paths are handled by **HAL** (`kernel.devices[xxx]`), bypassing MountManager. This is consistent with the "everything is a file" principle — devices are not vnode filesystems.

Flow / How It Works

Mount flow at boot

```
initializeSubsystems()
└─ new BKFS("system.db")           → root filesystem
└─ mount("/", bkfs, "bkfs", "system.db", false)
└─ processFstab()                 → baca /etc/fstab.conf
    └─ untuk tiap entry:
        active === false → skip
        pastikan dir mount point ada (mode ?? 0o755)
        pilih driver: bkfs → BKFS, ramfs → RamFS, else → HostVFS
        mountManager.mount(vfsPath, driver, type, hostPath, ...)
└─ mount() normalisasi path + sort descending by panjang
```

Resolution flow on a file syscall

```
Aplikasi → syscall file (mis. OPEN, path mentah)
→ MountManager.resolve(path)
    1. normalized = PathResolver.resolve("/", path)
    2. loop mounts (terpanjang dulu):
        exact match → relativePath = "/"
        prefix match → relativePath = sisa path
    3. hasil: { vfs, relativePath, mountPoint }
→ vfs.<op>(relativePath)           // backend mengeksekusi
Path /dev/* → kernel.devices[xxx] // HAL, bukan MountManager
```

Source Code

File	Role
src/kernel/MountManager.ts	mount/unmount/resolve/listMounts
src/common/PathResolver.ts	Path normalization (<code>///</code> , <code>.</code> , <code>..</code>)
src/kernel/Kernel.ts	<code>initializeSubsystems()</code> + <code>processFstab()</code> at boot
src/mirror/etc/fstab.conf	Default mount list
src/vfs/HostVFS.ts, RamFS.ts, BKFS.ts	Backends

Snippets (code level)

PathResolver.resolve()

```
public static resolve(cwd: string, targetPath: string): string {
  // 1. Jika path diawali '/', berarti absolute
  let absolutePath = targetPath.startsWith("/")
    ? targetPath
    : (cwd === "/" ? "/" + targetPath : cwd + "/" + targetPath);

  // 2. Normalisasi (Bersihkan //, ./ dan ..)
  const parts = absolutePath.split("/").filter(p => p.length > 0 && p !== ".");
  const stack: string[] = [];

  for (const part of parts) {
    if (part === "..") {
      stack.pop();
    } else {
      stack.push(part);
    }
  }

  return "/" + stack.join("/");
}
```

MountManager.resolve()

```
public resolve(path: string): {
  vfs: IVFS;
  relativePath: string;
  mountPoint: string;
} {
  // Normalize: robustly handle //, ./ and .. using PathResolver
  const normalized = PathResolver.resolve("/", path);

  for (const m of this.mounts) {
    // Case 1: Exact match
    if (normalized === m.vfsPath) {
      return { vfs: m.vfs, relativePath: "/", mountPoint: m.vfsPath };
    }

    // Case 2: Sub-path match
    // Special handling for root mount point "/"
    const prefix = m.vfsPath === "/" ? "/" : m.vfsPath + "/";
    if (normalized.startsWith(prefix)) {
      let relative = normalized.substring(
        m.vfsPath === "/" ? 0 : m.vfsPath.length,
      );
      if (!relative.startsWith("/")) relative = "/" + relative;
      return { vfs: m.vfs, relativePath: relative, mountPoint: m.vfsPath };
    }
  }

  throw new Error(`Path not found in any mounted filesystem: ${path}`);
}
```

MountManager.mount()

```
public mount(
  vfsPath: string,
  vfs: IVFS,
  type: string = "bkfs",
  source: string = "system.db",
  readOnly: boolean = false,
  uid?: number,
  gid?: number,
) {
```

```
// Normalisasi path menggunakan PathResolver agar konsisten
const cleanPath = PathResolver.resolve("/", vfsPath);

// Cek apakah sudah ada
const existing = this.mounts.find((m) => m.vfsPath === cleanPath);
if (existing) {
  this.logger.warn(`Mount point ${cleanPath} already exists. Replacing...`);
  existing.vfs = vfs;
  existing.type = type;
  existing.source = source;
  existing.readOnly = readOnly;
  existing.uid = uid;
  existing.gid = gid;
} else {
  this.mounts.push({
    vfsPath: cleanPath,
    vfs,
    type,
    source,
    readOnly,
    uid,
    gid,
  });
  // Sort by path length descending so longest match wins
  this.mounts.sort((a, b) => b.vfsPath.length - a.vfsPath.length);
}

this.logger.info(
  `Mounted ${type} filesystem at ${cleanPath} from ${source} (${readOnly ? "ro" : "rw"})`,
);
}
```

[!NOTE] `mount()` only sorts when a **new** entry is added. If the path already exists, `mount()` replaces its backend without touching the sort order.

Root mount in `initializeSubsystems()`

```
this.bkfs = new BKFS(cfg.kernel.database);
this.mountManager.mount("/", this.bkfs, "bkfs", cfg.kernel.database, false);
```

`processFstab()` — `fstab` processing loop

```
private async processFstab() {
  if (!this.bkfs) return;

  const fstabPath = "/etc/fstab.conf";
  if (!this.bkfs.exists(fstabPath)) return;

  try {
    const content = this.bkfs.read(fstabPath);
    if (!content) return;

    const entries = JSON.parse(content) as any[];
    for (const entry of entries) {
      const { vfsPath, hostPath, type, readOnly, uid, gid, active, mode } =
        entry;

      // Skip if explicitly marked inactive (default: active = true)
      if (active === false) {
        this.bootLogStart(`FSTAB: Skipping ${vfsPath} (inactive)`);
        this.bootLogEnd(true);
        continue;
      }

      // Ensure mount point exists with correct ownership & permissions
      const dirMode = mode ?? 0o755;
      if (!this.bkfs.exists(vfsPath)) {
```

```

        this.bkfs.mkdir(vfsPath, uid ?? 0, gid ?? 0, dirMode);
    } else {
        if (uid !== undefined || gid !== undefined) {
            // Re-apply ownership if dir already existed (e.g. created by ensureDefaultAuth)
            this.bkfs.chown(vfsPath, uid ?? 0, gid ?? 0);
        }
        if (mode !== undefined) {
            this.bkfs.chmod(vfsPath, dirMode);
        }
    }
}

let driver: IVFS;
if (type === "bkfs") {
    driver = new BKFS(
        path.resolve(process.cwd(), hostPath),
        readOnly || false,
        uid,
        gid,
        dirMode,
    );
} else if (type === "ramfs") {
    // RamFS tidak butuh hostPath — murni di RAM
    const label = vfsPath.replace(/\\/g, "_").replace(/^_/, "");
    driver = new RamFS(label, uid, gid, dirMode);
} else {
    driver = new HostVFS(hostPath, readOnly || false, uid, gid, dirMode);
}

this.bootLogStart(`FSTAB: Mounting ${vfsPath} (${type})`);
this.mountManager.mount(
    vfsPath,
    driver,
    type,
    hostPath,
    readOnly || false,
    uid,
    gid,
);
this.bootLogEnd(true);
}
} catch (e: any) {
    this.bootLog(`FSTAB: Error processing fstab: ${e.message}`, false);
}
}
}

```

Exercises / Practice

1. Run `mount` (or `cat /proc/mounts` if available) — observe the list of mount points and their order.
2. Write a file in `/tmp/x` then reboot — does the file still exist? (RamFS = volatile)
3. Read `src/kernel/Kernel.ts` — find `initializeSubsystems()` (root mount) and `processFstab()` (fstab).
4. Compute the result of `MountManager.resolve()` for `/mnt/shared/../../sbak/x` before running it. Compare with real behavior.
5. Modify `resolve()` to log the resolved path, then open a file from the shell.

References

- `wiki/Virtual-File-System.md` — VFS layer
- `wiki/course/00-overview.en.md` §4.4
- `src/kernel/MountManager.ts`, `src/common/PathResolver.ts`
- `src/kernel/Kernel.ts` — `initializeSubsystems()` & `processFstab()`
- `src/mirror/etc/fstab.conf` — default mount list

RFC-TSIX-EDU-002 | The eighth module of the TSIX curriculum. Understand the Virtual File System layer: one `IVFS` contract, three backends (`BKFS`/`RamFS`/`HostVFS`), and chunked I/O executed inside SQL.

Design key: **IVFS is one contract, many backends**. A syscall does not care whether the file lives in SQLite, RAM, or a host folder — it only calls `vfs.read()`. This enables "backend swapping" without changing the kernel.

Learning Objectives

- ☐ List the core methods of the `IVFS` contract
- ☐ Explain the differences between `BKFS`, `RamFS`, and `HostVFS`
- ☐ Explain why chunked I/O is executed inside SQL (`SUBSTR/CONCAT`)
- ☐ Explain the structure of the `vnodes` table in `BKFS`
- ☐ Explain the role of `getSize`, which does not fetch content
- ☐ Explain the flow of reading a single file: syscall `OPEN` → `VFS.read` → SQL

Core Concepts

The IVFS Contract

```
export interface IVFS {
  ls(path: string): any[];
  mkdir(path: string, uid?: number, gid?: number, mode?: number): boolean;
  read(path: string): string | null;
  touch(path: string, content?: string, uid?: number, gid?: number, mode?: number): boolean;
  stat(path: string): any;
  chmod(path: string, mode: number): boolean;
  chown(path: string, uid: number, gid: number): boolean;
  unlink(path: string): boolean;
  rmdir(path: string): boolean;
  exists(path: string, type?: VNodeType): boolean;
  getUsage(): Promise<{ size: number, files: number, dirs: number, diskSize?: number }>;
  append(path: string, content: string): boolean;

  // --- Chunked I/O (Progress-aware, untuk file besar) ---
  /** Membaca sebagian konten file (offset-based, return null jika di luar range) */
  readChunk(path: string, offset: number, length: number): string | null;
  /** Menulis (replace) sebagian konten file di offset tertentu */
  writeChunk(path: string, chunk: string, offset: number): boolean;
  /** Mendapatkan ukuran file dalam byte, atau -1 jika tidak ditemukan */
  getSize(path: string): number;
}
```

Three Backends — Comparison

Backend	Source	Persistence	Speed	Use case
BKFS	SQLite <code>system.db</code> , <code>vnodes</code> table	✔ persistent (disk)	moderate; chunked I/O optimized inside SQL	root filesystem <code>/</code> , backup <code>/mnt/sbak</code>
RamFS	memory (<code>VNode</code> tree)	✗ volatile (lost on restart)	⚡ very fast	<code>/tmp</code> (temporary data)
HostVFS	real folder on the host (<code>Node fs</code>)	✔ persistent (host files)	moderate (host syscall)	<code>/mnt/shared</code> (bridge + anti-escape)

Which backend is used is decided by `MountManager` at boot time (see `src/mirror/etc/fstab.conf`):

Mount point	Type	Source	Description
/	bkfs	system.db	persistent root filesystem
/tmp	ramfs	RAM	volatile, mode 1023 = 0o1777 (sticky)
/mnt/shared	host	folder shared	bridge to host, uid 1000
/mnt/sbak	bkfs	systembak.db	root backup in a separate SQLite file

[!NOTE] **"Swap backend" without changing the kernel** The kernel only holds an IVFS reference. Because every backend implements the same contract, swapping storage (e.g. /tmp from RamFS to BKFS) only requires changing an entry in `fstab.conf` — the syscall does not change at all.

 VFS routing: app → syscall → MountManager → BKFS/RamFS/HostVFS Source: wiki/diagram/Virtual-File-System-1.mmd

BKFS: vnodes structure

Files in BKFS are stored as **rows** in the `vnodes` table — a tree structure where `parent_id` points to the parent row (NULL for the root /):

```
CREATE TABLE IF NOT EXISTS vnodes (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  parent_id INTEGER,      -- FK ke id parent (NULL untuk root "/)
  name TEXT NOT NULL,
  type TEXT NOT NULL,     -- 'FILE' | 'DIRECTORY'
  content TEXT,           -- isi file (NULL untuk direktori)
  size INTEGER DEFAULT 0,
  uid INTEGER DEFAULT 0,  -- User ID (0 = root)
  gid INTEGER DEFAULT 0,  -- Group ID (0 = root)
  mode INTEGER DEFAULT 420, -- Permission desimal (octal 644 = 420, 755 = 493)
  created_at INTEGER,
  modified_at INTEGER,
  FOREIGN KEY (parent_id) REFERENCES vnodes(id),
  UNIQUE(parent_id, name)    -- satu nama per folder induk
);
```

Important notes:

- **uid/gid/mode are stored as decimal** by the kernel, not octal strings. Example: mode 420 = 0o644, 493 = 0o755. Root / is inserted at init with mode 493; /tmp in `fstab.conf` is written as mode = 0o1777 (decimal equivalent 1023, sticky bit).
- **Path navigation** is done row by row: each segment is looked up with `SELECT id FROM vnodes WHERE name = ? AND parent_id = ? AND type = 'DIRECTORY'`, then `parent_id` shifts to the resulting id. This happens in `getNodeId()` / `getNodeIdAndSize()`.
- **UNIQUE(parent_id, name)** prevents duplicate names in the same folder. This schema also migrates old DBs: the `uid/gid/mode/size/modified_at` columns are added via `ALTER TABLE` if they do not exist yet.

Chunked I/O in SQL — and why it matters

The main trick: **never pull the full content into JS memory**. For large files, reading everything at once is expensive: memory is used up and the UI feels "hung" while waiting. Chunked I/O breaks it into small pieces (`offset + length`) — used for progress-aware read/write, streaming, and OTA firmware.

All slicing is done **inside SQLite**:

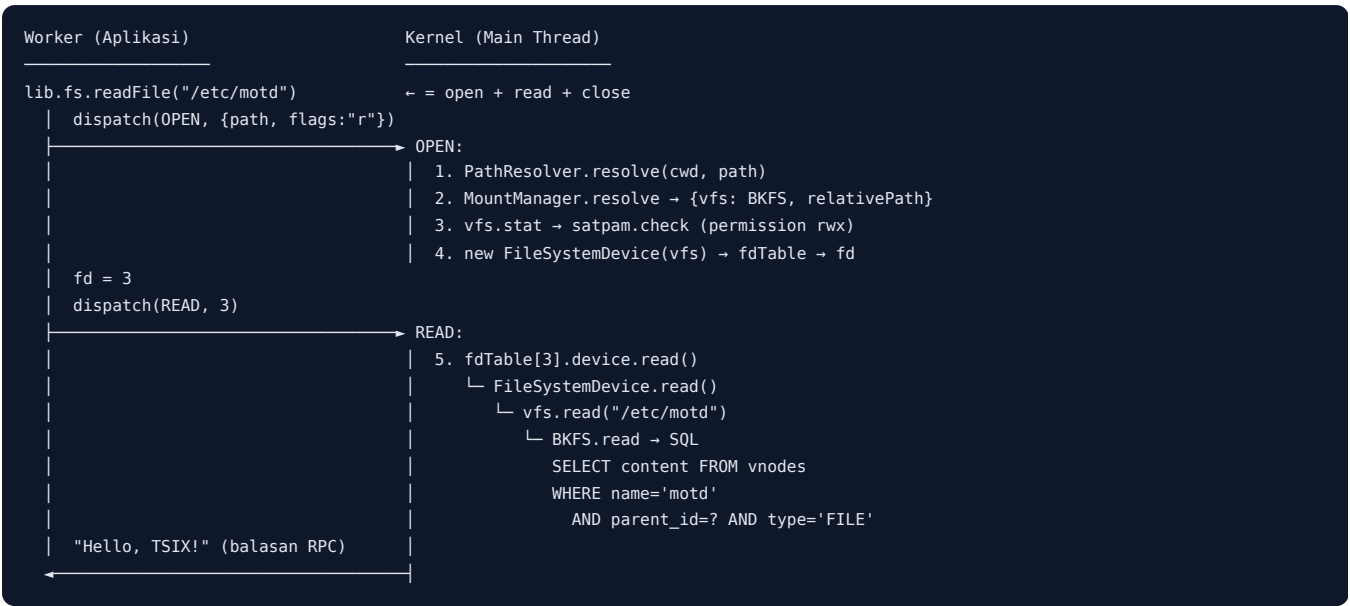
- `readChunk` → `SELECT SUBSTR(content, ? + 1, ?)` — only the requested bytes are read.
- `writeChunk` sequential append (`offset >= currentSize`) → simple `CONCAT ( IFNULL(content, '') || ? )` — **very fast**, does not rewrite the content.
- `writeChunk` random write in the middle → `SUBSTR(content, 1, ?) || ? || SUBSTR(content, ? + 1)` — overwrites a specific part.

- `getSize` → reads only the `size` column (via `getNodeIdAndSize`), **without fetching content** — cheap for checking progress before `readChunk`.

[!IMPORTANT] **Why SUBSTR(content, ? + 1, ?) ?** SQLite SUBSTR uses a 1-based index. To read from byte offset (0-based in the API), the first parameter must be `offset + 1`. Example: `readChunk(path, 2, 4) → SUBSTR(content, 3, 4)`.

Flow / How it works

All file access goes through one path: **syscall → MountManager → IVFS → backend**. Here is the full flow of reading `/etc/motd` from an application (Worker) down to SQLite:



Step by step:

#	Layer	What happens
1	UserLib (Worker)	<code>lib.fs.readFile('/etc/motd')</code> → <code>open</code> → send the <code>OPEN</code> syscall via <code>postMessage</code> (RPC).
2	Syscall OPEN	<code>PathResolver.resolve(pcb.cwd, path)</code> normalizes the path (<code>//</code> , <code>..</code> , <code>..</code>) → absolute.
3	<code>MountManager.resolve()</code>	Find the longest prefix mount point (<code>/</code> → <code>BKFS</code>) → { <code>vfs</code> , <code>relativePath</code> , <code>mountPoint</code> }.
4	PermissionManager	<code>vfs.stat()</code> fetches metadata (uid/gid/mode) → <code>satpam.check()</code> verifies read permission.
5	FD Table	The file is wrapped in <code>FileSystemDevice</code> , <code>setPath(relativePath, flags)</code> , enters <code>pcb.fdTable</code> → <code>fd</code> is returned.
6	Syscall READ	<code>pcb.fdTable[fd].device.read()</code> calls the driver.
7	<code>FileSystemDevice.read()</code>	Polymorphic delegation: <code>vfs.read(this.currentPath)</code> — the kernel does not care about the backend.
8	<code>BKFS.read()</code>	Tree navigation per segment in SQL → <code>SELECT content ...</code> → the string is sent back via RPC.

[!TIP] **Relative vs absolute paths** The syscall receives a path relative to `pcb.cwd`; `PathResolver.resolve()` normalizes it before `MountManager`. So `/etc/../etc/motd` still opens the same file.

Large file variant — the `READ_CHUNK` syscall:

Instead of `READ` (which pulls the whole content into memory), the application uses `lib.fs.readChunk(path, offset, length)`:

```
lib.fs.readChunk("/firmware.bin", 65536, 4096)
→ dispatch(READ_CHUNK, {path, offset, length})
→ MountManager.resolve → vfs.stat → satpam.check(READ)
→ vfs.readChunk(relativePath, offset, length)
  ↳ BKFS: SELECT SUBSTR(content, ? + 1, ?) ... -- hanya byte yang diminta
```

The result: JS memory never holds the whole file — slicing happens inside SQLite.

Source Code

File	Role
src/vfs/IVFS.ts	Filesystem contract
src/vfs/BKFS.ts	SQLite backend + chunked I/O
src/vfs/RamFS.ts	In-memory backend
src/vfs/HostVFS.ts	Host backend (anti-escape)

Snippets (code level)

BKFS.read — read full content

```
public read(path: string): string | null {
  const parts = path
    .split("/")
    .filter((p) => p.length > 0 && p !== "." && p !== "..");
  const fileName = parts.pop();
  if (!fileName) return null;

  let parentId = this.getRootId();
  for (const part of parts) {
    let node = this.db
      .prepare(
        "SELECT id FROM vnodes WHERE name = ? AND parent_id = ? AND type = 'DIRECTORY'",
      )
      .get(part, parentId) as { id: number } | undefined;
    if (!node) return null;
    parentId = node.id;
  }

  const file = this.db
    .prepare(
      "SELECT content FROM vnodes WHERE name = ? AND parent_id = ? AND type = 'FILE'",
    )
    .get(fileName, parentId) as { content: string } | undefined;

  return file ? file.content : null;
}
```

Key point: the path is split per segment, then each segment is looked up with `name + parent_id` (tree). The final query targets `type = 'FILE'` — directories are not read.

BKFS.touch — create / overwrite file (upsert)

```
public touch(
  path: string,
  content: string = "",
  uid: number = 0,
  gid: number = 0,
  mode: number = 420,
): boolean {
  if (this.readOnly) throw new Error("Read-only filesystem");
  const parts = path
```

```

    .split("/")
    .filter((p) => p.length > 0 && p !== "." && p !== "..");
const fileName = parts.pop();
if (!fileName) return false;

let parentId = this.getRootId();
// Navigasi ke folder tujuan
for (const part of parts) {
    let node = this.db
        .prepare(
            "SELECT id FROM vnodes WHERE name = ? AND parent_id = ? AND type = 'DIRECTORY'",
        )
        .get(part, parentId) as { id: number } | undefined;
    if (!node) return false;
    parentId = node.id;
}

// Cek apakah file sudah ada
const existing = this.db
    .prepare(
        "SELECT id FROM vnodes WHERE name = ? AND parent_id = ? AND type = 'FILE'",
    )
    .get(fileName, parentId);

if (existing) {
    this.db
        .prepare(
            "UPDATE vnodes SET content = ?, size = ?, modified_at = ? WHERE id = ?",
        )
        .run(content, content.length, Date.now(), (existing as any).id);
    return true;
}

const now = Date.now();
this.db
    .prepare(
        "INSERT INTO vnodes (parent_id, name, type, content, size, uid, gid, mode, created_at, modified_at) VALUES (?, ?, 'FILE', ?, ?, ?, ?, ?, ?, ?)",
    )
    .run(
        parentId,
        fileName,
        content,
        content.length,
        uid,
        gid,
        mode,
        now,
        now,
    );

this.logger.debug(
    `File created/updated in BKFS: ${fileName} (Mode: ${mode.toString(8)})`,
);
return true;
}

```

Key point: `touch` is an **upsert** — an existing file is `UPDATE` d (content + size), a missing one is `INSERT` ed. The default mode `420 = 0o644`. The `OPEN` syscall with the `w` flag uses this to create/truncate a file.

readChunk (BKFS) — content chunk

```

public readChunk(path: string, offset: number, length: number): string | null {
    const nodeId = this.getNodeId(path);
    if (nodeId < 0) return null;

    const row = this.db
        .prepare(
            "SELECT SUBSTR(content, ? + 1, ?) as chunk FROM vnodes WHERE id = ? AND type = 'FILE'",
        )
        .get(offset, length, nodeId) as { chunk: string | null } | undefined;
}

```

```
return row?.chunk ?? null;
}
```

writeChunk (BKFS) — two paths

```
if (offset >= currentSize) {
  // Sequential append — simple CONCAT (paling cepat)
  UPDATE vnodes SET content = IFNULL(content, '') || ?, size = ?, modified_at = ? WHERE id = ?;
} else {
  // Random write di tengah — SUBSTR + CONCAT
  UPDATE vnodes SET
    content = SUBSTR(content, 1, ?) || ? || SUBSTR(content, ? + 1),
    size = ?, modified_at = ? WHERE id = ?;
}
```

Exercises / Practice

1. Open `system.db` with the SQLite CLI: `.schema vnodes` — observe the table structure.
2. Read `src/vfs/HostVFS.ts` — find `toHostPath()` and check the anti-escape.
3. Write a large file (e.g. 1MB) then call `readChunk` with random offsets — compare the speed vs a full `read`.
4. Read `src/vfs/IVFS.ts` — recognize the whole contract a new backend must satisfy.
5. Place a breakpoint in the `OPEN` and `READ` cases of `src/kernel/Syscalls.ts`, then read `/etc/motd` from the shell — trace down to `BKFS.read` and observe its SQL query.

References

- [wiki/Virtual-File-System.md](#) — the full VFS layer
- [wiki/course/00-overview.en.md §5.1](#)
- `src/vfs/IVFS.ts`, `src/vfs/BKFS.ts`, `src/vfs/RamFS.ts`, `src/vfs/HostVFS.ts`
- `src/kernel/MountManager.ts` — backend selection from path (longest prefix)
- `src/kernel/Syscalls.ts` — syscall `OPEN`, `READ`, `READ_CHUNK`, `WRITE_CHUNK`, `GET_SIZE`
- `src/kernel/devices/FileSystemDevice.ts` — FD → IVFS bridge
- `src/mirror/etc/fstab.conf` — mount configuration at boot

Module 08 — complete. Continue to [Module 09 — FD Table & File Syscalls](#).

RFC-TSIX-EDU-002 | Ninth module of the TSIX curriculum. Understand "everything is a file" at the FD table level: how `open` → `FDEntry` → device, and how a regular file is wrapped by `FileSystemDevice`.

In TSIX, a **file descriptor is not an index into a file array** — it is an index into an `FDEntry { device, context, flags }` that points to any `IDevice` object. A regular file, TTY, pipe, socket — all of them are `IDevice`.

Learning Objectives

- ☐ Explain the structure of `FDEntry { device, context, flags }` and `fdTable`
- ☐ Explain the `OPEN` flow: resolve path → check permission → pick device → `fdTable.push`
- ☐ Explain how `READ/WRITE` dispatch to `entry.device`
- ☐ Explain "everything is a file": regular files vs `/dev/*`
- ☐ Explain pipe refcount via `ioctl ( INC_REF / DEC_REF )`
- ☐ Explain FD cleanup when a process exits

Core Concepts

FDEntry

```
export interface FDEntry {
  device: IDevice; // objek device nyata
  context: any;    // state per-FD
  flags?: string;  // "r" | "w" | "a" | "+"
}
```

`fdTable` lives in the PCB — each process has an array of `(FDEntry | null)[]`. FD 0/1/2 (stdin/stdout/stderr) are filled at spawn from the TTY device.

Everything is a File

Opened item	Device in the FD table
Regular file (<code>/etc/passwd</code>)	<code>FileSystemDevice</code> (wraps IVFS)
<code>/dev/tty1</code>	<code>TTYDevice</code>
<code>/dev/null</code>	<code>NullDevice</code>
Pipe (PIPE syscall)	<code>PipeDevice</code>
Socket (SOCKET syscall)	<code>SocketDevice</code>

There is no *type* difference between a file and a device in the FD table — both are merely objects that implement `IDevice ( read(), write(), ioctl() )`. Opening a regular file → the kernel creates a new `FileSystemDevice` that wraps the IVFS backend returned by `mountManager.resolve()`. Opening `/dev/*` → the kernel takes the device already registered in `kernel.devices` (TTY, null, stdin, etc.). This is the heart of "everything is a file": `read/write` are always polymorphic to `entry.device`.

Standard FDs (0/1/2)

When a process is created (`Scheduler.createProcess`) or on `EXEC`, the kernel fills the first three FDs from the stdio devices. The default at `EXEC` is the global device `kernel.devices.stdin/stdout/stderr`; if the process runs on a TTY (`pcb.ttyId`), all three are pointed at the same `tty{N}`.

FD	Name	flags	Device (common)
0	stdin	"r"	TTYDevice (ttyN) when the process has a TTY; fallback KeyboardDevice (kernel.devices.stdin)
1	stdout	"w"	TTYDevice (kernel.devices.stdout = tty1)
2	stderr	"w"	TTYDevice (kernel.devices.stderr = tty1)

[!NOTE] The flags of FD 0 is "r" (read), while FD 1 and 2 are "w" (write). This is what lets WRITE refuse to write to an FD that was not opened for writing ("Bad File Descriptor").

Pipe refcount

A pipe has a refcount on both the read and write sides. EOF occurs when `writeRefs == 0`. The kernel manages the refcount through the device `ioctl` — command numbers:

ioctl cmd	Meaning	Called from
10	INC_READ_REF	OPEN when flags contain "r"
20	INC_WRITE_REF	OPEN when flags contain "w" / "a"
11	DEC_READ_REF	CLOSE when flags contain "r"
21	DEC_WRITE_REF	CLOSE when flags contain "w" / "a"

Every `open` increments, every `close` decrements. `FileSystemDevice` does not use refcount — its `ioctl()` returns `-1` (no-op). Refcount only matters for real devices such as pipes, TTYs, and serial.

Flow / How It Works

OPEN: from path to FD

Steps in the `SyscallCode.OPEN` case (`src/kernel/Syscalls.ts`):

1. **Resolve path** — `PathResolver.resolve(pcb.cwd, args)` converts the path (relative/absolute) to an absolute one.
2. **Resolve VFS** — `mountManager.resolve(absolutePath) → { vfs, relativePath }`. The backend is chosen from the longest mount prefix (`/ → BKFS`, `/tmp → RamFS`, `/mnt/* → HostVFS`).
3. **Check permission** — `vfs.stat(relativePath)` to get the node. `requiredPerm = WRITE` if flags contain "w" / "+", otherwise `READ`.
 - Node exists → `satpam.check(pcb, node, requiredPerm)`; failure → `Permission Denied`.
 - Node does not exist → flags "r" → `File not found`; flags "w" / "+" → check the parent directory's permission, then `vfs.touch` creates the file (mode `0644`), and truncates when flags are "w" without "a".
4. **Pick device:**
 - Path `/dev/*` → look up `kernel.devices[devName]` (aliases: `screen/console/stdout/fb0` and `keyboard/stdin`; `tty → tty{pcb.ttyId}`). The refcount is incremented via `ioctl INC_*_REF`, then `device.open()` is called lazily.
 - Regular path → `new FileSystemDevice(vfs) + setPath(relativePath, flags)`.
5. **Insert into the table** — `const fd = pcb.fdTable.length; pcb.fdTable.push({ device, context, flags }); return fd;`. `fd` is a new index in the array, not a free number. For a regular file, `context` is set to `relativePath`; for `/dev/*`, `context` is set to the absolute path.

READ / WRITE: dispatch to entry.device

`READ` and `WRITE` do not care about the device type. They take `entry = pcb.fdTable[fd]`, then call the polymorphic method on `entry.device`:

- `READ(fd) → entry.device.read()` — a TTY reads its buffer, a file reads from the VFS.
- `WRITE(fd, content) → check entry.flags` (must contain "w" / "a" / "+", otherwise `Bad File Descriptor`), then `entry.device.write(content)`.

There is a special IPC route: `WRITE` with `{ pid, content }` writes to the target process's `fdTable[0]` (TTY → injection via `ioctl(0x2001)`); `READ` with `{ pid }` reads from the target process's `fdTable[1]` (TTY → `ioctl(0x2002)`).

Walkthrough: opening `/etc/passwd`

Trace a shell process that executes `cat /etc/passwd`:

```
cat /etc/passwd
|
| OPEN("/etc/passwd", "r")
v
+-----+
| case OPEN (kernel) |
| 1. PathResolver.resolve(cwd, path) |
|   -> "/etc/passwd" (absolut) |
| 2. mountManager.resolve("/etc/passwd") |
|   -> { vfs: BKFS, relativePath: "etc/passwd" } |
| 3. vfs.stat("etc/passwd") -> node ada |
|   satpam.check(pcb, node, READ) -> lolos |
| 4. const fsDriver = new FileSystemDevice(BKFS) |
|   fsDriver.setPath("etc/passwd", "r") |
| 5. const fd = pcb.fdTable.length // misal = 3 |
|   pcb.fdTable.push({ device: fsDriver, |
|                     context: "etc/passwd", |
|                     flags: "r" }) |
| return fd; // fd = 3 |
+-----+
|
| fd = 3, lalu READ(3)
v
+-----+
| case READ |
| const entry = pcb.fdTable[3]; |
| return entry.device.read(); |
|   -> FileSystemDevice.read() |
|   -> vfs.read("etc/passwd") // BKFS: SQL SELECT |
+-----+
|
| isi file /etc/passwd
v
| CLOSE(3)
v
+-----+
| case CLOSE |
| device.ioctl(11 / 21) // DEC_REF; FileSystemDevice no-op |
| pcb.fdTable[3] = null; |
+-----+
```

[!NOTE] `FileSystemDevice` returns `null` when `currentPath` is empty and `false` when writing without a path — a "fail safe" before touching the VFS.

Cleanup on exit

The kernel installs a global exit hook in the `SyscallHandler` constructor (`setOnProcessExit`). When a process EXITED:

1. **Close all FDs** — iterate `pcb.fdTable`; every entry that is not `null` is `CLOSE`-dispatched (triggering `DEC_*_REF` on devices that support it). Errors during mass cleanup are ignored.
2. **Port safety net** — `portManager.releasePortsByPid(pid)` releases all MQTNL ports owned by the process (in anticipation of sockets that escaped `CLOSE`).
3. **GUI cleanup** — `guiRegistry.destroyAllForPid(pid)` destroys all windows owned by the process and sends `GUI_REQ DESTROY_WINDOW` to the GUI daemon (`gued`).

Source Code

File	Role
src/kernel/Scheduler.ts	Defines <code>FDEntry</code> + <code>fdTable</code> in the PCB
src/kernel/devices/FileSystemDevice.ts	File ↔ IVFS bridge
src/kernel/devices/PipeDevice.ts	Pipe + refcount
src/kernel/Syscalls.ts	OPEN/READ/WRITE/CLOSE/PIPE + cleanup

Snippets (code level)

OPEN — resolve, permission, pick device, `fdTable.push`

```
case SyscallCode.OPEN: {
  // ... PathResolver.resolve + mountManager.resolve (lihat "Alur / Cara Kerja") ...

  // 1. Permission Check
  const requiredPerm =
    flags.includes("w") || flags.includes("+")
      ? Permission.WRITE
      : Permission.READ;

  // Existence Check
  const node = vfs.stat(relativePath);

  if (node) {
    // Check File Permission
    if (!this.satpam.check(pcb, node, requiredPerm)) {
      throw new Error(
        `Permission Denied: Cannot open ${absoluteOpenPath} for ${Permission[requiredPerm]}`,
      );
    }
  } else {
    // ... "File not found", cek parent-dir, touch buat file, truncate ...
  }

  // ... branch /dev/* -> kernel.devices[devName] + INC_REF + lazy open ...

  // File biasa -> bungkus FileSystemDevice
  const fsDriver = new FileSystemDevice(vfs);
  fsDriver.setPath(relativePath, flags);

  // TRUNCATE: If opened with 'w' (and not 'a'), we must clear content first
  if (
    flags &&
    flags.includes("w") &&
    !flags.includes("a") &&
    !flags.includes("+")
  ) {
    vfs.touch(relativePath, "", pcb.uid, pcb.gid, 420);
  }

  const fd = pcb.fdTable.length;
  pcb.fdTable.push({ device: fsDriver, context: relativePath, flags });
  return fd;
}
```

[!IMPORTANT] Note two details: (1) `fd` is `pcb.fdTable.length` — a new index at the end of the array, not a free number; (2) for a regular file `context` is set to `relativePath`, while for `/dev/*` it is set to the absolute path. `context` is freely interpreted by the device.

READ / WRITE — dispatch to `entry.device`

```
// WRITE
case SyscallCode.WRITE: {
  // ... route khusus pid -> targetPcb.fdTable[0] (TTY: ioctl(0x2001)) ...

  const { fd, content } = args;
  const entry = pcb.fdTable[fd];
  if (!entry) return false;

  // Check if FD was opened with Write permission
  const flags = entry.flags || "r";
  if (
    !flags.includes("w") &&
    !flags.includes("a") &&
    !flags.includes("+")
  ) {
    throw new Error("Bad File Descriptor: Not open for writing");
  }

  return entry.device.write(content);
}

// READ
case SyscallCode.READ: {
  // ... route khusus pid -> targetPcb.fdTable[1] (TTY: ioctl(0x2002)) ...

  const fd = args as number;
  const entry = pcb.fdTable[fd];
  if (!entry) throw new Error(`FD NOT FOUND: ${fd}`);
  return entry.device.read();
}
}
```

FileSystemDevice — file ↔ VFS bridge

```
export class FileSystemDevice implements IDevice {
  name = "FileSystem";
  private vfs: IVFS;
  private currentPath: string = "";
  private flags: string = "r";

  constructor(vfs: IVFS) {
    this.vfs = vfs;
  }

  public setPath(path: string, flags: string = "r") {
    this.currentPath = path;
    this.flags = flags;
  }

  read() {
    if (!this.currentPath) return null;
    return this.vfs.read(this.currentPath);
  }

  write(data: any) {
    if (!this.currentPath) return false;

    const isAppend = this.flags.includes("a");
    const isWrite = this.flags.includes("w");
    const isUpdate = this.flags.includes("+");

    if (isAppend || isWrite || isUpdate) {
      return this.vfs.append(this.currentPath, data);
    }

    // Fallback: Default Overwrite (Backward Compatibility)
    return this.vfs.touch(this.currentPath, data);
  }

  ioctl(cmd: number, arg: any) { return -1; }
}
```

FD cleanup on process exit

```
this.scheduler.setOnProcessExit(async (pid) => {
  const pcb = this.scheduler.getProcess(pid);
  if (pcb) {
    // Tutup semua FD — trigger DEC_REF via dispatch CLOSE
    for (let fd = 0; fd < pcb.fdTable.length; fd++) {
      if (pcb.fdTable[fd]) {
        try {
          await this.dispatch(pid, SyscallCode.CLOSE, fd);
        } catch (e) {
          // Ignore errors during mass cleanup
        }
      }
    }
  }

  // === Force release semua port milik PID ini ===
  try {
    this.kernel.getPortManager().releasePortsByPid(pid);
  } catch (_) {}

  // --- GUI Cleanup ---
  const guiRegistry = this.kernel.guiRegistry;
  if (guiRegistry) {
    const ownedWindows = guiRegistry.destroyAllForPid(pid);
    if (ownedWindows.length > 0) {
      const gudedPid = guiRegistry.getDaemonPid();
      if (gudedPid !== null) {
        for (const wid of ownedWindows) {
          this.scheduler.sendEvent(gudedPid, "gui_request", {
            syscall: "GUI_REQ",
            pid,
            wid,
            action: GUIAction.DESTROY_WINDOW,
          } as IGUIPayload);
        }
      }
    }
  }
}
});
```

Exercises / Practice

1. Read `src/kernel/devices/PipeDevice.ts` — explain when a pipe returns EOF (connect it to `writeRefs` and the `DEC_WRITE_REF` ioctl).
2. From the shell, `echo hello > /tmp/x` then `cat /tmp/x` — trace the `OPEN/WRITE/READ/CLOSE` syscalls in the log. Note the `fd` returned by `OPEN` and the device behind it.
3. Read `src/kernel/Syscalls.ts` — find the `OPEN` implementation and how it picks the device (`FileSystemDevice` vs `/dev/*`).
4. Read `createProcess` in `src/kernel/Scheduler.ts` — explain how `fds 0/1/2` (`stdin/stdout/stderr`) are filled from `options.fds`.

References

- `wiki/Virtual-File-System.md` — \$device model
- `wiki/course/00-overview.en.md` — \$5 VFS & Device Drivers
- `src/kernel/devices/FileSystemDevice.ts` — file ↔ IVFS bridge
- `src/kernel/Syscalls.ts` — `OPEN/READ/WRITE/CLOSE/EXIT` cleanup
- `src/kernel/Scheduler.ts` — `FDEntry` + `fdTable` in the PCB

RFC-TSIX-EDU-002 | Tenth module of the TSIX curriculum. Understand the `IDevice` contract, aux-devices plugins, udev-style configuration, and the virtual `/dev` device table.

`/dev` in TSIX is the **device registry in the kernel** (`kernel.devices[xxx]`) — not a filesystem vnode. Every device is an `IDevice` object that can be opened like a file.

Learning Objectives

- ☐ Explain the `IDevice` contract (read/write/ioctl + optional init/open/close/present)
- ☐ Explain why `/dev` is virtual — not a filesystem vnode
- ☐ Explain the device registration order at boot
- ☐ Explain the aux-devices plugin: `export default + static autoRegister` conventions
- ☐ Explain `applyDeviceConfigs()` (udev-style from `sysconfig.conf`)
- ☐ Explain `present()` for hotplug (udev-like)
- ☐ Explain the `DbLib` dual transport: `/dev/mysql` vs the `mysqld` daemon
- ☐ Write your own driver (10-step checklist)

Core Concepts

The IDevice Contract

Every device is an object that implements `IDevice`. The kernel only needs to call the three required methods — `read`, `write`, `ioctl` — without knowing what is inside. This is the HAL abstraction: one contract for all hardware.

```
export interface IDevice {
  name: string;
  read(offset?: number, length?: number): any;
  write(data: any, offset?: number): boolean;
  ioctl(cmd: number, arg: any): any;

  init?(ctx: KContext): void; // Opsional: Untuk terima 'suntikan' dari Kernel
  open?(): boolean; // Opsional: Lazy open device hardware
  close?(): boolean; // Opsional: Close device hardware (with refcounting)

  /**
   * Opsional: Apakah hardware benar-benar tersedia SEKARANG?
   * Dipakai `ls /dev` untuk menyembunyikan device yang tidak present
   * (udev-like hotplug). Jika tidak diimplementasikan → selalu present.
   */
  present?(): boolean;

  uid?: number;
  gid?: number;
  mode?: number;
  disabled?: boolean;
}
```

`KContext` is a utility injection from the kernel into the driver (delivered through `init`):

```
export interface KContext {
  syslog: (message: string) => void;
}
```

Member	Required	Meaning
name	✓	Driver name; becomes the path name <code>/dev/<name></code> (lowercase)
read(offset?, length?)	✓	Read data from the device
write(data, offset?)	✓	Write data; <code>boolean</code> success/failure
ioctl(cmd, arg)	✓	Control the device (clear, raw mode, refcount, etc.)
init(ctx)	☐	Called once at boot; <code>ctx.syslog()</code> for logging
open()	☐	Lazy open when the device is opened (e.g. <code>SerialDevice</code>)
close()	☐	Close the device with refcounting
present()	☐	Hotplug: <code>ls /dev</code> hides devices that are not present
uid/gid/mode	☐	Permission metadata (used by the <code>OPEN</code> permission check)
disabled	☐	<code>true</code> → skipped by the plugin loader at boot

/dev Virtual — Not a Vnode

`/dev` does **not** store device nodes in the VFS database. The `OPEN` code comment states it explicitly: "*we don't want to create device nodes in VFS database*". What exists is only the `/dev` directory (created by `bkfs.mkdir("/dev", 0, 0, 493)` at boot) and the `kernel.devices` table.

Operation	Handling in syscall
<code>ls /dev</code>	Special: enumerate <code>Object.keys(kernel.devices)</code> , filter by <code>present()</code> , report metadata
<code>open("/dev/xxx")</code>	<code>devName = path.replace("/dev/", "")</code> → <code>kernel.devices[devName]</code> → <code>FDEntry {device, ...}</code>
<code>stat("/dev/xxx")</code>	Special: fetch device metadata (<code>uid/gid/mode</code>)
<code>chmod / chown /dev/*</code>	Root only; set directly on the device object
<code>/dev/</code> without a driver	Falls back to the VFS node (if any) with a normal permission check

Aliases recognized by `OPEN`:

- `tty` → the calling process's TTY (`tty${pcb.ttyId || 1}`).
- `screen`, `console`, `stdout`, `fb0` → the caller's active TTY (fallback `fb0`).
- `keyboard`, `stdin` → `kernel.devices.stdin`.

The `OPEN` permission check uses device metadata with a **default** `0o600` (root only) when `mode` is not set:

```
const devPerm = {
  name: devName,
  uid: device.uid ?? 0,
  gid: device.gid ?? 0,
  mode: device.mode ?? 0o600, // Default: root only (rw-----)
};
```

The standard I/O devices (`stdout`/`stdin`/`fb0`/`console`/`screen`/`keyboard`) **bypass** this check. When opened, the kernel calls `ioctl(10)` (`INC_READ_REF`) / `ioctl(20)` (`INC_WRITE_REF`) for refcounting, then `device.open()` if available (lazy open). A failed `open()` → error Failed to open device `/dev/xxx`.

The /dev Device Table

/dev/	Class	Function
<code>stdin</code>	<code>KeyboardDevice</code>	Keyboard input (cooked/raw)
<code>fb0/stdout/stderr</code>	<code>TTYDevice</code>	Standard output → active TTY
<code>tty1..tty32</code>	<code>TTYDevice</code>	Virtual consoles

/dev/	Class	Function
null	NullDevice	Black hole
smqtnl0/1	SimpleMQTNLDriver	MQTNL network interface
randomdevice	RandomDevice	Random numbers (0-999 per read)
mcp23017	MCP23017Device	I2C GPIO (autoRegister)
joystick	JoystickDevice	USB HID gamepad; present() = hotplug
ttyUSB*	SerialDevice	Auto-detect serial
mysql (experimental)	MySQLDevice	External DB connection — disabled: true by default
(virtual)	PipeDevice / SocketDevice	Runtime instances, not paths

[!NOTE] MySQLDevice has disabled: true in the source code — so /dev/mysql is **not registered by default**. Enable it only when a device transport is needed. See the dual transport note below.

Device Registration at Boot

Kernel.boot() registers devices in a fixed order:

```
// 1. TTYManager(32) → tty1..tty32 (TTYDevice)
// 2. Map device inti:
this.devices = {
  stdin: new KeyboardDevice(),
  fb0: ttysDevs.tty1,    // Alias fb0 ke TTY1
  stdout: ttysDevs.tty1, // Alias stdout ke TTY1
  stderr: ttysDevs.tty1, // Alias stderr ke TTY1
  null: new NullDevice(),
  ...ttysDevs,
};
// 3. Interface jaringan (cfg.network.interfaces) → SimpleMQTNLDriver
// 4. loadAuxDevices() → plugin dari folder aux-devices
// 5. SerialDeviceManager → auto-detect ttyUSB*
// 6. applyDeviceConfigs() → udev-style dari sysconfig.conf
// 7. init() semua driver → suntikan KContext (syslog)
// 8. pastikan /dev ada di VFS (mkdir 493)
```

The order matters: applyDeviceConfigs() must run **after** all devices are registered, and init() after the configuration is applied.

aux-devices Plugins & the autoRegister Convention

The src/kernel/devices/aux-devices/ folder is scanned by loadAuxDevices() at boot. Two conventions:

1. **export default is required** — the loader reads module.default || module. Without a default export, the class is not recognized.
2. **static autoRegister(kernel) is optional** — for platform-specific hardware configuration (e.g. MCP23017 + I2C bus). Called after the instance is registered.

applyDeviceConfigs (udev-style)

The [device.<name>] sections in sysconfig.conf → "udev" rules: set mode/uid/gid per device name — exactly like Linux udev rules.

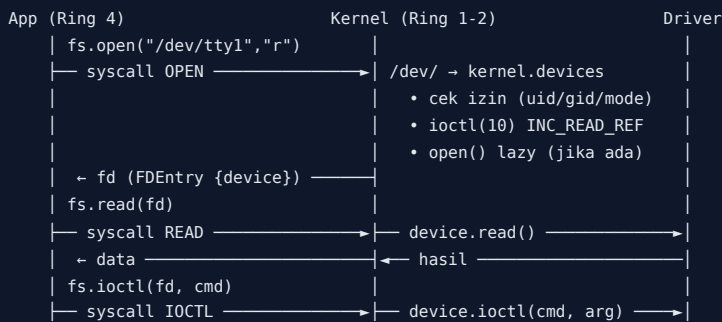
```
{
  "devices": {
    "randomdevice": {
      "mode": 438
    }
  }
}
```

```
}  
}
```

438 (decimal) = 0o666 (rw-rw-rw-), applied to /dev/randomdevice. See the `applyDeviceConfigs()` snippet.

Flow / How It Works

Device access flow (read/write)



Boot registration flow

1. TTYManager(32) creates 32 consoles → TTYDevice tty1..32.
2. The core map `this.devices: stdin/fb0/stdout/stderr/null + tty1..32`.
3. Network interfaces from `cfg.network.interfaces` → SimpleMQTNLDriver.
4. `loadAuxDevices()`: scan `aux-devices/` → export default → register → `autoRegister`.
5. `SerialDeviceManager` detects `ttyUSB*`.
6. `applyDeviceConfigs()`: apply `mode/uid/gid` from `sysconfig.conf`.
7. Loop `init()` over all drivers → inject `KContext.syslog`.
8. The `/dev` directory is guaranteed to exist in VFS.

The `ls /dev` flow (udev-like hotplug)

```
ls /dev
→ syscall LS dengan path /dev
→ enumerasi kernel.devices
→ filter: device dengan present() hanya tampil jika present() true
→ laporan {name, type:"DEVICE", size:0, uid, gid, mode}
```

Example: `JoystickDevice.present()` returns `this.connected` — when the gamepad is unplugged, `/dev/joystick` disappears from `ls /dev`; when plugged back in, it reappears.

Special Note: MySQLDevice & DbLib (Dual Transport)

[!IMPORTANT] `/dev/mysql` is not a hardware driver. `MySQLDevice` is the **first transport** for external database integration through the device model — an example of extending HAL in an unusual direction.

DbLib is already implemented (a UserLib sub-library, the `lib.fs/lib.net` pattern) with **pluggable dual transport**:

- **Device** (`/dev/mysql`): syscall `DB_*` (67–69) → kernel → device → `mysql2`
- **Service daemon** (`mysqld`): syscall `DB_*` → kernel → Ring 4 daemon → `mysql2`

The kernel routes **dynamically**: if `mysqld` is registered (syscall `DB_SERVICE_REGISTER=70`) → to the daemon; if not → to the device (fallback). The daemon replies via `DB_SERVICE_REPLY=71`. Applications only see

`db.connect()/query()/disconnect()` — the medium is invisible. The sandbox is safe because only the kernel or a privileged daemon touches `mysql2`.

DB syscall codes

Code	Name	Function
67	DB_CONNECT	Open a connection ({host, user, password, database})
68	DB_QUERY	Execute SQL (SELECT → rows; INSERT/UPDATE/DELETE → ResultSetHeader)
69	DB_DISCONNECT	Close the connection
70	DB_SERVICE_REGISTER	The <code>mysqld</code> daemon registers as a service transport
71	DB_SERVICE_REPLY	The daemon sends the result {requestId, result} to the kernel

Dynamic routing flow

```
App (Ring 4)  lib.db.query(sql)          ← DbLib (@tsix/DbLib)
  ▼
UserLib.dispatch(DB_QUERY = 68)
  ▼
Kernel (Syscalls.ts)
  | if (this.dbServicePid !== null)
  | — YA → forwardDbRequest(pid, "query", sql)
  |     → sendEvent(mysqld, "db_request") → daemon → mysql2
  |     → DB_SERVICE_REPLY=71 (requestId, result) → resolve pending
  | — TIDAK → /dev/mysql (MySQLDevice) → device.query(sql, pid) → mysql2
  ▼
resolve → App (rows)
```

The routing logic is identical in every `DB_CONNECT`/`DB_QUERY`/`DB_DISCONNECT` case: check `dbServicePid` first, then fall back to the device. When the daemon exits, `Syscalls.ts` resets `dbServicePid = null` so the route automatically returns to the device.

Implementation details

- **MySQLDevice** (`aux-devices/MySQLDevice.ts`) has `disabled: true` — **not registered by default**.
- **Multi-instance**: connections are stored per-PID (`Map<pid, Connection>`); each app has its own connection, and two apps can access different servers/databases in parallel. The raw `ioctl` path without a pid uses an anonymous slot (`ANON_PID = 0`): `ioctl(0x2001)` connect / `ioctl(0x2002)` disconnect.
- **release(pid)**: called by the kernel when a process dies → the connection is closed forcibly (anti-leak).
- **The `mysqld` daemon** (`src/mirror/etc/mysqld/mysqld.ts`): privileged Ring 4 — the host module allow-list includes `mysql2` and `mysql2/promise`.
- **Apps never touch `mysql2`** — only `lib.db.*` or `import { db } from "@tsix/Application"`.

Snippet (code level)

All snippets below are **exact copies from the source** (code is the truth).

The IDevice contract — `src/kernel/devices/IDevice.ts`

```
export interface IDevice {
  name: string;
  read(offset?: number, length?: number): any;
  write(data: any, offset?: number): boolean;
  ioctl(cmd: number, arg: any): any;

  init?(ctx: KContext): void; // Opsional: Untuk terima 'suntikan' dari Kernel
  open?(): boolean; // Opsional: Lazy open device hardware
  close?(): boolean; // Opsional: Close device hardware (with refcounting)
```

```

/**
 * Optional: Apakah hardware benar-benar tersedia SEKARANG?
 * Dipakai `ls /dev` untuk menyembunyikan device yang tidak present
 * (udev-like hotplug). Jika tidak diimplementasikan → selalu present.
 */
present?(): boolean;

uid?: number;
gid?: number;
mode?: number;
disabled?: boolean;
}

```

A small real driver — NullDevice (/dev/null)

```

// src/kernel/devices/NullDevice.ts
import { IDevice } from "../IDevice";

/**
 * NULL DEVICE (/dev/null)
 *
 * Lubang hitam kernel.
 * Apapun yang ditulis ke sini akan dibuang.
 * Apapun yang dibaca dari sini akan langsung EOF (null/empty).
 */
export class NullDevice implements IDevice {
    name = "NullDevice";

    read() {
        return "";
    }

    write(_data: any): boolean {
        return true;
    }

    ioctl(_cmd: number, _arg: any): any {
        return true;
    }
}

```

Note: IDevice.ts also contains a concise NullDevice (name "Null", ioctl → -1) as a learning example. The /dev/null driver used at boot is the NullDevice.ts above.

Driver with init + default export — RandomDevice (/dev/randomdevice)

```

// src/kernel/devices/aux-devices/RandomDevice.ts
import { IDevice, KContext } from "../IDevice";

/**
 * RANDOM DEVICE (/dev/random)
 *
 * Menghasilkan angka acak sebagai string.
 * Implementasi sederhana untuk demonstrasi plugin system.
 */
export class RandomDevice implements IDevice {
    name = "RandomDevice";
    private kctx: KContext | null = null;

    init(ctx: KContext) {
        this.kctx = ctx;
        this.kctx.syslog("Driver initialized and ready.");
    }

    read() {
        // Balikin angka acak 0-999 sebagai string
    }
}

```

```

    const val = Math.floor(Math.random() * 1000);
    if (this.kctx) {
        this.kctx.syslog(`Random number generated: ${val}`);
    }
    return val.toString() + "\n";
}

write(_data: any): boolean {
    // Menulis ke random device biasanya diabaikan atau buat seeding
    if (this.kctx) {
        this.kctx.syslog("Write attempt to RandomDevice ignored.");
    }
    return true;
}

ioctl(_cmd: number, _arg: any): any {
    return true;
}
}

// Plugin Export: Harus export default class yang implement IDevice
export default RandomDevice;

```

loadAuxDevices() — plugin loader (src/kernel/Kernel.ts)

```

private loadAuxDevices() {
    if (!this.devices) return;

    const auxPath = path.resolve(__dirname, "devices/aux-devices");
    if (!fs.existsSync(auxPath)) {
        this.logger.debug(`Auxiliary devices directory not found: ${auxPath}`);
        return;
    }

    const files = fs.readdirSync(auxPath);
    files.forEach((file) => {
        if (file.endsWith(".ts") || file.endsWith(".js")) {
            try {
                const fullPath = path.join(auxPath, file);
                const module = require(fullPath);
                const DeviceClass = module.default || module;

                if (typeof DeviceClass === "function") {
                    // 1. Try auto-loading as device instance (original behavior)
                    const instance = new DeviceClass() as IDevice;

                    // Check if device is explicitly disabled
                    if (instance.disabled === true) {
                        this.logger.debug(
                            `[Dynamic HAL] Kernel Plugin ${file} is disabled, skipping.`
                        );
                        return;
                    }

                    const devName = (
                        instance.name || file.replace(".ts", "").replace(".js", "")
                    ).toLowerCase();
                    this.devices![devName] = instance;
                    this.logger.info(
                        `[Dynamic HAL] Kernel Plugin Loaded: /dev/${devName}`
                    );

                    // 2. Check for static autoRegister method (new convention)
                    if (typeof (DeviceClass as any).autoRegister === "function") {
                        try {
                            (DeviceClass as any).autoRegister(this);
                            this.logger.debug(
                                `[Dynamic HAL] Auto-register called for ${file}`
                            );
                        } catch (e: any) {
                            this.logger.debug(
                                `[Dynamic HAL] Auto-register skipped for ${file}: ${e.message}`
                            );
                        }
                    }
                }
            } catch {
                // Silent failure
            }
        }
    });
}

```

```

        });
    }
} catch (e: any) {
    this.logger.error(`Failed to load aux device ${file}: ${e.message}`);
}
}
});
}

```

Note the `const DeviceClass = module.default || module;` line — this is why `export default` is **required** so the class can be instantiated.

The autoRegister convention — MCP23017Device

```

// src/kernel/devices/aux-devices/MCP23017Device.ts (bagian)

// Platform-specific hardware configuration
const HARDWARE_CONFIGS = [
    { bus: 2, address: 0x20, name: "mcp23017" } // Orange Pi 3B default
];

export class MCP23017Device implements IDevice {
    name: string;
    uid: number = 0;
    gid: number = 0;
    mode: number = 0x660;
    disabled: boolean = false;

    static autoRegister(kernel: any): void {
        for (const config of HARDWARE_CONFIGS) {
            try {
                const device = new MCP23017Device(config.bus, config.address, config.name);
                kernel.devices[config.name] = device;
            } catch (e: any) {
                // Silently skip if hardware not present or i2c-bus not available
            }
        }
    }
    // ... (pinMode/digitalWrite/digitalRead)
}

export default MCP23017Device;

```

applyDeviceConfigs() — udev-style (src/kernel/Kernel.ts)

```

private applyDeviceConfigs() {
    const cfg = Config.get();
    if (!cfg.devices) return;

    for (const devName in cfg.devices) {
        const device = this.devices[devName];
        if (device) {
            const devCfg = cfg.devices[devName];
            if (devCfg.mode !== undefined) device.mode = devCfg.mode;
            if (devCfg.uid !== undefined) device.uid = devCfg.uid;
            if (devCfg.gid !== undefined) device.gid = devCfg.gid;
            this.logger.info(
                `[udev] Configuration applied to / dev / ${devName}: mode = ${device.mode?.toString(8)}, uid = ${device.uid}, gid = ${device.gid}`,
            );
        }
    }
}

```

OPEN routing for /dev/* — src/kernel/Syscalls.ts (core excerpt)

```
if (absoluteOpenPath.startsWith("/dev/")) {
  const devName = absoluteOpenPath.replace("/dev/", "");
  let device: IDevice | null = null;
  if (this.kernel.devices && this.kernel.devices[devName]) {
    device = this.kernel.devices[devName];
  }
  // ... (alias: tty / screen, console, stdout, fb0 / keyboard, stdin — diringkas)
  if (device) {
    const devPerm = {
      name: devName,
      uid: device.uid ?? 0,
      gid: device.gid ?? 0,
      mode: device.mode ?? 0o600, // Default: root only (rw-----)
    };
    if (!this.satpam.check(pcb, devPerm, requiredPerm)) {
      throw new Error(
        `Permission Denied: You cannot access device /dev/${devName}`,
      );
    }
  }
  const f = flags || "r";
  if (f.includes("r") && device.ioctl) await device.ioctl(10, null); // INC_READ_REF
  if ((f.includes("w") || f.includes("a")) && device.ioctl)
    await device.ioctl(20, null); // INC_WRITE_REF
  if (device.open) {
    const opened = device.open();
    if (!opened) {
      throw new Error(`Failed to open device /dev/${devName}`);
    }
  }
  const fd = pcb.fdTable.length;
  pcb.fdTable.push({ device, context: absoluteOpenPath, flags });
  return fd;
}
```

Guide: Writing Your Own Driver

[!TIP] This answers the "Next steps" in the Module 10 ToC: *a tutorial for writing your own driver*. The smallest examples already exist: `NullDevice` and `RandomDevice`.

File locations

Driver type	Location	How to register
Core device (stdin, fb0, tty, null)	src/kernel/devices/*.ts	Directly in <code>Kernel.boot()</code> (the <code>this.devices</code> map)
Hardware / experimental	src/kernel/devices/aux-devices/<Name>Device.ts	Auto via <code>loadAuxDevices()</code> + export default
Network interface	—	Via <code>cfg.network.interfaces</code> in <code>sysconfig.conf</code>

The 10-step checklist

1. **Create the file:** `src/kernel/devices/aux-devices/EchoDevice.ts`.
2. **Import the contract:** `import { IDevice, KContext } from "../IDevice";`
3. **Write the class:** `export class EchoDevice implements IDevice { name = "echo"; ... }` — `name` determines the path `/dev/echo` (lowercase).
4. **Implement the required methods:** `read()`, `write()`, `ioctl()`. The kernel only calls these three.
5. **Optional lifecycle:** `init(ctx)` for syslog, `open()/close()` for lazy open, `present()` for hotplug.
6. **Optional metadata:** `uid/gid/mode` (default permission `0o600`), `disabled` to turn it off at boot.

7. `export default EchoDevice;` — REQUIRED. Without it `loadAuxDevices()` does not recognize the plugin (`module.default || module`).
8. **Optional** `static autoRegister(kernel)` — for platform-specific hardware configuration (bus, address, name).
9. **Set udev-style permissions:** `sysconfig.conf` → `[device.echo]` + `mode = 0o666`.
10. **Test:** boot → `ls /dev/echo` → read/write from the shell → check the syslog for `[Dynamic HAL] Kernel Plugin Loaded: /dev/echo`. Add a unit test in `src/kernel/devices/aux-devices/C10-AuxDevices.test.ts` (the C10.08-C10.12 pattern).

[!IMPORTANT] Default export convention: the loader uses `const DeviceClass = module.default || module;` then `if (typeof DeviceClass === "function")`. The driver class **must** be export default so it can be instantiated. `MCP23017Device` satisfies both: `export default` + `static autoRegister`.

Exercises / Practice

1. Read `wiki/DEVELOPER_GUIDE_DEVICES.md` — the driver-writing guide.
2. Read `src/kernel/devices/aux-devices/RandomDevice.ts` — the minimal device pattern (`export default + init`).
3. Read `src/kernel/devices/aux-devices/MCP23017Device.ts` — study `autoRegister` + disabled.
4. Read `src/kernel/devices/aux-devices/joystick.ts` — study `present()` (udev-like hotplug).
5. Run `ls /dev` after boot — notice `/dev/randomdevice` and its metadata.
6. Change `mode` in `sysconfig.conf` (the `[device.randomdevice]` section) → reboot → observe the `ls /dev` results.
7. (Challenge) Write the `/dev/echo` driver: `read()` returns the last text written, `write()` stores it. Follow the 10-step checklist above, then test it from the shell.

References

- `wiki/DEVELOPER_GUIDE_DEVICES.md` — driver-writing guide
- `wiki/mcp23017-registration.md` — plugin registration example
- `wiki/course/00-overview.en.md` §5.2 — device model (HAL) + DbLib notes
- `src/kernel/devices/IDevice.ts` — the `IDevice` + `KContext` contract
- `src/kernel/devices/NullDevice.ts`, `src/kernel/devices/aux-devices/RandomDevice.ts`, `src/kernel/devices/aux-devices/joystick.ts` — driver examples
- `src/kernel/devices/aux-devices/MCP23017Device.ts`, `src/kernel/devices/aux-devices/MySQLDevice.ts` — `autoRegister` & DB transport
- `src/kernel/Kernel.ts` — `boot`, `loadAuxDevices()`, `applyDeviceConfigs()`
- `src/kernel/Syscalls.ts` — `OPEN/LS/STAT/CHMOD/CHOWN` routing for `/dev`, `DB_*` 67–71
- `src/common/SyscallCode.ts` — `DB_*` syscall codes
- `src/mirror/lib/DbLib.ts` — the `db.connect/query/disconnect` API
- `src/mirror/etc/mysqld/mysqld.ts` — service daemon (alternative transport)
- `src/kernel/devices/aux-devices/C10-AuxDevices.test.ts` — driver tests (C10.08-C10.12)
- `src/sysconfig.conf` — the `[device.*]` sections (udev-style)

Module 10 — complete. Part III done. Continue to [Module 11 — Worker Thread & Sandbox](#).

RFC-TSIX-EDU-002 | Eleventh module of the TSIX curriculum. Understand how one worker thread becomes one TSIX process, and how the sandbox locks the doors to the host API.

`WorkerEntry.ts` is the **bootloader** that runs first inside every worker. It initializes `UserLib`, loads the app, then **locks the doors** (`restrictHostAPI`) before the app runs. This sandbox is one of TSIX's two real privilege boundary layers.

Learning Objectives

- ☐ Explain TSIX's two real privilege boundary layers (thread + `PermissionManager`, and the `WorkerEntry` sandbox)
- ☐ Explain `WorkerEntry`'s order of operations (bootloader)
- ☐ Explain the `Module._load` hijack and resolution of `@tsix/*` / `@common/*` from `vfsCache`
- ☐ Explain Direct Memory Execution (`_compile` with a dummy filename)
- ☐ Explain what is sabotaged in `process` (`exit` / `kill`)
- ☐ Explain the `isPrivileged` rule and the host module allow-list
- ☐ Explain why the `@tsix/*` framework is always accessible
- ☐ Explain unhandledRejection/uncaughtException handling
- ☐ Explain the weakness of the name-substring heuristic and defense in depth

Core Concepts

TSIX has **two real privilege boundary layers**. They work together, but each has different properties.

 Four layers of TSIX security Source: [wiki/diagram/Keamanan-dan-Sandboxing-1.mmd](#)

1. Thread + IPC + `PermissionManager` boundary (kernel)

The kernel runs on the **main thread**. Each app runs on its **own worker thread**. An app never touches kernel memory; the only bridge is `postMessage` (syscall request → response). On the kernel side, `PermissionManager` checks `root` (bypass → owner → group → others), `validateArgs` validates the syscall argument contract, and the `SETUID` bit allows root-only privilege elevation (e.g. `/bin/login`).

This layer **cannot be tricked** by app code — the app has no reference to kernel objects.

 Isolation: main thread (kernel) vs worker threads (apps) — IPC only Source: [wiki/diagram/Keamanan-dan-Sandboxing-2.mmd](#)

2. `WorkerEntry` sandbox (worker-local)

`WorkerEntry.ts` is the **bootloader** that runs first inside the worker. It sabotages dangerous host APIs and restricts `require`, so the app is **forced** to use syscalls through `UserLib`. Core mechanisms:

Mechanism	Purpose
Hijack <code>Module._load</code>	<code>@tsix/*</code> & <code>@common/*</code> are resolved from <code>vfsCache</code> (memory), not the filesystem
Direct Memory Execution (DME)	<code>_compile</code> content from memory with a dummy filename ; no disk hit
<code>restrictHostAPI(appName)</code>	Locks the doors: replaces <code>global.require</code> , sabotages <code>process.exit</code> / <code>process.kill</code>
Privileged check	App name substring (fragile): <code>server</code> , <code>daemon</code> , <code>dome</code> , <code>tbuild</code> , <code>vfs</code> , <code>mysqld</code>
Host module allow-list	Only privileged apps may <code>require</code> certain host modules
Fatal handler	<code>unhandledRejection</code> / <code>uncaughtException</code> → send <code>GUI_WINDOW_ERROR</code> to parent → <code>realExit(1)</code>

require behavior: non-privileged vs privileged

require request	Non-privileged app	Privileged app
Framework @tsix/*, @common/*	✔ always allowed	✔ always allowed
Path /lib/... or /common/...	✔ always allowed	✔ always allowed
Host allow-list modules (http, ws, path, fs, url, esbuild, crypto, os, bcryptjs, mysql2, mysql2/promise)	⊘ throw	✔ allowed
Other host modules (net, child_process, ...)	⊘ throw	⊘ throw (specific message)
process.exit / process.kill	⊘ throw	⊘ throw

[!IMPORTANT] The @tsix/* and @common/* frameworks are **always** accessible, even in a locked sandbox. Reason: the framework is the only legitimate bridge to syscalls — without it, an app cannot do anything useful.

Known weaknesses

[!WARNING] The privileged check is an **app name substring** heuristic — fragile. Anyone can name their app evil-daemon and gain the host module allow-list. This is **not** a real security boundary; it is only a convenience barrier. The real defense stays in the kernel: PermissionManager + validateArgs + SETUID. Long-term plan: replace this heuristic with *capability-based* (the app declares the permissions it needs).

Flow / How It Works

Short walkthrough

```
app dipanggil (tsh / rc.local / spawnProcess)
  | syscall EXEC
  ▼
Kernel.spawnWorker()
  | new Worker(WorkerEntry.ts, { workerData, execArgv })
  ▼
WorkerEntry.ts (bootloader di dalam worker)
1. simpan realExit = process.exit (sebelum disabotase)
2. hijack Module._load → @tsix/* & @common/* dari vfsCache
3. pasang unhandledRejection / uncaughtException
4. UserLib ← hijackRequire("@tsix/UserLib") dari memory cache
   lib = new UserLibClass(pid) → global._tsixLib
5. DME aplikasi: transpile appContent → _compile (dummy filename)
6. cari export main / Main / default
7. restrictHostAPI(appName) ← KUNCI PINTU (sebelum app jalan)
8. new AppClass() → await execute(lib, args)
9. hasil string → lib.std.print(); lalu lib.shell.exit(0)
  |
  ▼
KERNEL — tiap akses resource = syscall → PermissionManager
```

Detailed steps (matching the code)

- 1. **Kernel.spawnWorker** (src/kernel/Scheduler.ts) builds the workerData: WorkerInitData:

```
const workerData: WorkerInitData = {
  pid: pcb.pid,
  appName: options.appName || pcb.name,
  args: options.args || [],
  appPath: options.appPath,
```

```

stackBkfsPath: options.stackBkfsPath,
appContent: options.appContent,
env: pcb.env,
vfsCache: this.vfsCacheProvider ? this.vfsCacheProvider() : {}
};

```

`vfsCache` is the result of `rebuildVFSCache()`: the framework `/lib` is pre-compiled into memory at boot.

2. **execArgv** is chosen based on the target extension:

- `*.js` → **JS-Direct (FAST)**: only `--enable-source-maps`.
- otherwise (`.ts`) → **TS-Transpile**: adds `-r esbuild-register` and `-r tsconfig-paths/register`.

3. The **bootloader** (`WorkerEntry.ts`) immediately stores `realExit = process.exit.bind(process)` — used later to really exit even though `process.exit` has been sabotaged.

4. **Hijack Module._load**: all `require("@tsix/...")` / `require("@common/...")` calls (and relative aliases) are resolved from `vfsCache`. If found, the content is `_compiled` from memory (DME) with a **dummy filename**, then cached. If not found, it falls back to `originalLoad` (host `require`).

5. **UserLib is loaded from the memory cache**: `hijackRequire("@tsix/UserLib")` → `new UserLibClass(pid)` → `global._tsixLib`. This provides `lib.fs`, `lib.shell`, `lib.std`, `lib.net`, etc.

6. **App DME**: if there is no physical `appPath`, `appContent` (from `workerData`) is transpiled (TS → JS via `esbuild.transformSync`) then `_compiled` with **two filenames**:

- `moduleFilename` (physical) → so `require` can find `node_modules`.
- `stackFilename` (BKFS path, `.ts` → `.js`) → so the stack trace points to the correct BKFS path.

7. **Find AppClass**: `exports.main` || `exports.Main` || `exports.default` || the first exported function.

8. **restrictHostAPI(appName)** is called right after **AppClass is found and before** `new AppClass()` — "lock the doors before the app runs". From here on, `require` is restricted and `process.exit/process.kill` → `throw`.

9. **Run**: `new AppClass()` → `await app.execute(lib, args)`. If it returns a non-empty string, it is printed via `lib.std.print`, then `lib.shell.exit(0)`.

10. **If a runtime error occurs**: send `GUI_WINDOW_ERROR` to the parent (WM/Asteracea), print to `lib.std.error`, then `realExit(1)`.

Source Code

File	Role
<code>src/userland/WorkerEntry.ts</code>	Worker bootloader + sandbox (<code>restrictHostAPI</code>)
<code>src/kernel/Scheduler.ts</code>	<code>spawnWorker()</code> — builds <code>workerData</code> , <code>vfsCache</code> , <code>execArgv</code>
<code>src/common/IPCTypes.ts</code>	Types <code>WorkerInitData</code> & <code>SyscallResponse</code>

Snippets (code level)

[!NOTE] All snippets below are copied from `src/userland/WorkerEntry.ts` (verify directly against the code).

1. Hijack Module._load (framework resolution from vfsCache)

```

const originalLoad = Module._load;
const vfsCache = (workerData as any).vfsCache || {};
const moduleCache: Record<string, any> = {};

```

```

Module._load = function (request: string, parent: any, isMain: boolean) {
  let normalizedRequest = request;

  // Resolve relative paths
  if (request.startsWith(".")) {
    if (request.includes("/common/")) {
      normalizedRequest = "@common/" + request.split("/common/")[1];
    } else if (request.includes("/lib/")) {
      normalizedRequest = "@tsix/" + request.split("/lib/")[1];
    } else if (parent && parent.filename) {
      const basename = path!.basename(parent.filename);
      if (basename.startsWith("@tsix_") && request.startsWith("./")) {
        normalizedRequest = "@tsix/" + request.substring(2);
      } else if (basename.startsWith("@common_") && request.startsWith("./")) {
        normalizedRequest = "@common/" + request.substring(2);
      }
    }
  }

  // Cached Module
  if (moduleCache[normalizedRequest]) return moduleCache[normalizedRequest];

  // Resolusi Memory Framework (VFS)
  let vfsPath = null;
  if (normalizedRequest.startsWith("@tsix/")) {
    vfsPath = "/lib/" + normalizedRequest.substring(6) + ".ts";
  } else if (normalizedRequest.startsWith("@common/")) {
    vfsPath = "/lib/common/" + normalizedRequest.substring(8) + ".ts";
  }

  if (vfsPath && vfsCache[vfsPath]) {
    const content = vfsCache[vfsPath];
    const dummyFilename = path!.join(process.cwd(), normalizedRequest.replace("/", "_") + ".js");

    const newMod = new Module(dummyFilename, parent);
    newMod.filename = dummyFilename;
    newMod.paths = Module._nodeModulePaths(process.cwd());

    // Framework modules are now pre-compiled in Kernel.
    // Direct execution for maximum performance.
    (newMod as any)._compile(content, dummyFilename);

    moduleCache[normalizedRequest] = newMod.exports;
    return newMod.exports;
  }

  return originalLoad.apply(this, arguments);
};

```

Alias → VFS path mapping: @tsix/x → /lib/x.ts; @common/y → /lib/common/y.ts. Relative imports from inside the framework are also normalized (./z in @tsix_... → @tsix/z).

2. restrictHostAPI — locking the doors (full)

```

const restrictHostAPI = (appName: string) => {
  const forbidden = (msg: string = "Security Violation: Direct Host API access is forbidden in TSIX Sandbox.") => {
    throw new Error(msg);
  };

  const isPrivileged = appName.toLowerCase().includes("server") ||
    appName.toLowerCase().includes("daemon") ||
    appName.toLowerCase().includes("dome") ||
    appName.toLowerCase().includes("tbuild") ||
    appName.toLowerCase().includes("vfs") ||
    appName.toLowerCase().includes("mysqld");
  const allowedModules = ["http", "ws", "path", "fs", "url", "esbuild", "crypto", "os", "bcryptjs", "mysql2", "mysql2/promise"];

  const privilegedRequire = (mod: string) => {
    // Framework aliases are ALWAYS allowed, even in sandbox
    if (mod.startsWith("@tsix/") || mod.startsWith("@common/") || mod.includes("/lib/") || mod.includes("/common/")) {
      return hijackRequire(mod);
    }
  };
};

```

```

    }

    if (allowedModules.includes(mod)) {
        return hostRequire!(mod);
    }
    forbidden(`Security Violation: Module '${mod}' is not in the privileged allow-list.`);
};

// Sembunyikan require jika ada (tergantung module loader)
if (typeof require !== "undefined") {
    (global as any).require = isPrivileged ? privilegedRequire : (mod: string) => {
        // Even in sandbox, framework cores MUST be accessible
        if (mod.startsWith("@tsix/") || mod.startsWith("@common/") || mod.includes("/lib/") || mod.includes("/common/")) {
            return hijackRequire(mod);
        }
        forbidden();
    };
}

// Batasi akses process yang sensitif
const p = (global as any).process;
if (p) {
    p.exit = forbidden;
    p.kill = forbidden;
}
};

```

Note: `allowedModules` has **11 entries** — including both `mysql2` and `mysql2/promise`. The privileged check uses substring matching: `server`, `daemon`, `dome`, `tbuild`, `vfs`, `mysqld` (checked with `toLowerCase()`).

3. Direct Memory Execution — `_compile` the app (DME path)

```

// Module filename HARUS physical path untuk require() nemu node_modules
const moduleFilename = path!.join(process.cwd(), appName + ".js");
// stackBkfsPath = BKFS path untuk stack trace (/opt/test/gui-test.js)
const stackBkfsPath = (data as any).stackBkfsPath;
const stackFilename = stackBkfsPath
    ? stackBkfsPath.replace(/\.ts$/, '.js')
    : moduleFilename;
// sourcefile untuk esbuild sourcemap – cukup nama file aja (tanpa path)
const sourceFileName = (stackBkfsPath || moduleFilename).split(/[\\\/]).pop()!.replace(/\.js$/, '.ts');

// Jika content adalah TypeScript, transpile dulu ke JavaScript
if (isTypeScript) {
    const esbuild = hostRequire!("esbuild");
    const result = esbuild.transformSync(content, {
        loader: "ts",
        format: "cjs",
        target: "node18",
        sourcemap: "inline",
        sourcefile: sourceFileName,
    });
    content = result.code;
}

const appModule = new Module(moduleFilename, module.parent);
appModule.filename = stackFilename; // __filename = BKFS path
appModule.paths = Module._nodeModulePaths(path!.dirname(moduleFilename));

// _compile dengan stackFilename agar stack trace nunjuk BKFS path
(appModule as any)._compile(content, stackFilename);

```

4. Fatal error handling

```

process.on("unhandledRejection", (reason) => {
    const msg = reason instanceof Error ? reason.message : String(reason);
    console.error("[Worker Fatal] Unhandled Rejection:", msg);
    trySendErrorToParent(msg);
    realExit(1);
});

```

```
});

process.on("uncaughtException", (err) => {
  const msg = err instanceof Error ? err.message : String(err);
  console.error("[Worker Fatal] Uncaught Exception:", msg);
  trySendErrorToParent(msg);
  realExit(1);
});
```

`trySendErrorToParent` uses `global._tsixLib: lib.getParentPid()` then `lib.shell.send(parentPid, { type: "GUI_WINDOW_ERROR", ... })` — so the error shows up in an Asteracea window.

Exercises / Practice

1. Read `src/userland/WorkerEntry.ts` — find the line where `restrictHostAPI(appName)` is called. Note: it is called after `AppClass` is found, before `new AppClass()`.
2. Write an app that calls `process.exit(0)` — observe the "Security Violation" error. Compare with `lib.shell.exit(0)` (the correct way).
3. Write an app that does `require("fs")` — compare the error between a normal app vs an app named `my-daemon` (privileged).
4. Write an app named `evil-daemon` that does `require("mysql2")` — prove the substring heuristic can be bypassed. Then explain why this is not actually dangerous (the kernel still guards syscalls).
5. Explain why the `@tsix/*` framework must always be accessible, even in the sandbox.

References

- [wiki/Keamanan-dan-Sandboxing.md](#) — full sandbox model
- [wiki/course/00-overview.en.md §6](#) — Worker Thread & Sandboxing (summary)
- `src/userland/WorkerEntry.ts` — bootloader + sandbox (main source code)
- `src/kernel/Scheduler.ts` — `spawnWorker()` (`workerData` + `vfsCache`)
- `src/common/IPCTypes.ts` — `WorkerInitData`

Module 11 — done. Continue to [Module 12 — Module Resolution & DME](#).

RFC-TSIX-EDU-002 | The twelfth module of the TSIX curriculum. It uncovers one of the most "magical" parts of TSIX: the kernel pre-compiles `/lib` into memory, workers execute from memory with no disk hits, and the filename-alias trick keeps relative imports working.

DME (Direct Memory Execution) is how TSIX loads the `@tsix/*` and `@common/*` frameworks **without touching the filesystem** at runtime. The kernel pre-compiles `/lib` once at boot, the cache is copied to each worker via `workerData`, then `WorkerEntry` compiles modules from an in-memory string with a **fake filename** (dummy filename) so relative imports keep working.

Learning Objectives

- ☐ Explain the DME (Direct Memory Execution) concept
- ☐ Explain the `Module._load` hijack and the `@tsix/*` / `@common/*` alias rewrite
- ☐ Explain the dummy-filename trick for relative `./x` imports
- ☐ Explain the dual identity filename (physical vs BKFS)
- ☐ Explain `moduleCache` and why the framework feels instant
- ☐ Explain the difference between the TS-transpile and JS-Direct paths

Core Concepts

What is Direct Memory Execution

DME answers two problems:

1. **Performance** — a normal `require()` resolves and reads files from disk every time a module is loaded. For a framework used by every app, this is expensive. DME replaces disk reads with a **lookup in a JavaScript object** (`vfsCache`).
2. **Alias consistency** — every app must see the **exact same** framework (`@tsix/*` / `@common/*`) with the same behavior, regardless of the VFS backend.

```

rebuildVFSCache()
  fetchDir("/lib") ← rekursif dari BKFS
    baca /lib/UserLib.ts, /lib/emerald.ts, /lib/common/*.ts, ...
    .ts → esbuild.transformSync() → JS (CJS, target node18)
    simpan cache["/lib/...ts"] = kode JS (objek di memori)

vfsCache = { "/lib/UserLib.ts": "...js...", "/lib/Application.ts": "...js..." }

scheduler.setVFSCacheProvider(() => vfsCache) ← snapshot per spawn
|
| workerData.vfsCache (postMessage)
▼
WORKER (WorkerEntry.ts – bootloader)

Module._load = hijack(request, parent, isMain)
require("@tsix/UserLib")
  → vfsPath = "/lib/UserLib.ts"
  → vfsCache["/lib/UserLib.ts"] ketemu? → YA →
  → dummyFilename = "<cwd>/@tsix_UserLib.js"
  → new Module(dummyFilename, parent)
  → _compile(kodeJS, dummyFilename) ← compile dari MEMORI
  → moduleCache["@tsix/UserLib"] = exports
  → return exports (TANPA hit filesystem)

```

```
require("@tsix/UserLib") berikutnya → moduleCache (cached)
```

Two cache layers: `vfsCache` vs `moduleCache`

Cache	Where	Key	Content	Lifecycle
<code>vfsCache</code>	Kernel (<code>Kernel.ts</code>)	BKFS path (<code>/lib/*.ts</code>)	pre-compiled JS code	once per boot
<code>moduleCache</code>	per worker (<code>WorkerEntry.ts</code>)	alias (<code>@tsix/UserLib</code>)	<code>module exports</code> object	once per worker

`vfsCache` only contains **code strings**. `moduleCache` contains **ready-to-use exports objects** — created by the worker the first time it loads a module. A framework file is pre-compiled once (by the kernel), but it is `_compiled` and cached once **per worker**.

Alias and path mapping

Alias	Rewritten to	<code>vfsPath</code> (<code>vfsCache</code> key)
<code>@tsix/UserLib</code>	—	<code>/lib/UserLib.ts</code>
<code>@tsix/emerald</code>	—	<code>/lib/emerald.ts</code>
<code>@common/GUITypes</code>	—	<code>/lib/common/GUITypes.ts</code>
<code>./UserLib</code> (parent <code>@tsix_*</code>)	<code>@tsix/UserLib</code>	<code>/lib/UserLib.ts</code>
<code>./GUITypes</code> (parent <code>@common_*</code>)	<code>@common/GUITypes</code>	<code>/lib/common/GUITypes.ts</code>

[!IMPORTANT] The `@tsix/X` → `/lib/X.ts` mapping always appends the `.ts` extension. That is why framework files in BKFS **must** be `.ts` — a convention enforced by `vfs-bootstrap`. The kernel also caches `.js`/`.json` files under `/lib`, but the DME `@tsix/*` path only targets `.ts`.

Why this matters

1. **Performance: zero FS hits per require.** The framework is loaded once at boot, then executed from memory in every worker. Without DME, every app that does `import "@tsix/UserLib"` would have to read and re-compile the file from VFS — expensive for GUIs/daemons that must start fast.
2. **Alias consistency.** The relative rewrite `./x` → `@tsix/x` / `@common/x` ensures that internal framework code (e.g. `Application` using `UserLib`) always loads the **same instance** from the cache — no duplication of physical vs VFS paths.
3. **Single source of truth.** The kernel is the only one that reads `/lib` from BKFS. Workers have no direct access to the framework disk — this reinforces the sandbox boundary (see [Module 11](#)).

[!TIP] DME is an **architectural technique** that speeds up framework (`/lib`) execution — it complements, rather than replaces, TSIX's five core principles. For the list of core principles, see [wiki/course/00-overview.md §1](#) and [§6.2](#).

Two app execution paths

Path	Condition	Method
DME (VFS)	<code>appPath</code> empty, <code>appContent</code> present	<code>_compile</code> from a string; <code>.ts</code> files transpiled by esbuild in the worker
Physical (host)	<code>appPath</code> filled in	<code>hostRequire(finalAppPath)</code> — normal Node.js path, <code>-r esbuild-register</code>

Type detection: `isTypeScript = !(appPath || appName || "").toLowerCase().endsWith(".js")`.

Flow / How it works

1. **Kernel boot** (`Kernel.initializeSubsystems`) calls `rebuildVFSCache()`.
2. `rebuildVFSCache()` runs recursively (`fetchDir`) from `/lib` in BKFS; `.ts` files are transpiled by `esbuild` → `vfsCache`.
3. The kernel connects the scheduler: `scheduler.setVFSCacheProvider(() => vfsCache)`.
4. On `spawnWorker()`, the scheduler copies the cache: `vfsCache: this.vfsCacheProvider()` goes into `workerData`.
5. `new Worker(workerPath, { workerData, execArgv })` — a new thread with a cache snapshot.
6. `WorkerEntry.ts` (module scope) saves `originalLoad = Module._load`, then replaces `Module._load` with the hijacked version.
7. `main()` loads `UserLib`: `hijackRequire("@tsix/UserLib")` → through `Module._load` → `vfsCache` → `_compile` from memory.
8. The app is loaded via DME (if the content comes from VFS) or physically (if `appPath`).
9. The sandbox is enabled (`restrictHostAPI`) — see [Module 11](#) — before the app runs.

Source Code

File	Role
<code>src/kernel/Kernel.ts</code>	<code>rebuildVFSCache()</code> — pre-compile <code>/lib</code> → <code>vfsCache</code>
<code>src/kernel/Scheduler.ts</code>	<code>setVFSCacheProvider()</code> + <code>spawnWorker()</code> — send the cache via <code>workerData</code>
<code>src/userland/WorkerEntry.ts</code>	Hijack <code>Module._load</code> , DME, dummy filename, dual identity

Snippets (code level)

1. Kernel: pre-compile `/lib` into memory

`rebuildVFSCache()` in `src/kernel/Kernel.ts` — copied exactly from the source:

```
private rebuildVFSCache() {
  this.bootLogStart(
    "VFS: Pre-compiling framework libraries (Memory Cache)... ",
  );
  try {
    const esbuild = require("esbuild");
    const cache: Record<string, string> = {};
    const fetchDir = (dir: string) => {
      if (!this.bkfs!.exists(dir)) return;
      const items = this.bkfs!.ls(dir);
      for (const item of items) {
        const p = `${dir}/${item.name}`;
        if (item.type === "DIRECTORY") {
          fetchDir(p);
        } else if (
          item.type === "FILE" &&
          (item.name.endsWith(".ts") ||
            item.name.endsWith(".js") ||
            item.name.endsWith(".json"))
        ) {
          let content = this.bkfs!.read(p);
          if (!content) continue;

          let code = content;

          // Transpile TS to JS for framework files
          if (item.name.endsWith(".ts")) {
            try {
              const result = esbuild.transformSync(code, {
                loader: "ts",
                format: "cjs",
                target: "node18",
                sourcemap: "inline",
              });
              code = result.code;
            } catch {
              // ignore
            }
          }
          cache[p] = code;
        }
      }
    };
    fetchDir("/lib");
  } catch {
    // ignore
  }
}
```

```

        } catch (err: any) {
            this.logger.error(`Failed to pre-compile ${p}: ${err.message}`);
        }
    }

    cache[p] = code;
}
}
};
fetchDir("/lib");
this.vfsCache = cache;
this.bootLogEnd(true, "OK");
} catch (e: any) {
    this.bootLogEnd(false, `Error: ${e.message}`);
}
}
}

```

Things to note:

- `fetchDir` is **recursive** — it includes subfolders like `/lib/common/`.
- `.ts` files are transpiled via `esbuild.transformSync` (loader `ts`, format `cjs`, target `node18`). `.js/.json` files are copied as-is.
- The cache key = **BKFS path** (`/lib/UserLib.ts`), the value = **a JS code string**.

2. Kernel → Scheduler: cache channel

```

// Kernel.ts — saat boot
this.rebuildVFSCache();
this.scheduler.setVFSProvider(() => {
    return this.vfsCache;
});

// Scheduler.ts — spawnWorker()
const workerData: WorkerInitData = {
    pid: pcb.pid,
    appName: options.appName || pcb.name,
    args: options.args || [],
    appPath: options.appPath,
    stackBkfsPath: options.stackBkfsPath,
    appContent: options.appContent,
    env: pcb.env,
    vfsCache: this.vfsCacheProvider ? this.vfsCacheProvider() : {}
};

```

`setVFSProvider()` stores a *callback*, not a snapshot. Each `spawnWorker()` calls that callback to fetch the **current** cache and copy it into `workerData`.

3. Worker: hijack `Module._load` + dummy filename + `_compile`

In `src/userland/WorkerEntry.ts` — copied exactly from the source:

```

const originalLoad = Module._load;
const vfsCache = (workerData as any).vfsCache || {};
const moduleCache: Record<string, any> = {};

Module._load = function (request: string, parent: any, isMain: boolean) {
    let normalizedRequest = request;

    // Resolve relative paths
    if (request.startsWith(".")) {
        if (request.includes("/common/")) {
            normalizedRequest = "@common/" + request.split("/common/")[1];
        } else if (request.includes("/lib/")) {
            normalizedRequest = "@tsix/" + request.split("/lib/")[1];
        } else if (parent && parent.filename) {
            const basename = path!.basename(parent.filename);
            if (basename.startsWith("@tsix_") && request.startsWith("./")) {
                normalizedRequest = "@tsix/" + request.substring(2);
            }
        }
    }
}

```

```

    } else if (basename.startsWith("@common_") && request.startsWith("./")) {
        normalizedRequest = "@common/" + request.substring(2);
    }
}

// Cached Module
if (moduleCache[normalizedRequest]) return moduleCache[normalizedRequest];

// Resolusi Memory Framework (VFS)
let vfsPath = null;
if (normalizedRequest.startsWith("@tsix/")) {
    vfsPath = "/lib/" + normalizedRequest.substring(6) + ".ts";
} else if (normalizedRequest.startsWith("@common/")) {
    vfsPath = "/lib/common/" + normalizedRequest.substring(8) + ".ts";
}

if (vfsPath && vfsCache[vfsPath]) {
    const content = vfsCache[vfsPath];
    const dummyFilename = path!.join(process.cwd(), normalizedRequest.replace("/", "_") + ".js");

    const newMod = new Module(dummyFilename, parent);
    newMod.filename = dummyFilename;
    newMod.paths = Module._nodeModulePaths(process.cwd());

    // Framework modules are now pre-compiled in Kernel.
    // Direct execution for maximum performance.
    (newMod as any)._compile(content, dummyFilename);

    moduleCache[normalizedRequest] = newMod.exports;
    return newMod.exports;
}

return originalLoad.apply(this, arguments);
};

```

The hijack steps, one by one:

1. **Normalize the request** — relative `./x / .../lib/x` paths are rewritten to aliases.
2. **Check moduleCache** — if already loaded, return exports directly (no re-compilation).
3. **Map alias → vfsPath** (`@tsix/X → /lib/X.ts`, `@common/X → /lib/common/X.ts`).
4. **Check vfsCache** — if found, take the JS code from memory.
5. **Dummy filename** — `<cwd>/@tsix_X.js` (see below).
6. **_compile(content, dummyFilename)** — Node.js compiles the module from a string, **without reading a file**.
7. **Save to moduleCache** and return exports.
8. If not in cache → **fallback** to `originalLoad.apply(...)` (normal Node.js path).

4. Dummy filename trick — framework module identity

The kernel only stores code in `vfsCache` keyed by the BKFS path (`/lib/UserLib.ts`). The `@tsix_*.js` names are **created when loaded in the worker**, not during pre-compilation:

```
const dummyFilename = path!.join(process.cwd(), normalizedRequest.replace("/", "_") + ".js");
```

Alias	replace("/", "_")	dummyFilename (basename)
@tsix/UserLib	@tsix_UserLib	<cwd>/@tsix_UserLib.js
@tsix/Application	@tsix_Application	<cwd>/@tsix_Application.js
@common/GUITypes	@common_GUITypes	<cwd>/@common_GUITypes.js

Why are the `@tsix_` / `@common_` prefixes important? Because relative rewrites inside the framework happen **based on the parent's basename**:

```
const basename = path!.basename(parent.filename);
if (basename.startsWith("@tsix_") && request.startsWith("./")) {
```

```
normalizedRequest = "@tsix/" + request.substring(2);
}
```

This means that if `Application.ts` (loaded with filename `<cwd>/@tsix_Application.js`) does `import x from './UserLib'`, then `./UserLib` → `@tsix/UserLib` → `/lib/UserLib.ts`. The alias cycle stays consistent across the whole framework. This is what makes `@tsix/*` look like an instant package.

[!NOTE] So the "dual identity" of a framework module is: **content identity** = the BKFS path (`/lib/UserLib.ts`, the `vfsCache` key), while the **filename identity** seen by Node = the dummy path (`<cwd>/@tsix_UserLib.js`).

5. Dual identity filename for applications

When the worker loads an **app** via DME (the block in `main()`), two paths are created:

```
let content = (data as any).appContent;
const isTypeScript = !(appPath || appName || "").toLowerCase().endsWith(".js");
// Module filename HARUS physical path untuk require() nemu node_modules
const moduleFilename = path!.join(process.cwd(), appName + ".js");
// Stack filename = BKFS path biar stack trace bener (/opt/test/gui-test.js)
// stackBkfsPath = BKFS path untuk stack trace (/opt/test/gui-test.js)
const stackBkfsPath = (data as any).stackBkfsPath;
const stackFilename = stackBkfsPath
  ? stackBkfsPath.replace(/\.ts$/, '.js')
  : moduleFilename;
// sourcefile untuk esbuild sourcemap - cukup nama file aja (tanpa path)
const sourceFileName = (stackBkfsPath || moduleFilename).split(/[\\\/]).pop()!.replace(/\.js$/, '.ts');

// Jika content adalah TypeScript, transpile dulu ke JavaScript
if (isTypeScript) {
  try {
    const esbuild = hostRequire!("esbuild");
    const result = esbuild.transformSync(content, {
      loader: "ts",
      format: "cjs",
      target: "node18",
      sourcemap: "inline",
      sourcefile: sourceFileName,
    });
    content = result.code;
  } catch (transpileErr: any) {
    console.error(`[Worker ${pid}] TS Transpile Error: ${transpileErr.message}`);
    throw transpileErr;
  }
}

// Create a new module instance with physical path (for node_modules resolution)
const appModule = new Module(moduleFilename, module.parent);
appModule.filename = stackFilename; // __filename shows BKFS path
appModule.paths = Module._nodeModulePaths(path!.dirname(moduleFilename));

// _compile dengan stackFilename agar stack trace nunjuk BKFS path
(appModule as any)._compile(content, stackFilename);

AppClass = appModule.exports.main || appModule.exports.Main || appModule.exports.default || appModule.exports;
```

So:

- **moduleFilename** (physical: `<cwd>/<appName>.js`) is used for `new Module(...)` and `Module._nodeModulePaths(...)` — so internal `require("node_modules/...")` calls in the app can resolve.
- **stackFilename** (BKFS: `/opt/test/gui-test.js`) is set as `appModule.filename` and passed to `_compile` — so `__filename` and the **stack trace** point to the correct location in VFS.

This is the answer to exercise #4: a single module has **two identities** because of two different needs (module resolution vs error reporting).

Exercises / Practice

1. Read `rebuildVFSCache()` in `src/kernel/Kernel.ts`. Trace how `/lib` is read recursively, how `.ts` files are transpiled, and how the results are stored in `vfsCache`. What are the cache's keys and values?
 2. Add a log in `Module._load` (`WorkerEntry.ts`) when `vfsPath` is found. Run an app that does `import ... from "@tsix/UserLib"`. Observe that there is no filesystem hit — and that the second log comes from `moduleCache` (not a `re-compile`).
 3. Create an app that throws an error. Inspect its stack trace — does it point to the BKFS path (`/opt/...`) or the physical path (`<cwd>/...`)? Match it against `stackBkfsPath`.
 4. Add a log for `basename(parent.filename)` in the relative-rewrite branch. Look at the `@tsix_*` value when internal framework code uses `./x`. Explain its relationship to the dummy filename.
 5. Explain why the two filename identities are needed (require vs stack trace), and why the `@tsix_` / `@common_` prefixes determine the relative rewrite.
 6. Compare the TS-transpile vs JS-Direct paths: when is `-r esbuild-register` used in `execArgv`, and when is `esbuild` called manually in the worker? (See [Module 11](#).)
-

References

- [wiki/course/00-overview.md §1, §6.2](#) — DME as a core principle & bootstrap summary
 - [wiki/course/11-worker-thread-sandbox.md](#) — `WorkerEntry` as bootloader + `restrictHostAPI` (Module 11)
 - [wiki/course/05-syscall-ipc.md](#) — syscall & IPC: why workers communicate through syscall, not direct `require` (Module 05)
 - `src/kernel/Kernel.ts` — `rebuildVFSCache()`, `setVFSCacheProvider()`
 - `src/kernel/Scheduler.ts` — `spawnWorker()`, `workerData.vfsCache`
 - `src/userland/WorkerEntry.ts` — hijack `Module._load`, DME, dummy filename
 - `src/common/IPCTypes.ts` — the `WorkerInitData` type (the `workerData` contract)
-

Module 12 — done. Part IV is complete. Continue to [Module 13 — TTY & Virtual Console](#).

RFC-TSIX-EDU-002 | Thirteenth module of the TSIX curriculum. Understand the TTY as a full PTY emulation: screen buffer, ANSI subset, raw/cooked mode, and master-side ioctl.

The TSIX TTY is not just a "text console" — it is a **full PTY emulation**. A remote application (e.g. `pixelterm`) can control the TTY from the master side through `ioctl 0x2001 / 0x2002` (inject input / read output).

Learning Objectives

- ☐ Explain the role of `TTYManager` and the allocation of consoles 1–32
- ☐ Explain the TTY components (buffer, cursor, ANSI parser)
- ☐ Explain the important TTYDevice ioctls (`clear`, `raw`, `0x2001`, `0x2002`, `winsize`)
- ☐ Explain TTY allocation to processes via `ttyId` in `EXEC`
- ☐ Explain keyboard behavior (`Ctrl+C`, `Alt+F1-6`)

Core Concepts

Components

Component	File	Role
TTYManager	<code>src/kernel/tty/TTYManager.ts</code>	Manages 32 virtual consoles (<code>Map<number, TTY></code>), <code>activeId</code> , TTY switching, <code>onSwitch/onInterrupt</code> callbacks
TTY	<code>src/kernel/tty/TTY.ts</code>	Buffer <code>char[y][x]</code> , cursor, ANSI subset parser, raw/cooked mode, re-rendering
TTYDevice	<code>src/kernel/devices/TTYDevice.ts</code>	<code>/dev/ttyN</code> — wraps a TTY into an <code>IDevice</code> (read/write/ioctl)

A TTY is used as three devices at once for a process "attached" to a console: FD 0 (stdin), FD 1 (stdout), and FD 2 (stderr) all point to the same `TTYDevice`. `init` (PID 1) is created with `fds: [tty1, tty1, tty1]` and `ttyId: 1`.

Buffer & Modes

- Screen buffer:** `charBuffer: string[][]` holds the characters, `attrBuffer: string[][]` holds the ANSI styles (e.g. `"31;1"`). Both are indexed `[y][x]` (row first, then column). Default size is `80×24`, but at boot `TTYManager` uses the host terminal size (`process.stdout.rows / columns`).
- Cooked mode** (default): input is line-buffered. The TTY performs echo, `Enter` moves the line into `inputLines[]`, `Backspace` deletes the last character, and `Ctrl+C` produces a signal (it does not enter the buffer).
- Raw mode** (`setRawMode(true)`): each character goes straight into `inputBuffer[]` without echo and without line editing. Suitable for line editors / shells.
- Output buffer:** every `write()` also records data into `outputBuffer` — this is what the master reads via `ioctl 0x2002` (`getOutput()` also empties it).

TTYDevice ioctl

cmd	Name	Meaning	Implementation
1	<code>CLEAR_SCREEN</code>	clear the TTY screen	<code>tty.clear()</code> ; if active, reset the host stdout (<code>\x1bc</code>)
2	<code>SWITCH_TTY</code>	switch console	return <code>-1</code> — handled in <code>Syscalls.ts</code> via <code>Kernel.ttyManager.switch()</code>
10	<code>SET_RAW_MODE</code>	set raw/cooked mode	<code>tty.setRawMode(!arg)</code>
0x2001	<code>INJECT_INPUT</code>	master types into the slave stdin	<code>tty.pushInput(arg)</code>
0x2002	<code>READ_OUTPUT</code>	master reads the slave stdout	<code>tty.getOutput()</code> (and empty the buffer)

cmd	Name	Meaning	Implementation
4	TIOCGWINSZ	window size	{ lines: tty.height, columns: tty.width }

[!NOTE] The codes `0x2001` / `0x2002` are also used by `MySQLDevice` (`MYSQL_IOCTL_CONNECT` / `DISCONNECT`). The meaning of an ioctl depends on the driver — in `TTYDevice` both are PTY input injection / output reading.

ANSI Subset Handled

The TSIX ANSI parser is simple: a sequence `ESC [ ...` is parsed until a command letter (`cmd`) is found, then the arguments are split by `;`. The following subset is implemented in `TTY.handleANSI()`:

Sequence	Name	Behavior
<code>ESC[nA</code>	Cursor Up	move cursor up <code>n</code> rows (clamp 0)
<code>ESC[nB</code>	Cursor Down	move cursor down <code>n</code> rows (clamp <code>height-1</code>)
<code>ESC[nC</code>	Cursor Forward	move cursor forward <code>n</code> columns (clamp <code>width-1</code>)
<code>ESC[nD</code>	Cursor Backward	move cursor backward <code>n</code> columns (clamp 0)
<code>ESC[r;cH</code> / <code>ESC[r;cF</code>	Cursor Position	move cursor to absolute position (1-based)
<code>ESC[...m</code>	SGR (color/attribute)	store into <code>currentAttr</code> (default <code>"0"</code>)
<code>ESC[2J</code>	Erase Display	<code>clear()</code> the whole screen
<code>ESC[0J</code>	Erase Display (partial)	erase from cursor to bottom
<code>ESC[K</code>	Erase Line	erase from cursor to end of line
<code>ESC[nS</code>	Scroll Up (SU)	shift rows up by <code>n</code> rows
<code>ESC[nT</code>	Scroll Down (SD)	shift rows down by <code>n</code> rows
<code>ESC[nL</code>	Insert Line (IL)	insert <code>n</code> blank lines at the cursor row
<code>ESC[nM</code>	Delete Line (DL)	delete <code>n</code> lines at the cursor row
<code>ESC[s</code>	Save Cursor (DECSC)	save cursor position
<code>ESC[u</code>	Restore Cursor (DECRC)	restore cursor position

In addition, `write()` handles basic character controls: `\n` (new line), `\r` (`cursorX = 0`), `\b` (one column back). `render()` redraws the whole screen using absolute cursor positioning per row (`ESC[r;cH` + `ESC[2K`) so that it is not corrupted by host terminal wrapping.

TTY Allocation to Processes

Unlike `PortManager` (networking), TTY allocation to processes is done **via `ttyId` in the `EXEC` syscall** — the kernel overrides FD 0/1/2 to `tty{n}`:

- The `ttyId` argument is valid only for the range **1-12**.
- If valid, `stdin` / `stdout` / `stderr` are set to the device `tty{ttyId}`.
- If `ttyId` is not given, the process **inherits** `ttyId` from its parent (`pcb.ttyId`).
- `createProcess()` stores `ttyId` in the PCB; `runInit()` creates PID 1 with `ttyId: 1`, then calls `setForegroundProcess(pid, 1)`.

The scheduler maintains the mapping `ttyForegroundPids: Map<ttyId, pid>` — one foreground process per TTY. When a process **detaches** (daemonizes) or **exits**, it is removed from the foreground mapping and its `ttyId` is cleared. Example usage in userland: `init` (PID 1) on `tty1`, then `login` is spawned for other TTYs with a target `ttyId`.

Keyboard

- **Hotkey** Alt+F1-F6 → `TTYManager.switch`. Console switching also works with **Alt+1-6** and **Ctrl+Alt+1-6** on every terminal (both arrive as ESC+digit; `␣™£¢∞§` on macOS Terminal) plus **Ctrl+1-6** where the terminal reports modifiers (CSI-u / modifyOtherKeys); `Ctrl+4-6` is additionally recognised via caret notation (FS/GS/RS). Bare `Ctrl+1-6` cannot work: GNOME Terminal uses it to switch tabs and VS Code to focus editor groups, and the bytes no longer carry modifier information.
- **Ctrl+C** → `SIGINT` to the foreground process of the active TTY.
- **Console switch** → `SIGWINCH` to the foreground process of the newly active TTY.
- **Host window resize** → update `LINES/COLUMNS` in the env of all processes + `SIGWINCH` broadcast + `TTYManager.handleResize()`.

Flow / How It Works

Input: keystroke → foreground process

```

Host keyboard (process.stdin)
  | data mentah (utf8)
  ▼
KeyboardDevice (data handler kernel)
  | hotkey? (Alt+F1..6) → TTYManager.switch(...) → STOP
  ▼
Kernel: ttyManager.getActiveTTY().pushInput(data)
  |
  ▼
TTY.pushInput(data)
  | cooked : echo + lineBuffer → Enter → inputLines[]
  | raw    : tiap char → inputBuffer[]
  |
  ▼
Proses foreground → syscall READ(fd 0) → TTYDevice.read()
  | raw → tty.read() → inputBuffer.shift()
  | cooked → tty.read() → inputLines.shift()

```

Output: application → TTY → render

```

Aplikasi → syscall WRITE/PRINT(fd 1) → TTYDevice.write(data)
  |
  ▼
TTY.write(data)
  1. outputBuffer += data          (untuk master ioctl 0x2002)
  2. onWrite(data)                 (master remote ikut menerima)
  3. parse char: \n \r \b, ESC[...] (subset ANSI), karakter biasa
      | karakter biasa → putchar() → charBuffer[y][x] + attrBuffer[y][x]
      ▼
TTY aktif? → tty.onWrite → process.stdout.write(data) (layar host)
Master?    → syscall READ(pid) → ioctl 0x2002 → getOutput() (pixelterm)

```

Boot wiring (Kernel.boot)

```

new TTYManager(32)          → 32× TTY (ukuran host)
  └ 32× TTYDevice "tty1..tty32" → /dev/ttyN (isActive = aktifId == N)
devices: stdin=KeyboardDevice, fb0/stdout/stderr → tty1, null, tty1..32
kbd.setDataHandler          → getActiveTTY().pushInput(data)
kbd.setHotkeyHandler        → handleKeyboardHotkey (Alt+F1..6)
kbd.setInterruptHandler     → handleHostInterrupt (Ctrl+C)
ttyManager.setOnSwitchCallback → SIGWINCH ke foreground TTY baru
ttyManager.setOnInterruptCallback → SIGINT ke foreground TTY
runInit() → createProcess("init", fds:[tty1×3], ttyId:1)
          → setForegroundProcess(pid, 1)

```

Console switch (Alt+F2)

```
Host kirim seq "\x1b00" (atau "\x1b[1;3Q")
→ KeyboardDevice.onHotkey → handleKeyboardHotkey(seq) → true (stop)
→ ttyManager.switch(2)
  1. activeId = 2
  2. reset host terminal: "\x1bc\x1b[3J"
  3. banner "TSIX VIRTUAL CONSOLE [ TTY 2 ]" 500ms (jika visualIdentity ada)
  4. render() buffer TTY2 → process.stdout
  5. onSwitchCallback(2) → SIGWINCH ke foreground PID TTY2
```

Ctrl+C → SIGINT

```
Host/kbd deteksi "\u0003"
→ pushInput: cooked → onInterrupt() (tidak masuk buffer); raw → onInterrupt() + tetap masuk buffer
→ TTYManager.onInterruptCallback(activeId)
→ Kernel → SIGINT ke foreground process TTY aktif
→ Scheduler.sendInterruptSignal(ttyId) → kill(fgPid, 2)
  → event "signal" SIGINT → grace 100ms → worker.terminate() jika tak ditangani
```

Source Code

File	Role
src/kernel/tty/TTY.ts	TTY emulation: buffer, cursor, ANSI subset, raw/cooked, render
src/kernel/tty/TTYManager.ts	32 console allocation, activeId, switch, callbacks
src/kernel/devices/TTYDevice.ts	Device wrapper /dev/ttyN + ioctl
src/kernel/devices/KeyboardDevice.ts	Keyboard driver + hotkey/interrupt detection
src/kernel/Scheduler.ts	Foreground per TTY (ttyForegroundPids), SIGINT/SIGWINCH
src/kernel/Syscalls.ts	EXEC ttyId, ioctl 0x2001/0x2002, switch/resize
src/kernel/Kernel.ts	Boot wiring, hotkey handler, resize listener, runInit

Snippet (code level)

TTY input buffer — push/pop (src/kernel/tty/TTY.ts)

```
public pushInput(data: string) {
  if (this.rawMode) {
    // RAW MODE: langsung ke inputBuffer (tiap karakter)
    for (const char of data) {
      if (char === "\u0003") { // Ctrl+C
        if (this.onInterrupt) {
          this.onInterrupt();
        }
        // Tetap push untuk app yang mau menangani sendiri
        this.inputBuffer.push(char);
      } else {
        this.inputBuffer.push(char);
      }
    }
  } else {
    // COOKED MODE: echo + backspace + line buffering
    for (const char of data) {
      const code = char.charCodeAt(0);
      if (char === "\u0003") { // Ctrl+C → interrupt, jangan masuk buffer
        if (this.onInterrupt) this.onInterrupt();
        continue;
      }
      if (char === "\r" || char === "\n") {
        this.inputLines.push(this.lineBuffer + "\n");
        this.lineBuffer = "";
      }
    }
  }
}
```

```

        this.write("\n");          // echo newline
        continue;
    }
    if (code === 127 || code === 8) { // Backspace
        if (this.lineBuffer.length > 0) {
            this.lineBuffer = this.lineBuffer.slice(0, -1);
            this.write("\b \b");      // echo hapus visual
        }
        continue;
    }
    this.lineBuffer += char;
    // ... filter escape sequence untuk echo (NORMAL/ESC/CSI),
    //     lalu echo hanya char printable (0x20-0x7E) + \n \r \t
}
}
}

public read(): string | null {
    if (this.rawMode) {
        if (this.inputBuffer.length === 0) return null;
        return this.inputBuffer.shift() || null;
    } else {
        if (this.inputLines.length === 0) return null;
        return this.inputLines.shift() || null;
    }
}

public setRawMode(enabled: boolean) {
    this.logger.debug(`setRawMode(${enabled}) - previously ${this.rawMode}`);
    this.rawMode = enabled;
    // Apps yang pindah ke raw mode biasanya tidak ingin input cooked tertunda
    if (enabled) {
        this.lineBuffer = "";
    }
}
}

```

ioctl TTYDevice (src/kernel/devices/ttyDevice.ts)

```

public ioctl(cmd: number, arg: any): any {
    if (cmd === 1) { // 1 = CLEAR_SCREEN
        this.tty.clear();
        if (this.isActive()) {
            process.stdout.write("\x1bc");
        }
        return 0;
    }
    if (cmd === 2) { // 2 = SWITCH_TTY
        return -1; // Handled in Syscalls.ts via Kernel.ttyManager
    }
    if (cmd === 10) { // 10 = SET_RAW_MODE
        this.tty.setRawMode(!arg);
        return 0;
    }
    if (cmd === 0x2001) { // INJECT_INPUT (Master typing into slave stdin)
        this.tty.pushInput(arg as string);
        return true;
    }
    if (cmd === 0x2002) { // READ_OUTPUT (Master reading slave stdout)
        return this.tty.getOutput();
    }
    if (cmd === 4) { // 4 = TIOCGWINSZ (Get Window Size)
        return { lines: this.tty.height, columns: this.tty.width };
    }
    return -1;
}
}

```

TTYManager.switch (src/kernel/tty/ttyManager.ts)

```

public async switch(id: number, forceRedraw: boolean = false) {
    if (!this.ttys.has(id)) return;
}

```

```

if (id === this.activeId && !forceRedraw) return;

this.logger.info(`Switching from TTY${this.activeId} to TTY${id}`);
this.activeId = id;

// Reset total terminal host agar tidak ada sisa dari TTY lama
process.stdout.write("\x1bc\x1b[3J");

// Banner visual terpusat (500ms) jika ada visual identity
if (this.visualIdentity && !forceRedraw) {
  const rows = process.stdout.rows || 24;
  const cols = process.stdout.columns || 80;
  const text = `TSIX VIRTUAL CONSOLE [ TTY ${id} ]`;
  const barLines = this.visualIdentity.split("\n");
  const bannerWidth = 32;
  const bannerHeight = 1 + barLines.length;
  const startRow = Math.max(1, Math.floor((rows - bannerHeight) / 2));
  const startCol = Math.max(1, Math.floor((cols - bannerWidth) / 2));
  process.stdout.write("\x1b[?25l");
  process.stdout.write(`\x1b[${startRow};${startCol}H\x1b[97m  ${text}\x1b[0m`);
  barLines.forEach((line, index) => {
    process.stdout.write(`\x1b[${startRow + 1 + index};${startCol}H${line}`);
  });
  await new Promise(resolve => setTimeout(resolve, 500));
  process.stdout.write("\x1bc\x1b[3J\x1b[?25h");
}

const content = this.getActiveTTY().render();
process.stdout.write(content);

// Notify foreground process di TTY yang baru aktif
if (this.onSwitchCallback) {
  this.onSwitchCallback(id);
}
}

```

ttyId allocation in EXEC (src/kernel/Syscalls.ts)

```

// --- TTY REDIRECTION SUPPORT ---
// Jika TTY tertentu diminta, override I/O default ke TTY itu
if (ttyId !== undefined && ttyId >= 1 && ttyId <= 12) {
  const targetTtyDevice = this.kernel.devices[`tty${ttyId}`];
  if (targetTtyDevice) {
    stdinDevice = targetTtyDevice;
    stdoutDevice = targetTtyDevice;
    stderrDevice = targetTtyDevice;
  }
} else {
  // Warisan normal jika tidak ada TTY khusus
  if (stdoutFd !== undefined && pcb.fdTable[stdoutFd]) {
    stdoutDevice = pcb.fdTable[stdoutFd]!.device;
  }
  if (stdinFd !== undefined && pcb.fdTable[stdinFd]) {
    stdinDevice = pcb.fdTable[stdinFd]!.device;
  }
}

const newPcb = this.scheduler.createProcess(binaryName, {
  fds: [stdinDevice, stdoutDevice, stderrDevice],
  // ... appName, args, env, uid/gid, dst
  ttyId: ttyId !== undefined ? ttyId : pcb.ttyId, // target TTY atau warisi dari parent
  ppid: pid,
});

```

Console-switch hotkeys (Alt+F1..6, Alt+1..6, Ctrl+Alt+1..6) (src/kernel/Kernel.ts)

```

private handleKeyboardHotkey(seq: string): boolean {
  const hotkeys: Record<string, number> = {
    // Alt+F1..F6 (xterm, iTerm2, Terminal.app)
    "\x1b\x1b0P": 1,  "\x1b[1;3P": 1,  "\x1b[11;3~": 1,

```

```

"\x1b\x1b0Q": 2, "\x1b[1;3Q": 2, "\x1b[12;3~": 2,
"\x1b\x1b0R": 3, "\x1b[1;3R": 3, "\x1b[13;3~": 3,
"\x1b\x1b0S": 4, "\x1b[1;3S": 4, "\x1b[14;3~": 4,
"\x1b\x1b[15~": 5, "\x1b[15;3~": 5, "\x1b[1;3;15~": 5,
"\x1b\x1b[17~": 6, "\x1b[17;3~": 6, "\x1b[1;3;17~": 6,
// Option+1..6 (macOS Terminal with "Option as Meta" OFF)
"i": 1, "m": 2, "f": 3, "c": 4, "w": 5, "s": 6,
// Ctrl+4..6 – caret notation (FS/GS/RS)
"\x1c": 4, "\x1d": 5, "\x1e": 6,
};
for (let d = 1; d <= 6; d++) {
  const cp = 48 + d; // '1' = 49 ... '6' = 54
  hotkeys["\x1b" + d] = d; // Alt+digit & Ctrl+Alt+digit
  hotkeys["\x1b[${cp};3u"] = d; // Alt+digit (CSI-u)
  hotkeys["\x1b[${cp};5u"] = d; // Ctrl+digit (CSI-u)
  hotkeys["\x1b[${cp};7u"] = d; // Ctrl+Alt+digit (CSI-u)
  hotkeys["\x1b[27;3;${cp}~"] = d; // Alt+digit (modifyOtherKeys)
  hotkeys["\x1b[27;5;${cp}~"] = d; // Ctrl+digit (modifyOtherKeys)
  hotkeys["\x1b[27;7;${cp}~"] = d; // Ctrl+Alt+digit (modifyOtherKeys)
}
const targetTtyId = hotkeys[seq];
// typeof check: words like "constructor"/"toString" must not be swallowed
if (typeof targetTtyId === "number") {
  // FIRE AND FORGET: don't await, otherwise keyboard input gets blocked
  this.ttyManager?.switch(targetTtyId);
  return true; // handled
}
return false;
}
}

```

Callback & foreground wiring (src/kernel/Kernel.ts + src/kernel/Scheduler.ts)

```

// Kernel: daftarkan callback TTY
this.ttyManager.setOnSwitchCallback((ttyId: number) => {
  const fgPid = this.scheduler?.getForegroundProcess(ttyId);
  if (fgPid) {
    this.logger.debug(`Sending SIGWINCH to PID ${fgPid} (TTY${ttyId} activated)`);
    this.scheduler?.sendEvent(fgPid, "signal", "SIGWINCH");
  }
});
this.ttyManager.setOnInterruptCallback((ttyId: number) => {
  const fgPid = this.scheduler?.getForegroundProcess(ttyId);
  if (fgPid) {
    this.logger.info(`Sending SIGINT to PID ${fgPid} (TTY${ttyId} Ctrl+C)`);
    this.scheduler?.sendEvent(fgPid, "signal", "SIGINT");
  }
});

// Scheduler: satu foreground process per TTY
public setForegroundProcess(pid: number | null, ttyId?: number) {
  if (pid !== null && !ttyId) {
    const pcb = this.getProcess(pid);
    ttyId = pcb?.ttyId || 1;
  }
  const targetTty = ttyId || 1;
  if (pid === null) {
    this.ttyForegroundPids.delete(targetTty);
  } else {
    this.ttyForegroundPids.set(targetTty, pid);
  }
}
}

```

Exercises / Practice

1. Read `src/kernel/tty/TTY.ts` — explain the buffer structure `char[y][x]` + `attr[y][x]` and the ANSI subset in `handleANSI()`.

2. Run `pixelterm` (if available) — observe how it uses `ioctl 0x2001` (inject input) and `0x2002` (read output) from the master side.
3. From another TTY, press `Alt+F2` — observe the console switch and the visual banner.
4. Read `src/kernel/devices/TTYDevice.ts` — explain each `ioctl (1, 2, 10, 0x2001, 0x2002, 4)`.
5. Trace the `EXEC` syscall in `src/kernel/Syscalls.ts` — what is different when `ttyId` is set vs not? When does a process inherit the parent's `ttyId`?
6. Press `Ctrl+C` on a foreground process — explain the signal path from `pushInput` → `onInterrupt` → `SIGINT`.

References

- `wiki/Kernel-dan-Scheduler.md §TTY`
- `wiki/course/00-overview.en.md §8`
- `src/kernel/tty/TTY.ts`, `src/kernel/tty/TTYManager.ts`, `src/kernel/devices/TTYDevice.ts`
- `src/kernel/Scheduler.ts (§ ttyForegroundPids, setForegroundProcess, sendInterruptSignal)`
- `src/kernel/Syscalls.ts` (syscall `EXEC`, `SEND/READ` via `ioctl 0x2001/0x2002`, `SWITCH_TTY`, `WINSIZE`)

Module 13 — complete. Continue to [Module 14 — Userland](#).

RFC-TSIX-EDU-002 | Fourteenth module of the TSIX curriculum. Understand the application layer: PID 1 (init), login, shell, and the two styles of writing TSIX applications.

All userland runs in a Worker Thread (Ring 4) — including PID 1. There is no inittab: the init logic is **hardcoded** in `init.ts`.

Learning Objectives

- Explain the role of init (PID 1) without inittab
- Explain the login flow: `passwd+shadow(bcrypt) → setgroups→setgid→setuid`
- Explain the two application styles: `IProgram` vs `Program()` wrapper
- Explain the "error = window" concept (`std.error`)
- Explain the role of `monitorProcess` (respawn login)

Core Concepts

Init (PID 1)

There is no `/etc/inittab` — all init decisions are hardcoded in `src/mirror/bin/init.ts` (class `Init`). Execution order:

1. **Enforce setuid** — `chmod("/bin/passwd.js", 2541)` and `chmod("/bin/sudo.js", 2541)`. The value 2541 (decimal) equals `0o4755` (setuid root + `rw-r-xr-x`). The target is `.js`, not `.ts`, because the runtime executes the `.js` sidecar.
2. **RSA identity** — check `/etc/keys/rsa/id_rsa.pub`. If it is missing, `SecurityAgent.generateKeyPair()` creates a new key (RSA 2048-bit) and saves it to `/etc/keys/rsa/`. If it exists, the key is verified. The SHA256 fingerprint is taken via `shell.getFingerprint()`, and the identity color bar (`SecurityAgent.generateVisualIdentity()`) is sent to the TTY via `std.ioctl(1, 33, colorBar)` — `ioctl 33 = SET_VISUAL_IDENTITY`.
3. **Exec rc.local** — run `/etc/rc.local.js` then `waitpid`. Exit code 0 = successful startup; anything else is logged as a warning.
4. **Spawn login TTY2-6** — `for (let i = 2; i <= 6; i++) spawnLogin(i)`. Each login process is *monitored* by `monitorProcess` (respawn + 1s delay anti-crash-loop).
5. **Spawn login TTY1** (foreground) — done after the banner and fingerprint.
6. **Loop forever** — `while (true) { await new Promise(r => setTimeout(r, 10000)); }`. Init must not exit; if PID 1 dies, the kernel calls `process.exit(exitCode)` (`keepAlive, 1 = reboot`).

Login

`src/mirror/bin/login.ts` — one login process per TTY, spawned by init. Core flow:

1. Read `/etc/passwd` → find the user entry → take `uid` (field 3), `gid` (field 4), `home` (field 6), `shell` (field 7).
2. Read `/etc/shadow` → take the `bcrypt` hash (field 2) → `bcrypt.compareSync(password, hash)`.
3. Read `/etc/group` → collect *supplementary groups* (groups that contain the username, besides the primary gid).
4. `setgroups → setgid → setuid` — **mandatory POSIX order**. The GID must be changed before the UID; after the UID becomes non-root, the process loses the privilege to change the GID.
5. Set env `USER/HOME`, `chdir(home)`, then `exec` the shell — the shell is taken from `/etc/passwd` field 7 (usually `tsh`).

WM Login (Asteracea) — GUI mode

TTY login (`login.ts`) is re-spawned as a fresh root process per TTY, so it can always read `/etc/shadow` & `setuid`. In contrast, **the WM (Asteracea) is a SINGLE process that is both login manager and desktop session**:

- The WM starts as root (spawned by init), then after the first login it **permanently drops privileges** to the logged-in user.

- On logout → login again (e.g. as root), the WM is already non-root → **two problems**:
 - It can no longer read `/etc/shadow` (0640 root) → password verification fails.
 - The kernel rejects `setgroups` / `setgid` / `setuid` for non-root → cannot switch users.
- Solution (see Snippet):
 - Verify via a SetUID-root helper** — `login.js --verify <user> <pass> <file>` (login.js is already SetUID root) → result is written to a temp file (`OK/FAIL:...`) and read back by the WM. A file channel is used because the child's *exit code* is unreliable (WorkerEntry always finishes with `exit(0)`).
 - Saved UID (kernel)** — `pcb.suid`. When the WM drops from root, the kernel keeps `suid=0`; on re-login the WM may `setuid(0)` to **restore** root, then drop to the target user. This is the Unix Saved UID mechanism (`seteuid` / `setresuid`).
 - Credential order in the WM** — if the WM is not root yet → `setuid(0)` first (restore), then `setgroups` → `setgid` → `setuid(target user)` (POSIX order for dropping).

[!NOTE] **Saved UID safety** — `suid` defaults to the process's own UID in `createProcess`, so **regular apps do NOT inherit saved root** and cannot escalate. Only a process that dropped from root (i.e. the WM) keeps the "ticket back to root".

Shell (tsh)

`src/mirror/bin/tsh.ts` — interactive shell, written in the `IProgram` style (`export class main implements IProgram`). Main features:

- Prompt `user@hostname:cwd#/$` customizable via the `PROMPT_FORMAT` env (`&username`, `&hostname`, `&cwd`, `&usertype`).
- Line editor: raw mode, history (up/down arrows), Tab completion, Ctrl+U, Home/End.
- Before running a command, the shell switches to cooked mode so foreground commands can receive SIGINT.
- `tsh <username>` — delegates to `/bin/login.js` (login is SetUID root, so it can authenticate & switch users).
- Resize (SIGWINCH / `RESIZE` event) updates the `LINES` / `COLUMNS` env without overwriting the foreground application's screen.

Two application styles

There are two styles for writing TSIX applications:

1. Class `IProgram` (mostly legacy) — `export class main` that implements `IProgram`, with the method `execute(lib, args)`. All APIs are accessed through the `lib` parameter:

```
export class main implements IProgram {
  async execute(lib: OSContext, args: string[]) {
    // ...
  }
}
```

Examples: `ls`, `cat`, `ps`, `kill`, `tsh`.

2. `Program()` wrapper + proxy singletons (new) — import the `std`, `fs`, `shell`, `net`, `db` singletons from `@tsix/Application`, then wrap the main function with `Program(fn)`:

```
import { Program, std, fs, shell, net } from "@tsix/Application";

export default Program(async (args) => {
  // ...
});
```

Examples: `esp-send`, `dome`, `iot-dashboard`.

[!NOTE] **How `Program()` works** — `Program(fn)` returns an anonymous class that implements `IProgram`. The `execute` method stores the `OSContext` into `global._tsix0sc`, calls `fn(args)`, and if `fn` throws an error, calls

`std.error(error.stack, appName)` then re-throws. The `std/fs/shell/net/db` singletons are Proxy objects that forward access to `_tsixLib` on the current thread. `os.pid` and `os.rand` are also available.

Comparison of the two styles:

Aspect	Class <code>IProgram</code> (legacy)	<code>Program()</code> wrapper (new)
Shape	Class <code>main</code> + method <code>execute()</code>	Anonymous function wrapped in <code>Program(fn)</code>
Signature	<code>execute(lib: OSContext, args: string[])</code>	<code>fn(args: string[])</code>
API access	Through the <code>lib</code> parameter (<code>lib.std, lib.fs, ...</code>)	Import proxy singletons (<code>std, fs, shell, net, db</code>)
Arguments	Second parameter	Single function parameter
Error handling	Manual (own try/catch)	Automatic → <code>std.error()</code> then re-throw
Implementation	implements <code>IProgram</code> directly	<code>Program()</code> returns a class implements <code>IProgram</code>
Examples	<code>ls, cat, ps, kill, tsh</code>	<code>esp-send, dome, iot-dashboard</code>

Error = "Window"

`std.error()` does 4 things:

1. Write to `/var/log/syslog`.
2. Find `fileHint` from the stack trace (skipping `UserLib, Application, emerald`).
3. Broadcast to the parent via IPC — message `{ type: "GUI_WINDOW_ERROR", wid, pid, file, error, context, timestamp }`.
4. Print to the TTY in red `[ERROR]`.

So when an application crashes, **an error window appears on the desktop** — not just text in the terminal.

Flow / How It Works

Boot: Kernel → init (PID 1)

```
Kernel (main.ts) – kernel.runInit()
  resolve /bin/init.js
  createProcess("init", fds:[tty1×3]) → PID 1
  setForegroundProcess(1, 1)

[init.ts – Worker Thread, Ring 4]
  1. Enforce setuid
     chmod /bin/passwd.js 2541 (0o4755)
     chmod /bin/sudo.js 2541
  2. RSA identity
     cek /etc/keys/rsa/id_rsa.pub
     ada → verify + fingerprint
     tidak ada → generateKeyPair() → simpan id_rsa + id_rsa.pub
     fingerprint + color bar → ioctl(1, 33, colorBar) (SET_VISUAL_IDENTITY)
  3. Exec /etc/rc.local.js + waitpid
  4. for (i = 2; i <= 6; i++) spawnLogin(i)
     exec /bin/login.js (ttyId=i)
     monitorProcess → respawn saat login mati (delay 1s anti-crash-loop)
  5. Banner ASCII + fingerprint color bar
  6. spawnLogin(1) ← foreground TTY1
  7. while (true) { sleep 10s } ← init tidak boleh exit
```

Login: TTY → shell

```
login.ts (satu proses per TTY)
  1. username = args[0] || readLine("Username: ")
  2. password = readPassword("Password: 🐣") ← masked, raw mode
  3. /etc/passwd → uid(3), gid(4), home(6), shell(7)
  4. /etc/shadow → hash bcrypt (field 2)
```

```

5. bcrypt.compareSync(password, hash)
   gagal → "Login: Invalid username or password" → ulangi dari langkah 1
6. MOTD: /etc/motd + frasa acak /etc/motd.json
7. /etc/group → supplementary gids (grup yang memuat username)
8. setgroups(gids) → setgid(gid) → setuid(uid) ← urutan POSIX
9. setenv USER / HOME → chdir(home)
10. exec(shell dari /etc/passwd) → waitpid → break (satu sesi)

```

WM Login: GUI → desktop

```

asteracea.ts (single process = login manager + desktop)
1. WM starts as root (spawned by init)
2. Show GUI login screen
3. Verify: spawn /bin/login.js --verify <user> <pass> <file> ← SetUID root
   read result file: "OK" / "FAIL:..."
4. /etc/passwd → uid(3), gid(4), home(6) ← world-readable, read directly
5. /etc/group → supplementary gids
6. If the WM is not root yet (already logged in non-root) → setuid(0) ← restore via Saved UID
7. setgroups(gids) → setgid(gid) → setuid(uid) ← POSIX order (drop)
8. setenv USER / HOME → chdir(home)
9. Show desktop — the WM BECOMES that user (taskbar, launcher, wallpaper)
10. Logout → back to step 2 (WM stays alive; identity is reset via Saved UID)

```

Source Code

File	Role
src/mirror/bin/init.ts	PID 1 — hardcoded init (setuid, RSA, rc.local, spawn login)
src/mirror/bin/login.ts	passwd+shadow (bcrypt) authentication → setgroups/setgid/setuid → exec shell
src/mirror/bin/tsh.ts	Interactive shell (IProgram style)
src/mirror/opt/asteracea/asteracea.ts	WM + GUI login manager — verify via login.js --verify , re-elevate via Saved UID
src/mirror/lib/UserLib.ts	Framework: std, fs, shell, net, db (sub-library)
src/mirror/lib/Application.ts	Program() wrapper + proxy singleton
src/mirror/lib/IProgram.ts	IProgram & OSContext interfaces

Snippet (code level)

All snippets below are copied from the source — *code is the truth*.

Init — enforce setuid

```

// Runtime mengeksekusi sidecar .js (bukan source .ts),
// jadi chmod harus di .js. Nilai 2541 = 0o4755 (setuid root).
await lib.fs.chmod("/bin/passwd.js", 2541);
await lib.std.print(`${ok} [INIT] SetUID bit applied to: /bin/passwd.js\n`);
await lib.fs.chmod("/bin/sudo.js", 2541);
await lib.std.print(`${ok} [INIT] SetUID bit applied to: /bin/sudo.js\n`);

```

ok is the green [OK] prefix defined at the start of execute() in init.ts.

Init — spawnLogin (TTY2-6)

```

const spawnLogin = async (ttyId: number) => {
  try {
    if (ttyId > 1)
      await lib.std.print(`${ok} Initializing session on TTY${ttyId}...\n`);
  }
}

```

```

    const result = await lib.shell.exec("/bin/login.js", [], undefined, undefined, ttyId);
    if (result && result.pid) {
      terminals.set(ttyId, result.pid);
      await lib.std.log(`Login service spawned on TTY${ttyId} (PID ${result.pid})`, "init");
      // Kita nungguin di background (thread terpisah di Worker)
      this.monitorProcess(lib, ttyId, result.pid, spawnLogin);
    }
  } catch (e: any) {
    await lib.std.print(`Init: Error spawning login on TTY${ttyId}: ${e.message}\n`);
  }
};

// Start login on all TTYs (TTY2-6)
for (let i = 2; i <= 6; i++) {
  await spawnLogin(i);
}

```

Init — monitorProcess (respawn anti-crash-loop)

```

private async monitorProcess(
  lib: UserLib, ttyId: number, pid: number,
  respawn: (tty: number) => Promise<void>,
) {
  const exitCode = await lib.shell.waitpid(pid);
  await lib.std.print(
    `Init: Process on TTY${ttyId} (PID ${pid}) exited with code ${exitCode}. Respawning...\n`,
  );
  // Delay sedikit biar nggak spam kalau crash loop
  await new Promise(r => setTimeout(r, 1000));
  await respawn(ttyId);
}

```

Login — bcrypt verification

```

// 1. Cari user di /etc/passwd → uid, gid, home, shell
const parts = userEntry.split(":");
const uid = parseInt(parts[2]);
const gid = parseInt(parts[3]);
const home = parts[5];
const userShell = parts[6]; // shell dari /etc/passwd field 7

// 2. Ambil hash dari /etc/shadow → bcrypt.compareSync
const hash = shadowEntry.split(":")[1];
const match = bcrypt.compareSync(password, hash);
if (!match) {
  await lib.std.print("Login: Invalid username or password\n");
  continue; // kembali ke loop → prompt ulang
}

```

Login — setgroups → setgid → setuid (POSIX order)

```

// 3.5. Resolve Supplementary Groups dari /etc/group
const supplementaryGids: number[] = [gid]; // dimulai dari primary gid
try {
  const groupContent = await lib.fs.readFile("/etc/group") || "";
  const groupLines = groupContent.split("\n")
    .map(l => l.trim()).filter(l => l.length > 0);
  for (const gLine of groupLines) {
    const gParts = gLine.split(":");
    const groupGid = parseInt(gParts[2]);
    const groupUsers = gParts[3] ? gParts[3].split(",") : [];
    if (groupUsers.includes(username.trim()) && groupGid !== gid) {
      supplementaryGids.push(groupGid);
    }
  }
} catch (e) {
  // Ignore group errors
}

```

```
// Set Identity – GID harus SEBELUM UID.
// Setelah UID menjadi non-root, proses kehilangan privilege untuk mengubah GID.
await lib.shell.setgroups(supplementaryGids);
await lib.shell.setgid(gid);
await lib.shell.setuid(uid);

// Set Env + exec shell
await lib.shell.setenv("USER", username.trim());
await lib.shell.setenv("HOME", home);
await lib.shell.chdir(home);
const result = await lib.shell.exec(userShell); // shell dari /etc/passwd
if (result && result.pid) {
  await lib.shell.waitpid(result.pid);
}
```

WM Login — verify via login.js --verify (file channel)

```
// asteracea.ts – verify password via a SetUID-root helper.
// A non-root WM cannot read /etc/shadow (0640 root) → delegate to login.js.
const verifyOut = `/tmp/verify-${Date.now()}-${Math.random().toString(36).slice(2, 8)}.txt`;
let authOk = false;
const vp = await shell.exec("/bin/login.js", ["--verify", u, password, verifyOut]);
if (vp && vp.pid) {
  await shell.waitpid(vp.pid); // wait until done
  const res = (await fs.readFile(verifyOut)) || "";
  authOk = String(res).trim() === "OK"; // "OK" / "FAIL:..."
}
await fs.unlink(verifyOut); // cleanup temp file
```

```
// login.ts – helper side (runs as SetUID root, so it can read shadow)
if (args[0] === "--verify") {
  const vUser = (args[1] || "").trim();
  const vPass = args[2] || "";
  const vOut = args[3] || "/tmp/verify-result.txt";
  let ok = false, errMsg = "";
  try {
    const shadow = (await lib.fs.readFile("/etc/shadow")) || "";
    const entry = shadow.split("\n").map(l => l.trim()).filter(l => l)
      .find(l => l.split(":")[0] === vUser);
    const hash = entry ? entry.split(":")[1] : "";
    ok = !!hash && bcrypt.compareSync(vPass, hash);
    if (!hash) errMsg = "account not found or no password";
  } catch (e) { errMsg = e.message || "verify error"; }
  await lib.fs.writeFile(vOut, ok ? "OK" : "FAIL:" + errMsg);
  return; // done – WorkerEntry finishes the process
}
```

Saved UID (kernel) — setuid may restore to suid

```
// Syscalls.ts – SETUID with Saved UID (mirrors Unix setuid/seteuid)
if (this.isRoot(pcb)) {
  pcb.suid = pcb.uid; // root may change UID freely; keep the way back (0 for root)
  pcb.uid = newUid;
  pcb.ruid = newUid;
} else if (newUid === pcb.suid) {
  pcb.uid = newUid; // non-root may only RESTORE to its Saved UID (e.g. back to root)
  pcb.ruid = newUid;
} else {
  throw new Error("Permission Denied: Only root or root group members can change UID");
}

// Scheduler.ts – createProcess: suid defaults to the process's own UID
suid: options.suid ?? options.uid ?? 0, // regular apps CANNOT escalate
```

App — IProgram style (legacy)

```
// greet.ts — gaya klasik: class main implements IProgram
import { IProgram, OSContext } from "@tsix/IProgram";

export class main implements IProgram {
  async execute(lib: OSContext, args: string[]): Promise<string | void> {
    const name = args[0] || "dunia";
    await lib.std.print(`Halo, ${name}!\n`);
    await lib.std.print(`CWD: ${await lib.shell.getcwd()}\n`);
  }
}
```

App — Program() wrapper style (new)

```
// halo.ts — gaya baru: Program() wrapper + proxy singleton
import { Program, std, os } from "@tsix/Application";

export default Program(async (args) => {
  const name = args[0] || "dunia";
  await std.print(`Halo, ${name}!\n`);
  await std.print(`PID: ${os.pid}\n`);
  return "Selesai.";
});
```

Exercises / Practice

1. Log in as `root`, then run `ps` — identify the login process per TTY (`/bin/login.js`).
2. From the root shell, run `tsh tamu` — observe the login delegation (`setuid root`) and the new shell with a different user.
3. Read `src/mirror/bin/login.ts` — trace the `setgroups/setgid/setuid` order and explain why this order is mandatory.
4. Read `src/mirror/bin/init.ts` — find `monitorProcess`; kill the login process on a TTY (kill `<pid>` from root) then observe init respawning it within about 1 second.
5. Write an app in the `IProgram` style and an app in the `Program()` style — compare their structures (see Snippet).
6. Create an app that calls `std.error()` — observe the error window appearing on the desktop (`GUI_WINDOW_ERROR` message).

References

- [wiki/Memulai.md](#), [wiki/Perintah-Sistem.md](#), [wiki/Panduan-Developer.md](#)
- [wiki/course/00-overview.en.md §9](#) (Userland: init, shell, applications)
- [src/mirror/bin/init.ts](#), [src/mirror/bin/login.ts](#), [src/mirror/bin/tsh.ts](#)
- [src/mirror/lib/UserLib.ts](#), [src/mirror/lib/Application.ts](#), [src/mirror/lib/IProgram.ts](#)

Module 14 — done. Part V complete. Continue to [Module 15 — Networking MQTNL](#).

RFC-TSIX-EDU-002 | Fifteenth module of the TSIX curriculum. Understand MQTNL (MQTT Network Layer) — the TSIX networking protocol that uses MQTT pub/sub as the wire, instead of IP/routing.

Instead of TCP/IP, MQTNL uses **MQTT pub/sub as the wire**. Address = **node name** (string), port = **application endpoint** (PortManager). Connections are connectionless/UDP-like — there is no real listen/accept, only polling emulation.

Learning Objectives

- ☐ Explain the concept of address = node name, port = application endpoint
- ☐ Explain the socket flow: bind → sendto → recvfrom
- ☐ Explain the role of PortManager and releasePortsByPid
- ☐ Explain why there is no listen/accept (polling emulation)
- ☐ List the networking syscalls (30–34) + SECAGENT_LIST (39)

Core Concepts

Model: MQTT pub/sub is the "wire"

MQTNL does not use TCP/IP. There is no IP address, no routing, no stateful connection. Instead, TSIX uses **MQTT pub/sub as the transport medium (wire)**:

- **Every node** connects to the **same MQTT broker**.
- **Address = node name** (string): "tsix", "tsix-node-2", "esp32S3". This name is unique per device.
- **Topic** = `mqtnl@1.0/<dstAddress>` (JSON protocol) / `mqtnl@1.1/<dstAddress>` (Binary protocol). The topic content is an **MQTNL packet** (header + payload).
- **All nodes subscribe to** `mqtnl@1.x/#`, then **filter** packets by `dstAddress` (or `"*"` for broadcast). Packets for other nodes are dropped.

Sending is fire-and-forget without a session — that is why this model is **connectionless / UDP-like**.

Address & Port

Concept	Value	TCP/IP analog
Address	Node name (string)	IP address
Port	Application endpoint (0–65535)	TCP/UDP port
Wire	MQTT pub/sub topic	IP packet
Connection	Connectionless (fire-and-forget)	UDP

Each `sendto()` creates a self-contained packet. The packet header carries the destination address & port, so no connection needs to be kept.

Three Main Components

1. **SimpleMQTNLDriver** — *network interface* (`/dev/smqtnl0`, `/dev/smqtnl1`). Keeps the connection to the MQTT broker, publishes outbound packets, receives & filters inbound packets, 32KB fragmentation, reassembly, and decryption. Each instance has its own `localAddress` (node name).
2. **SocketDevice** — the abstraction of one socket: inbound data buffer + waiter list. Created per `SOCKET` syscall and stored in the FD table. It is passive — inbound data is **pushed** by the driver, and the application reads it via `recvfrom`.

3. **PortManager** — keeper of virtual ports 0–65535. `allocatePort()` for explicit bind, `allocateRandomPort(10000-20000)` for `bind(0)`, `releasePortsByPid()` when a process exits.

Per-port handler

When an application `bind(port)`s, the kernel calls `driver.registerHandler(port, cb)`. This handler routes inbound data to the socket owned by the application:

```
bind(port)
  → PortManager.allocatePort(port, pid)
  → socket.setPort(port); socket.driver = targetDriver
  → driver.registerHandler(port, (data) => socket.push(data))
```

When a packet arrives, the driver calls the handler on `dstPort`. This is **per-port dispatch**: one driver serves many ports at once.

Socket flow (summary)

```
App → socket() → bind(port) → driver.registerHandler(port)
  → sendto(addr, port, data) → driver.send → publish topic
  → MQTT broker → semua node subscribe 'mqtnl@1.x/#'
  → filter dstAddress → reassembly → decrypt → socket.push()
  → recvfrom() → buffer.shift()
```

Syscall

```
SOCKET=30, BIND=31, SENDTO=32, RECVFROM=33, NETSTAT=34, SECAGENT_LIST=39
```

There is no `listen/accept` at the syscall level. UserLib emulates it: `listen() = socket() + bind(); accept() = polling recv()` until the first packet arrives.

`SECAGENT_LIST` (39) is used by the `secagent` tool to list the encryption agents registered in the kernel.

Flow / How It Works

Sending flow from **App A** on node "tsix" (port N) to **App B** on node "esp32S3" (port M) via the MQTT broker:



Detailed steps:

1. **SOCKET** — App A and App B call `socket()`. The kernel creates a new `SocketDevice` and inserts it into the FD table (context: "socket").
2. **BIND** — App A calls `bind(fd, N)`. The kernel determines the target driver (default `mqtnl0`), reserves port N via `PortManager.allocatePort(N, pid)`, then `registerHandler(N, cb)` routes data to socket A.
3. **SENDTO** — App A calls `sendto(fd, "esp32S3", M, data)`. The kernel calls `driver.send("esp32S3", M, data, FLAG_DATA, N)`.
4. **Fragmentation** — `send()` splits the payload into 32KB chunks when needed, then wraps each chunk in a packet with a 9-field header.
5. **Publish** — the driver publishes each chunk to the topic `mqtnl@1.0/esp32S3` (qos: 0, retain: false).
6. **Broker & filter** — all nodes subscribed to `mqtnl@1.x/#` receive the packet. Each driver filters: a packet is processed only when `dstAddress` equals its `localAddress` (or `"*"`). Other packets are dropped. Packets sent by the node itself (src = local) are also ignored.
7. **Reassembly & decryption** — the destination driver reassembles the chunks (session per `srcAddr:srcPort` → `dstAddr:dstPort`, TTL 30s), then decrypts the payload via the agent attached to the port (default `SecurityAgent/"chacha20"`; it can be upgraded to another agent via `upgradeSecurity(key, { agent })`, e.g. `"aes-gcm"`).
8. **Dispatch** — the driver calls the handler on `dstPort` → `socket.push({ src, port, localPort, data, isBinary, ts })` → data enters the socket buffer.
9. **RECVFROM** — App B calls `recv(fd)`. The kernel reads the buffer (`read() = buffer.shift()`). If empty, the kernel waits event-driven (`waitForData`) until data is pushed.

[!NOTE] **Local loopback.** If `dstAddress` equals the sender node's own address (or `"localhost"`), the packet does not go out to the broker — the driver forwards it directly to the local driver instance (`SimpleMQTNLDriver.findLocal()`), similar to `localhost` on a real OS.

Source Code

File	Role
<code>src/kernel/devices/SimpleMQTNLDriver.ts</code>	MQTNL network interface
<code>src/kernel/devices/SocketDevice.ts</code>	Socket = device (everything is a file)
<code>src/kernel/PortManager.ts</code>	Virtual port allocation
<code>src/mirror/lib/NetworkLib.ts</code>	<code>lib.net</code> API + <code>NetSocket</code> component (single source of truth)
<code>src/mirror/lib/UserLib.ts</code>	Imports & re-exports <code>NetworkLib</code> from <code>./NetworkLib</code>
<code>src/mirror/lib/Application.ts</code>	<code>net proxy</code> + exports <code>NetSocket</code> / <code>NetPacket</code>
<code>src/common/ISecurityAgent.ts</code>	Encryption agent contract (pluggable)
<code>src/common/AesGcmAgent.ts</code>	Example custom AES-256-GCM agent
<code>src/kernel/Syscalls.ts</code>	SOCKET-NETSTAT syscall implementation (30-34) + SECAGENT_LIST (39)
<code>src/mirror/sbin/secagent.ts</code>	Tool to list registered agents
<code>src/sysconfig.conf</code>	Default network interface configuration

[!NOTE] **Single NetworkLib.** The `NetworkLib` class is now the single source of truth in `NetworkLib.ts` and accepts either `dispatch` or `OSContext` (backward compatible). For new apps, prefer `NetSocket` — a high-level API wrapping lifecycle + events + security.

Default network interfaces (sysconfig.conf)

At boot, the kernel reads `cfg.network.interfaces` and creates one `SimpleMQTNLDriver` per entry (`Kernel.ts` → "Initialize Network Interfaces"). Each interface has a **deviceName** (`/dev name`), an **address** (node name), and a

broker:

deviceName	address (node name)	broker	defaultPort
smqtnl0	tsix	mqtt://192.168.1.204	1883
smqtnl1	tsix-node-2	mqtt://192.168.1.204	1883

`cfg.network.defaultDevice = smqtnl0`. If `bind()` is called without an `address` argument, the kernel uses this default interface.

Snippet (code level)

All snippets below are **copied from the source** — "code is truth". Trimmed parts are marked `// ...`.

NetworkLib — usage from the app side (NetworkLib.ts)

`lib.net` is a `NetworkLib` instance (single source of truth in `NetworkLib.ts`). Its constructor accepts either `dispatch` (used by `UserLib`) or `OSContext` (used by `ping.ts/network-traffic.ts`). A socket is returned as an **fd** (number), not an object:

```
export class NetworkLib {
  // Accepts `dispatch` (UserLib) OR `OSContext` (legacy apps) — unified into
  // a single source of truth (resolveDispatch picks the right dispatcher).
  constructor(source: DispatchFn | OSContext) {
    /* resolveDispatch(source) */
  }

  public async socket(): Promise<number> {
    return await this.dispatch(SyscallCode.SOCKET, null);
  }

  public async bind(
    fd: number,
    port: number,
    address?: string,
  ): Promise<boolean> {
    return await this.dispatch(SyscallCode.BIND, { fd, port, address });
  }

  // listen() = socket + bind — emulasi, tidak ada syscall LISTEN
  public async listen(port: number): Promise<number> {
    const fd = await this.socket();
    const ok = await this.bind(fd, port);
    return ok ? fd : -1;
  }

  // accept() = polling recv() sampai ada paket pertama
  public async accept(serverFd: number): Promise<any> {
    while (true) {
      const pkt = await this.recv(serverFd);
      if (pkt) {
        return {
          fd: serverFd,
          src: pkt.src,
          port: pkt.port,
          localPort: pkt.localPort || 0,
          firstPkt: pkt,
        };
      }
    }
    await new Promise((r) => setTimeout(r, 100));
  }

  public async sendto(
    fd: number,
    address: string,
    port: number,
```

```

data: any,
flag: number = 0,
srcPort: number = 0,
): Promise<boolean> {
  return await this.dispatch(SyscallCode.SENDTO, {
    fd,
    address,
    port,
    data,
    flag,
    srcPort,
  });
}

public async recv(fd: number): Promise<any> {
  return await this.dispatch(SyscallCode.RECVFROM, fd);
}
}

```

Usage example (server-client pattern):

```

// Server di node "tsix"
const fd = await lib.net.listen(8080); // socket + bind(8080)
const client = await lib.net.accept(fd); // polling recv → { src, port, firstPkt }
const data = await lib.net.recv(fd); // baca data

// Client mengirim ke node "esp32S3", port 5000
await lib.net.sendto(fd, "esp32S3", 5000, JSON.stringify({ cmd: "ping" }));

```

NetSocket — high-level API à la Cashew (NetworkLib.ts)

NetSocket wraps syscalls + security + lifecycle into a component: instantiate → event → open → close. Security is not automatic — switch explicitly via `upgradeSecurity(key, { agent })`:

```

const sock = new NetSocket({ port: 8080, key: KEY_HEX });

sock.onData = (pkt) => std.println(`[${pkt.src}:${pkt.port}] ${pkt.data}`);
sock.onError = (err) => std.println(`ERR: ${err.message}`);

await sock.open(); // socket + bind (plain first)
await sock.upgradeSecurity(KEY_HEX); // switch to chacha20 (default)
await sock.upgradeSecurity(KEY_HEX, { agent: "aes-gcm" }); // or a custom agent
await sock.sendto("esp32S3", 5000, JSON.stringify({ cmd: "ping" }));

await sock.waitClosed(); // keep the process alive
await sock.close(); // release port + normalize agent

```

PortManager — bind & release (PortManager.ts)

```

public allocatePort(port: number, pid?: number): boolean {
  if (port < 0 || port > 65535) return false;
  if (this.usedPorts.has(port)) return false;

  this.usedPorts.add(port);
  if (pid !== undefined) this.portOwner.set(port, pid);
  return true;
}

public allocateRandomPort(min: number = 10000, max: number = 20000): number | null {
  for (let i = 0; i < 100; i++) {
    const port = Math.floor(Math.random() * (max - min + 1)) + min;
    if (!this.usedPorts.has(port)) {
      this.usedPorts.add(port);
      return port;
    }
  }
  return null;
}

```

```

public releasePortsByPid(pid: number): void {
  const toRelease: number[] = [];
  this.portOwner.forEach((owner, port) => {
    if (owner === pid) toRelease.push(port);
  });
  for (const port of toRelease) {
    this.usedPorts.delete(port);
    this.portOwner.delete(port);
  }
}

```

[!TIP] `releasePortsByPid()` is called when a process exits — a **safety net** if an application forgets to release its ports.

SocketDevice — buffer & handler dispatch (SocketDevice.ts)

```

public push(data: any) {
  this.buffer.push(data);
  // Bangunkan semua reader yang sedang nunggu data (event-driven)
  while (this.waiters.length > 0) {
    const w = this.waiters.shift()!;
    w();
  }
}

public read(): any {
  return this.buffer.shift() || null;
}

public async waitForData(timeoutMs: number): Promise<boolean> {
  if (this.buffer.length > 0) return true;
  return new Promise<boolean>((resolve) => {
    let timer: ReturnType<typeof setTimeout> | undefined;
    const onData = () => {
      if (timer) clearTimeout(timer);
      const idx = this.waiters.indexOf(onData);
      if (idx >= 0) this.waiters.splice(idx, 1);
      resolve(true);
    };
    timer = setTimeout(() => {
      const idx = this.waiters.indexOf(onData);
      if (idx >= 0) this.waiters.splice(idx, 1);
      resolve(false);
    }, timeoutMs);
    this.waiters.push(onData);
  });
}

```

SimpleMQTNLDriver — init & send (SimpleMQTNLDriver.ts)

```

public init(ctx: KContext): void {
  this.client = mqtt.connect(this.brokerUrl);

  this.client.on("connect", () => {
    for (const proto of this.protocols) {
      const prefix = proto.getTopicPrefix();
      this.client?.subscribe(`${prefix}#`); // subscribe "mqtnl@1.0/#" & "mqtnl@1.1/#"
    }
  });

  this.client.on("message", (topic, message) => {
    this.handleIncomingMessage(topic, message);
  });
}

```

```

public async send(address: string, port: number, data: any, flag: PacketFlags = PacketFlags.FLAG_DATA, srcPort: number = 0) {
    // [LOOPBACK] Reserved alias "localhost" selalu menunjuk ke interface
    // pengirim sendiri – mirip localhost di OS sungguhan.
    if (address === "localhost") {
        address = this.localAddress;
    }

    // [LOOPBACK] Kalau tujuan adalah alamat lokal (milik node ini), kirim
    // langsung ke receiver tanpa keluar ke broker MQTT.
    const localTarget = SimpleMQTNLDriver.findLocal(address);

    // Cek koneksi hanya untuk paket yang benar-benar keluar ke broker.
    if (!localTarget && (!this.client || !this.client.connected)) return false;

    // ... (normalisasi payload, pemilihan protokol JSON/Binary, enkripsi,
    //      fragmentasi 32KB, header 9 field) – lihat file sumber

    const topic = `${prefix}${address}`; // mis. "mqtnl@1.0/esp32S3"

    for (let i = 0; i < packetCount; i++) {
        // ... (bungkus packet, protocol.pack)
        if (localTarget) {
            localTarget.handleIncomingMessage(topic, payloadBuffer); // LOOPBACK lokal
        } else {
            await new Promise((resolve) => {
                this.client!.publish(topic, payloadBuffer, { qos: 0, retain: false }, (err) => {
                    if (err) this.logger.error(`MQTT Publish error: ${err.message}`);
                    resolve(true);
                });
            });
        }
    }
    return true;
}

```

BIND — connecting a socket to the driver & port (Syscalls.ts)

```

case SyscallCode.BIND: {
    const { fd, port, address } = args as {
        fd: number;
        port: number;
        address?: string;
    };

    const entry = pcb.fdTable[fd];
    if (!entry || !(entry.device instanceof SocketDevice))
        throw new Error("Invalid Socket FD");

    const socket = entry.device as SocketDevice;
    if (socket.bound) throw new Error("Socket already bound");

    // ... pilih targetDriver (dari `address`, atau cfg.network.defaultDevice)

    const portMgr = this.kernel.getPortManager();
    let actualPort = port;

    if (port === 0) {
        const randomPort = portMgr.allocateRandomPort();
        if (randomPort === null) throw new Error("No random ports available");
        actualPort = randomPort;
    } else {
        if (!portMgr.allocatePort(port, pcb.pid))
            throw new Error(`Port ${port} already in use`);
    }

    socket.setPort(actualPort);
    socket.driver = targetDriver;

    // Register handler
    targetDriver.registerHandler(actualPort, (data) => {
        socket.push(data);
    });
}

```

```
    return true;
}
```

Exercises / Practice

1. Read `wiki/Networking-MQTNL.md` — the complete flow and topics.
 2. Read `src/kernel/PortManager.ts` — understand `allocateRandomPort` and `releasePortsByPid`.
 3. Run two nodes (e.g. `tsix` and `esp3253` on an MQTT broker) — send a message between them.
 4. Read `src/kernel/devices/SocketDevice.ts` — explain how a socket becomes an `IDevice`.
 5. From an app, run `lib.net.netstat()` — compare the result with the interface table in `sysconfig.conf`.
-

References

- `wiki/Networking-MQTNL.md` — full MQTNL documentation
 - `wiki/course/00-overview.md §7` — Networking MQTNL
 - `src/kernel/devices/SimpleMQTNLDriver.ts` — network interface
 - `src/kernel/devices/SocketDevice.ts` — socket = device
 - `src/kernel/PortManager.ts` — virtual port allocation
 - `src/kernel/Syscalls.ts` — SOCKET-NETSTAT syscall implementation (30-34) + SECAGENT_LIST (39)
 - `src/sysconfig.conf` — default network interfaces
-

Module 15 — done. Continue to [Module 16 — Wire Protocol MQTNL](#).

RFC-TSIX-EDU-002 | Sixteenth module of the TSIX curriculum. Understand the data format on top of MQTT: the JSON vs Binary dual protocol, magic byte detection, and fragmentation.

MQTNL has **two wire protocols**: JSON v1.0 (readable) and Binary v1.1 (compact, without encryption — for byte-exact OTA). The driver detects the protocol from the **first magic byte** per srcAddress.

Learning Objectives

- ☐ Explain the `IMQTNLProtocol` contract
- ☐ Distinguish the JSON v1.0 vs Binary v1.1 protocol
- ☐ Explain magic byte detection
- ☐ Explain 32KB fragmentation and 30s TTL reassembly
- ☐ Explain why userland never touches the protocol directly

Core Concepts

The IMQTNLProtocol Contract

All MQTNL wire protocols implement a single contract. This interface is the "only door" between the driver and the wire format — adding a new protocol (e.g. v2) only requires implementing a new class, without changing the driver.

```
export interface IMQTNLProtocol {
  /** Nama protokol (misal: "JSON", "Binary") */
  getName(): string;
  /** Byte pertama penanda protokol dalam byte stream.
   * JSON: '[' (0x5B). Binary: 'B' (0x42). */
  getMagicChars(): number[];
  /** MQTT Topic Prefix (misal: "mqtnl@1.0/", "mqtnl@1.1/") */
  getTopicPrefix(): string;
  /** Objek packet internal → Buffer/String untuk dikirim ke broker MQTT */
  pack(packet: any): Buffer | string;
  /** Raw data yang diterima → objek packet internal */
  unpack(data: Buffer | string): any;
}
```

Packet header (9 fields + payload)

The internal packet structure (the object that is `pack ed`/`unpack ed`) is identical for both protocols. The only difference is how fields are written to the wire.

Field	JSON (index array)	Binary (offset)	Type
srcAddress	0	S_ADDR_LEN + S_ADDR	string
srcPort	1	S_PORT	u16 LE
dstAddress	2	D_ADDR_LEN + D_ADDR	string
dstPort	3	D_PORT	u16 LE
packetCount	4	PACKET_COUNT	u16 LE
packetIndex	5	PACKET_INDEX	u16 LE
dataSize	6	DATA_SIZE	u32 LE
packetHeaderFlag	7	FLAG	u8

Field	JSON (index array)	Binary (offset)	Type
forwarded	8	FORWARDED	u8
payload	9	PAYLOAD (remaining buffer)	string / Buffer

[!NOTE] `packetCount` and `packetIndex` are part of the header (not a separate feature). `packetCount` = the total number of chunks, `packetIndex` = the chunk number (0-based). `dataSize` = the **total** payload size before it is split (not the size per chunk).

Dual protocol JSON vs Binary

Aspect	JSON v1.0	Binary v1.1
Class	<code>MQTNLProtocolJSON</code>	<code>MQTNLProtocolBinary</code>
Topic prefix	<code>mqtnl@1.0/<addr></code>	<code>mqtnl@1.1/<addr></code>
Magic byte	<code>[(0x5B)</code>	<code>B (0x42) + ver 0x01</code>
Wire format	JSON array (text, readable)	Compact binary buffer, byte-exact
Readability	easy for humans to read	hard to read (binary)
Size	larger (text overhead)	smaller & precise
Encryption	RSA handshake + ChaCha20-Poly1305	no encryption (plain, for OTA)
Used for	general communication (default legacy)	byte-exact OTA firmware transfer

 RSA handshake: SYN → SYN-ACK → ACK → encrypted DATA Source: [wiki/diafram/Networking-MQTNL-2.mmd](https://wiki.diafram.com/Networking-MQTNL-2.mmd)

Binary v1.1 frame format

Byte-by-byte diagram of the binary frame (N = length of `srcAddress`, M = length of `dstAddress`, L = payload length):

```

| 0x42 | 0x01 | len | <src addr> | srcPort | len | <dst addr> | dstPort |
| MAGIC| VER | sAL=N | srcAddr (N b) | u16LE | dAL=M | dstAddr (M b) | u16LE |

| pktCount | pktIndex | dataSize | flag | fwd | <payload> |
| u16LE | u16LE | u32LE | u8 | u8 | payload (L b) |

```

- Fixed header size = 18 bytes + N + M (2 + 1 + N + 2 + 1 + M + 2 + 2 + 2 + 4 + 1 + 1).
- Example: `srcAddress="tsix"` (N=4), `dstAddress="esp32S3"` (M=7) → header = 29 bytes.
- All fields are read in a deterministic order (no delimiter), so the address lengths must be sent (`S_ADDR_LEN/D_ADDR_LEN`).
- There is no CRC/checksum** inside the frame. OTA integrity is protected by `dataSize` (u32) and the byte-exact raw path.

Magic byte detection

The driver does not guess the protocol from the topic. It reads the **first byte** of the MQTT payload and matches it against `getMagicChars()` of each registered protocol:

```

// Detect protocol by Magic Byte (di SimpleMQTNLDriver.handleIncomingMessage)
const magicByte = Buffer.isBuffer(message) ? message[0] : (message as string).charCodeAt(0);
const proto = this.protocols.find(p => p.getMagicChars().includes(magicByte));
if (!proto) {
  this.logger.error(`Protocol mismatch on ${topic}. Magic: 0x${message[0].toString(16)}`);
  return;
}
packet = proto.unpack(message);

```

```
// [MULTI-PROTO] Register protocol untuk pengirim ini (per srcAddress)
this.protocolRegistry.set(packet.header.srcAddress, proto);
```

The result is stored in `protocolRegistry` per `srcAddress`. When sending back to that address, the driver uses the same protocol (default: JSON).

Fragmentation & Reassembly

- **Fragmentation:** the payload is split into 32KB chunks (`packetSize = 32768`). Each chunk is sent as a separate MQTT packet with `packetCount / packetIndex` filled in.
- **Reassembly:** the receiver collects chunks in a session per `src:port->dst:port`. The session is complete when the number of unique chunks equals `packetCount`, then they are merged in order (`Buffer.concat` for Buffer, `join("")` for strings).
- **30 second TTL:** sessions that are incomplete within 30 seconds are dropped (`cleanupExpiredPackets`, run every 60 seconds).



OTA firmware streaming via MQTNL Binary v1.1 Source: wiki/diagram/mqtnl-binary-ota-1.mmd

Send path (TX) — SimpleMQTNLDriver.send()

1. Determine the protocol: `protocolRegistry.get(address) || activeProtocol` (default JSON).
2. `useRaw = (protocol.getName() === "Binary")` — if Binary, **skip encryption** (`securedPayload = payload`), so the byte length is exact for OTA.
3. If JSON, encrypt the payload via `SecurityAgent.securePacketOut()` (RSA handshake → ChaCha20-Poly1305).
4. **Fragmentation:** split `securedPayload` into 32KB chunks (`this.packetSize`).
5. For each chunk `i`: create a packet object (`packetCount = chunks.length`, `packetIndex = i`, `dataSize = securedPayload.length`), `protocol.pack(packet) → Buffer`, then `publish` to the topic `${prefix}${address}` (or local loopback if the address is this node).

Receive path (RX) — SimpleMQTNLDriver.handleIncomingMessage()

1. Detect the protocol from the **first magic byte** of the MQTT payload.
2. `proto.unpack(message) → packet object`. Store the protocol in `protocolRegistry` per `srcAddress`.
3. **Packet filter:** drop packets whose `dstAddress` is not a local address (except `"*"`).
4. **Reassembly:** store `packet.payload` in the session `src:port->dst:port`. If not complete yet → wait for the next chunk.
5. If complete: merge the chunks in order → **decrypt** (Binary: `securePacketInRaw + plain fallback`; JSON: `securePacketIn`).
6. **Dispatch:** send `{src, port, localPort, data, isBinary, ts}` to the destination port handler (or serve PING/broadcast).

Snippet (code level)

All snippets below are **VERIFIED** — copied directly from the source.

JSON v1.0 — pack/unpack (MQTNLProtocolJSON.ts)

```
export class MQTNLProtocolJSON implements IMQTNLProtocol {
  getName(): string { return "JSON"; }
  getMagicChars(): number[] { return [0x5B]; } // '['
  getTopicPrefix(): string { return "mqtnl@1.0/"; }

  pack(packet: any): string {
    return JSON.stringify([
      packet.header.srcAddress, packet.header.srcPort,
      packet.header.dstAddress, packet.header.dstPort,
      packet.header.packetCount, packet.header.packetIndex,
      packet.header.dataSize, packet.header.packetHeaderFlag,
      packet.header.forwarded, packet.payload,
    ]);
  }

  unpack(data: Buffer | string): any {
    const str = Buffer.isBuffer(data) ? data.toString("utf8") : data;
    const packed = JSON.parse(str);
    return {
      header: {
        srcAddress: packed[0], srcPort: packed[1],
        dstAddress: packed[2], dstPort: packed[3],
        packetCount: packed[4], packetIndex: packed[5],
        dataSize: packed[6], packetHeaderFlag: packed[7],
        forwarded: packed[8],
      },
      payload: packed[9],
    };
  }
}
```

Binary v1.1 — pack/unpack (MQTNLProtocolBinary.ts)

```
export class MQTNLProtocolBinary implements IMQTNLProtocol {
  getName(): string { return "Binary"; }
  getMagicChars(): number[] { return [0x42]; } // 'B'
  getTopicPrefix(): string { return "mqtnl@1.1/"; }

  pack(packet: any): Buffer {
    const srcAddr = Buffer.from(packet.header.srcAddress || "", "utf8");
    const dstAddr = Buffer.from(packet.header.dstAddress || "", "utf8");
    const payload = Buffer.isBuffer(packet.payload)
      ? packet.payload
      : Buffer.from(packet.payload || "", "utf8");

    const headerSize = 2 + 1 + srcAddr.length + 2 + 1 + dstAddr.length
      + 2 + 2 + 2 + 4 + 1 + 1;
    const buf = Buffer.alloc(headerSize + payload.length);
    let offset = 0;
    buf.writeUInt8(0x42, offset++); // Magic 'B'
    buf.writeUInt8(0x01, offset++); // Proto Ver 1
    buf.writeUInt8(srcAddr.length, offset++);
    srcAddr.copy(buf, offset); offset += srcAddr.length;
    buf.writeUInt16LE(packet.header.srcPort, offset); offset += 2;
    buf.writeUInt8(dstAddr.length, offset++);
    dstAddr.copy(buf, offset); offset += dstAddr.length;
    buf.writeUInt16LE(packet.header.dstPort, offset); offset += 2;
    buf.writeUInt16LE(packet.header.packetCount, offset); offset += 2;
    buf.writeUInt16LE(packet.header.packetIndex, offset); offset += 2;
    buf.writeUInt32LE(packet.header.dataSize, offset); offset += 4;
    buf.writeUInt8(packet.header.packetHeaderFlag, offset++);
  }
}
```

```

        buf.writeUInt8(packet.header.forwarded, offset++);
        payload.copy(buf, offset);
        return buf;
    }

    unpack(data: Buffer | string): any {
        const buffer = Buffer.isBuffer(data) ? data : Buffer.from(data, "utf8");
        let offset = 0;
        if (buffer.readUInt8(offset++) !== 0x42) throw new Error("Invalid Magic Byte");
        if (buffer.readUInt8(offset++) !== 0x01) throw new Error("Invalid Protocol Version");

        const srcAddrLen = buffer.readUInt8(offset++);
        const srcAddress = buffer.subarray(offset, offset + srcAddrLen).toString("utf8");
        offset += srcAddrLen;
        const srcPort = buffer.readUInt16LE(offset);
        offset += 2;

        const dstAddrLen = buffer.readUInt8(offset++);
        const dstAddress = buffer.subarray(offset, offset + dstAddrLen).toString("utf8");
        offset += dstAddrLen;
        const dstPort = buffer.readUInt16LE(offset);
        offset += 2;

        const packetCount = buffer.readUInt16LE(offset);
        offset += 2;
        const packetIndex = buffer.readUInt16LE(offset);
        offset += 2;
        const dataSize = buffer.readUInt32LE(offset);
        offset += 4;
        const packetHeaderFlag = buffer.readUInt8(offset++);
        const forwarded = buffer.readUInt8(offset++);
        const payload = buffer.subarray(offset);

        return {
            header: { srcAddress, srcPort, dstAddress, dstPort,
                    packetCount, packetIndex, dataSize,
                    packetHeaderFlag, forwarded },
            payload: payload,
        };
    }
}

```

Fragmentation — 32KB loop (SimpleMQTNLDriver.send())

```

// --- FRAGMENTATION LAYER ---
const chunks: (string | Buffer)[] = [];
if (securedPayload.length === 0) {
    chunks.push(useRaw ? Buffer.alloc(0) : "");
} else {
    for (let i = 0; i < securedPayload.length; i += this.packetSize) {
        if (typeof securedPayload === "string") {
            chunks.push(securedPayload.substring(i, i + this.packetSize));
        } else {
            chunks.push(securedPayload.subarray(i, i + this.packetSize));
        }
    }
}

const packetCount = chunks.length;
const topic = `${prefix}${address}`;

for (let i = 0; i < packetCount; i++) {
    const packet = {
        header: {
            srcAddress: this.localAddress,
            srcPort: srcPort,
            dstAddress: address,
            dstPort: port,
            packetCount: packetCount, // jumlah total chunk
            packetIndex: i,           // chunk ke-i (0-based)
            dataSize: securedPayload.length, // ukuran TOTAL payload
            packetHeaderFlag: flag,

```

```

        forwarded: 0,
      },
      payload: chunks[i],
    });
    const payloadBuffer = protocol.pack(packet);
    this.txBytes += payloadBuffer.length;
    // publish ke topic (atau loopback lokal jika alamat = node ini)
  }

```

Reassembly — merging chunks (SimpleMQTNLDriver.handleIncomingMessage())

```

// --- REASSEMBLY LAYER ---
const key = `${packet.header.srcAddress}:${packet.header.srcPort}->${packet.header.dstAddress}:${packet.header.dstPort}`;

if (!this.receivedPackets.has(key)) {
  this.receivedPackets.set(key, {
    total: packet.header.packetCount,
    received: new Map(),
    timestamp: Date.now(),
  });
}

const session = this.receivedPackets.get(key)!;
session.received.set(packet.header.packetIndex, packet.payload);
session.timestamp = Date.now();

if (session.received.size < session.total) {
  return; // belum lengkap — tunggu chunk berikutnya
}

// Semua chunk tiba!
this.receivedPackets.delete(key);

const entries = Array.from(session.received.entries()).sort((a, b) => a[0] - b[0]);
const assembledPayload = Buffer.isBuffer(entries[0][1])
  ? Buffer.concat(entries.map(e => e[1] as Buffer))
  : entries.map(e => e[1] as string).join("");

```

30 second TTL — expired session cleanup

```

private cleanupExpiredPackets() {
  const now = Date.now();
  const ttl = 30000; // 30 detik untuk chunk yang ditinggalkan
  for (const [key, session] of this.receivedPackets.entries()) {
    if (now - session.timestamp > ttl) {
      this.logger.warn(`[REASSEMBLY] Dropped incomplete packet from ${key} (TTL Expired)`);
      this.receivedPackets.delete(key);
    }
  }
}

```

Exercises / Practice

1. Read `src/common/protocols/MQTNLProtocolJSON.ts` — understand the JSON packet structure (10-element array).
2. Read `src/common/protocols/MQTNLProtocolBinary.ts` — compute the binary header size for `srcAddress="tsix"` and `dstAddress="esp32S3"` (answer: 29 bytes).
3. Read `src/kernel/devices/SimpleMQTNLDriver.ts` — trace the TX path (send) and RX path (handleIncomingMessage), then find `packetSize` and `ttl`.
4. Read `wiki/mqtnl_binary_ota.md` — how Binary is used for byte-exact OTA.
5. Explain why Binary v1.1 deliberately has no encryption.
6. Experiment: send a payload larger than 32KB from an application, observe the `[FRAGMENTATION]` and `[REASSEMBLY]` logs.

References

- [wiki/mqtnl-ota.md](#), [wiki/mqtnl_binary_ota.md](#) — OTA via MQTNL
 - [wiki/course/00-overview.md](#) §7
 - [src/common/protocols/IMQTNLProtocol.ts](#), [MQTNLProtocolJSON.ts](#), [MQTNLProtocolBinary.ts](#)
-

Module 16 — done. Part VI complete. Continue to [Module 17 — PixelSpace Protocol](#).

RFC-TSIX-EDU-002 | Seventeenth module of the TSIX curriculum. Understand the GUI data contract: the Worker→Kernel→DOME→Browser data flow, GUIRegistry as the window ownership authority, and the security sanctions.

PixelSpace is the **constitution of TSIX's GUI**. Its contract (`GUITypes.ts`) must not be changed carelessly — every layer above it (DOME, Emerald, Cashew, Asteracea) depends on the shape of this data.

Learning Objectives

- ☐ Explain the UI mounting and event data flow
- ☐ Explain the role of GUIRegistry (auth wid↔pid)
- ☐ Mention the main GUIActions
- ☐ Explain the security sanctions (SIGKILL / SIGSEGV)
- ☐ Explain why the kernel overrides `payload.pid`

Core Concepts

Data flow

```
Worker (app) → GUI_REQ (61) → Kernel (GUIRegistry auth pid↔wid)
→ DOME daemon (WS broadcast) → Browser (DOM)
Browser → event → DOME → Kernel (SEND_MSG) → Worker (callback)
```

UI mounting flow

```
Worker App → GUI_REQ (MOUNT_NODE) → Kernel → gui_request event → DOME → WS → Browser (createElement)
```

Event flow (click/input)

```
Browser → WS → DOME → SEND_MSG ke pid → Kernel → ipc_message event → Worker → callback → updateProps
```

GUIAction (GUI operations)

Action	Purpose
CREATE_WINDOW / DESTROY_WINDOW	Create / destroy a window
MOUNT_NODE / UNMOUNT_NODE	Mount / unmount an element
UPDATE_PROPS	Change element properties
MINIMIZE_WINDOW / RESTORE_WINDOW	Hide / restore
MAXIMIZE_WINDOW / UNMAXIMIZE_WINDOW	Fullscreen / normal

GUIRegistry (kernel)

The **single** authority for window ownership:

- `wid → pid` (primary map) + `pid → Set<wid>` (reverse map, fast lookup on exit)
- Z-index auto-increment (starts at 100)
- `registerDaemon(pid)` — only "gued" may receive forwarded GUI_REQ
- Automatic cleanup when a process dies (`destroyAllForPid`)

Security

Violation	Sanction
Malformed payload format	SIGKILL — process is killed
Accessing a window owned by another PID	SIGSEGV — segmentation fault

[!IMPORTANT] **The kernel always overrides `payload.pid`**. Never trust the `pid` sent by the application — the kernel sets the real identity from the process context.

Source Code

File	Role
<code>src/common/GUITypes.ts</code>	Data contract (IDOMNode, IGUIPayload, GUIAction, IBrowserEvent, IGUIEventIPC)
<code>src/kernel/GUIRegistry.ts</code>	Window authority + auth
<code>src/kernel/Syscalls.ts</code>	GUI_REQ handler + security
<code>src/mirror/root/ps-sample1.ts</code>	Raw protocol practice

Snippet (code level)

GUIRegistry.createWindow

```
public createWindow(wid: string, pid: number, title: string = "Untitled"): IWindowEntry {
  if (this.windows.has(wid)) {
    throw new Error(`GUIRegistry: Window '${wid}' already exists.`);
  }
  const entry: IWindowEntry = { wid, pid, title,
    zIndex: this.nextZIndex++, focused: true, createdAt: Date.now() };
  this.windows.set(wid, entry);
  // Update reverse map pid → Set<wid>
  if (!this.pidToWids.has(pid)) this.pidToWids.set(pid, new Set());
  this.pidToWids.get(pid)!.add(wid);
  // Defocus window lain
  return entry;
}
```

Exercise / Practice

1. Read `src/common/GUITypes.ts` — understand the entire "constitution" interface.
2. Read `src/mirror/root/ps-sample1.ts` — practice the raw protocol without a toolkit.
3. Run a GUI app then `ps` — find the `gued/DOME` PID. Read `wiki/identity_guid_ipc_walkthrough.md` for the identity flow.
4. Modify a window owned by another PID via `GUI_REQ` — observe `SIGSEGV`.

References

- `wiki/PIXELSPACE_DEVELOPER_GUIDE.md` §1-2, 10
- `wiki/course/00-overview.en.md` §10
- `src/common/GUITypes.ts`, `src/kernel/GUIRegistry.ts`, `src/kernel/Syscalls.ts`

Module 17 — complete. Continue to [Module 18 — DOME Engine](#).

RFC-TSIX-EDU-002 | Eighteenth module of the TSIX curriculum. Understand DOME: WebSocket relay + primitive DOM producer + compositor (titlebar, drag, resize, focus, replay).

DOME is the TSIX display server — analogous to X11/Wayland, but rendered as **DOM in the host browser** via WebSocket :8080. The server is monolithic (single daemon, relay + compositor) because of drag/resize latency considerations (a separate compositor = ~2–4ms overhead). The browser client is instead modular: `dome.ts` serves static `dome-client-*.js` modules via `/dome/*.js`.

Learning Objectives

- ☐ Explain DOME's role as a relay + compositor
- ☐ Explain the DOME specification & features
- ☐ Explain the monolithic vs separate-compositor design tradeoff
- ☐ Explain the role of `windowStates` and replay
- ☐ Explain the enforcement of the `maximizable` flag on all maximize paths
- ☐ Explain navigation protection, per-app traffic, DDC modules, and boot readiness
- ☐ Explain the DOME ↔ Kernel ↔ Browser relationship

Core Concepts

Specification

Aspect	Description
Location	<code>src/mirror/opt/dome/dome.ts</code> (server) + thin <code>dome-client.html</code> + <code>dome-client-*.js</code> modules (browser)
Port	WebSocket :8080
Role	Display server (Ring 4 daemon)
Model	WS relay + primitive DOM producer + compositor

Features

- Window registry, Z-index, event relay
- State Relay** (`windowStates`): all `MOUNT_NODE` & `UPDATE_PROPS` are stored, then re-sent when the browser reconnects (F5)
- State Pruning** (`pruneWindowState`): when a node is unmounted, DOME cleans up the state — prevents orphan state from leaking into replay
- Orphan Discard** (browser-side): `MOUNT_NODE` whose parent is not found is discarded — prevents dialog/modal residue after F5
- Overlay Layer Search**: `findElementById` searches 3 levels — `win.el` → `__global_start_menu__` → `__tsix_overlay_layer__`
- Modular client**: browser logic is split into static modules `dome-client-core.js`, `-term.js`, `-codemirror.js`, `-chart.js`, `-dom.js`, `-windows.js`, `-ui.js`, `-ddc.js`, `-res.js`; `dome.ts` reads them from VFS `/opt/dome/*.js` at startup and serves them via `/dome/*.js`
- maximizable flag**: honored on all maximize paths — double-click titlebar, "Maximize" context menu, server guard `maximize_window`, and syscall guard `MAXIMIZE_WINDOW`
- Maximize while minimized**: the window is first shown at its last position before the rect is measured; restore uses `_origRect` if `_savedRect` is zero
- Navigation protection**: refresh (F5/Ctrl+R/Cmd+R) and back/forward require confirmation (`protectNavigation()` + `beforeunload`)

- **Per-app traffic accounting:** `TRAFFIC_QUERY` excludes the asking app's own traffic and reports `selfTxBytes / selfTxPkts`
- **DDC module:** `dome-client-ddc.js` hosts Native-JS apps (`Fabric.js` / `Three.js` via CDN) in a Shadow DOM; DOME relays `DDC_MSG/DDC_RESIZE/DDC_STOP`; `destroyDDCByWid(wid)` is called when a window is closed
- **Boot readiness:** DOME writes `/var/run/dome.ready`; `rc.local` polls the marker (10s timeout, 200ms interval) before starting Asteracea — replaces `sleep 1s`
- **`ensureListener()`**: attaches a listener once per element per event — replaces the `cloneNode` pattern that dropped old listeners

Design tradeoff: why monolithic?

	Separate compositor (X11-style)	Monolithic DOME
Overhead	~2-4ms per drag/resize operation	direct, no extra IPC
Advantage	modularity	low latency

Because the main target is **latency-sensitive drag/resize/focus interaction**, DOME chose to be monolithic. There is a design note & refactor plan in the wiki.

Replay Flow

```

Browser F5 → WebSocket reconnect
→ DOME kirim CREATE_WINDOW (semua window aktif)
→ DOME kirim semua stored states (MOUNT_NODE + UPDATE_PROPS)
→ Jika window sedang maximized, DOME kirim MAXIMIZE_WINDOW ulang
→ Browser build ulang UI + terima event dari user
  
```

Snippets (code level)

State pruning (condensed)

```

const allIds = new Set<string>([targetId]);
for (const state of states) {
  if (state?.type === "MOUNT_NODE" && state.node) {
    const nodeIds = new Set<string>();
    collectNodeIds(state.node, nodeIds);
    if (nodeIds.has(targetId)) for (const id of nodeIds) allIds.add(id);
  }
}
// Hapus MOUNT_NODE (parent atau node di dalam tree) & UPDATE_PROPS (target exact match)
  
```

Browser orphan discard

```

if (parent) {
  parent.appendChild(domEl);
} else {
  console.log('[DEBUG] Orphan discarded:', node.id, targetId); // DISCARD
}
  
```

Maximize guard — all paths honor the `maximizable` flag

```

// dome-client-windows.js – double-click titlebar & handleMaximizeWindow
if (!w._isMaximized && w.maximizable === false) return; // titlebar dbl-click
if (win.maximizable === false) return;                  // handleMaximizeWindow
  
```

```
// dome.ts — guard otoritatif sisi server (event browser + syscall)
if (event.eventType === "maximize_window") {
  if (entry.maximizable === false) return;
}
// syscall GUIAction.MAXIMIZE_WINDOW
if (windows.get(wid)?.maximizable === false) break;
```

Safe restore — fallback when the saved rect is zero

```
// dome-client-windows.js — handleRestoreWindow
let saved = win._savedRect;
if (saved && (saved.width <= 0 || saved.height <= 0)) {
  saved = win.el._origRect || null; // jangan restore ke (0,0,0,0)
}
```

ensureListener — listeners are not duplicated and not lost

```
// dome-client-dom.js — dipakai buildDOM() & handleUpdateProps()
function ensureListener(el, event, listener) {
  if (!el.__tsixL) el.__tsixL = Object.create(null);
  if (!el.__tsixL[event]) {
    el.addEventListener(event, listener);
    el.__tsixL[event] = true;
  }
}
```

Per-app traffic — the observer does not count itself

```
// dome.ts — TRAFFIC_QUERY meng-exclude traffic app penanya
const self = appTraffic.get(payload.pid) || { txBytes: 0, txPkts: 0 };
const stats = {
  ...wsTraffic,
  txBytes: Math.max(0, wsTraffic.txBytes - self.txBytes),
  txPkts: Math.max(0, wsTraffic.txPkts - self.txPkts),
  selfTxBytes: self.txBytes, // traffic milik app penanya
  selfTxPkts: self.txPkts,
};
```

DDC cleanup — stop the RAF loop when the window is closed

```
// dome-client-windows.js — handleDestroyWindow (safety net)
if (typeof TSIX.destroyDDCByWid === "function") {
  TSIX.destroyDDCByWid(wid); // stop semua runtime DDC window ini
}
```

Practice / Hands-on

1. Open `http://localhost:8080` — observe the DOME client in the browser.
2. Press F5 while several windows are open — observe the navigation confirmation prompt and the state replay.
3. Read `src/mirror/opt/dome/dome.ts` — find `windowStates`, `pruneWindowState`, the `maximizable` guard, the `DDC_*` relay, and `TRAFFIC_QUERY`.
4. Read `src/mirror/opt/dome/dome-client-windows.js` — find `handleMaximizeWindow` (maximize-while-minimized) and `handleRestoreWindow`.
5. Read `src/mirror/opt/dome/dome-client-core.js` — find `protectNavigation()`.
6. Read `src/mirror/opt/dome/dome-client-ddc.js` — find `initDDC` and `destroyDDCByWid`.
7. Read `src/mirror/opt/dome/dome-client-dom.js` — find `ensureListener`.
8. Read `src/mirror/etc/rc.local.ts` — find the polling of `/var/run/dome.ready` before starting Asteracea.

References

- [wiki/PIXELSPACE_DEVELOPER_GUIDE.md](#) §2, §11
 - [wiki/course/00-overview.en.md](#) §10
 - [wiki/changelogs/dome.md](#) — DOME change history
 - `src/mirror/opt/dome/dome.ts` + `src/mirror/opt/dome/dome-client-*.js`
 - `src/mirror/etc/rc.local.ts` — polling of `/var/run/dome.ready`
-

Module 18 — done. Continue to [Module 19 — Emerald Widget Toolkit](#).

RFC-TSIX-EDU-002 | Module nineteen of the TSIX curriculum. Understand the GUI toolkit layer on top of the stable protocol: the Screen wrapper, factory functions, and self-rendering connected widgets.

Emerald (`@tsix/emerald`) is the GTK/Qt of TSIX. All UI is built as a **Virtual DOM Tree** (`IDOMNode`) and sent via syscall — applications have **no access** to `document`, `window`, or any browser API.

Learning Objectives

- ☐ Explain the position of Emerald in the GUI stack
- ☐ Explain factory functions (`div`, `button`, `input`, ...)
- ☐ Explain the Screen wrapper & the mount → bind → loop pattern
- ☐ Explain connected widgets (self-rendering), including `ConnectedDataGrid`
- ☐ Explain `ensureListener()` — listener props persistent across `UPDATE_PROPS` (`cloneNode` bug fixed)

Core Concepts

Position in the architecture

BROWSER → DOME (display server) → KERNEL (auth) → EMERALD (kamu di sini)

Basic app structure

```
import { Program } from "@tsix/Application";
import { Screen, div, h1, paragraph } from "@tsix/emerald";

export const main = Program(async (args: string[]) => {
  // 1. Buat Screen (jendela)
  // 2. Bangun Virtual DOM tree dengan factory functions
  // 3. Mount, bind event, loop
});
```

Factory functions

Factories produce `IDOMNode`. The full list is in `src/mirror/lib/emerald.ts`:

- Basic: `text()`, `div()`, `span()`, `h1()/h2()/h3()`, `paragraph()`, `button()`, `input()`, `textarea()`, `selectBox()`, `image()`
- IoT: `lineChart()`, `radialGauge()`, `verticalGauge()`, `sevenSegment()`, `indicatorLamp()`, `toggleSwitch()`, `sensorCard()`, `relayCard()`, `badge()`, `taskbarButton()`
- Table: `dataGrid()` (static), `slider()`

`alert()`, `confirm()`, `question()`, `openFileDialog()`, `saveFileDialog()` are **not factories** — they are methods on `Window/Screen` that show modal overlay dialogs, not functions that produce `IDOMNode`.

Basic pattern: Mount → Bind → Loop

```
const screen = new Screen({ title: "App", width: 400, height: 300 });
await screen.mount(
  div({ id: "root", style: { padding: "16px" } },
    h1({ text: "Hello" }),
    button({ id: "btn", text: "Klik" }),
  ),
);
// bind event → updateProps (di-batch & auto-flush) → loop
```

```
await screen.on("btn", "click", async () => {
  await screen.update("btn", { text: "✅ Diklik!" });
});
```

Connected widgets (self-rendering)

Connected widgets render themselves and update the screen on their own — a higher-level pattern than plain factories. The common pattern is `build()` → `mount(screen)` → `setData()/setValue()`, using **targeted updates** (not a full `setContent()`).

The `Connected*` classes in `emerald.ts`:

- `ConnectedLineChart`, `ConnectedRadialGauge`, `ConnectedSevenSegment`, `ConnectedIndicatorLamp`
- `ConnectedVerticalGauge`, `ConnectedToggle`, `ConnectedSensorCard`, `ConnectedRelayCard`
- `ConnectedDataGrid` — interactive data table: asc/desc sort, column resize (native drag), row selection based on stable keys, incremental `appendData()` & `maxRows` option. One scroll container (sticky `th` header) — horizontal & vertical scroll automatically stay in sync.

[!IMPORTANT] **Listener props & `ensureListener()` (`cloneNode` bug fixed).** Four listener props — `onClickId`, `onContextMenuId`, `onInputId`, `onKeyDownId` — are installed by the DOME client through the `ensureListener()` helper in `src/mirror/opt/dome/dome-client-dom.js` (used in `buildDOM()` and `handleUpdateProps()`), **once per element per event type** (tracked via `el.__tsixL`). Previously `handleUpdateProps` cloned nodes to "clean up old listeners" — `cloneNode` does not copy listeners, so if one batch of `UPDATE_PROPS` carries several listener props at once (e.g. `onInputId` + `onKeyDownId` on a password field), a newly installed listener could be lost. With `ensureListener`, listeners are **persistent across `UPDATE_PROPS`** — not lost and not duplicated. Best practice still applies: set listener props (`onClickId`, `onInputId`, etc.) **at mount time** (in `build()/mount()`), then register callbacks via `screen.on()` / `win.bindHandler()`.

Source Code

File	Role
<code>src/mirror/lib/emerald.ts</code>	Widget toolkit @tsix/emerald
<code>src/mirror/lib/theme.ts</code>	Theme
<code>src/mirror/lib/cashew.ts</code>	Layer on top of Emerald (Module 20)
<code>src/mirror/opt/dome/dome-client-dom.js</code>	DOME client DOM engine — <code>buildDOM()</code> , <code>handleUpdateProps()</code> , <code>ensureListener()</code>
<code>src/mirror/root/ps-sample2.ts</code> , <code>ps-sample3.ts</code>	Practice

Exercises / Practice

1. Read `wiki/emerald-in-a-nutshell.md` — work through Hello World up to the complete case study.
2. Build a form with an input + button; bind a click event that changes the text.
3. Use `alert()/confirm()` — observe how it appears as a window overlay.
4. Read `src/mirror/lib/emerald.ts` — find the `Screen` and `setContent` implementations.
5. Build a `ConnectedDataGrid` with column sort & resize; use `appendData()` for real-time data and `maxRows` to limit rows.

References

- `wiki/emerald-in-a-nutshell.md`, `wiki/cashew-in-a-nutshell.md`
- `wiki/PIXELSPACE_DEVELOPER_GUIDE.md` §5-7
- `src/mirror/lib/emerald.ts`, `src/mirror/lib/theme.ts`

Module 19 — done. Continue to [Module 20 — Cashew Component Framework](#).

RFC-TSIX-EDU-002 | Twentieth module of the TSIX curriculum. Understand the OOP/Delphi-style layer above Emerald: `TForm`, `TButton`, `TEdit`, auto-bind lifecycle, and complex widgets.

Cashew (@tsix/cashew) adopts the component pattern of **Delphi / Turbo Pascal** — components are **stateful classes** with properties & events, not nested functions. It is the VCL analogue on top of GTK.

Learning Objectives

- ☐ Explain the OOP/Delphi-style philosophy of Cashew
- ☐ Explain the `TForm.run()` lifecycle (auto-bind)
- ☐ Explain the basic components: `TForm`, `TPanel`, `TLabel`, `TButton`, `TEdit`
- ☐ Explain why `onClickId`/`onInputId` are set at build time (mount-time)
- ☐ Name the complex widgets (`TChart`, `TSevenSegment`, `TSensorCard`, etc.)

Core Concepts

Philosophy

In Emerald you write UI with nested functions (`div([button(...)])`). In Cashew, you create **class objects**:

```
const form = new TForm("My App", 400, 300);
const btn = new TButton("btn-click");
btn.caption = "Klik";
btn.onClick = () => { count++; lblCounter.caption = "Count: " + count; };
form.add(btn);
await form.run();
```

TForm.run() lifecycle

`run()` has an **auto-bind lifecycle**: `bindEventHandler` + `refresh` per component. After `run()`, property changes (`caption`, `text`, etc.) automatically sync to the screen.

The sequence in `TForm.run()` (`src/mirror/lib/cashew.ts`):

- Theme** — load the active theme before building so the CSS color variables are correct.
- Build & mount** — build the `IDOMNode` tree via `this.build()` then `_screen.mount(...)`.
- Auto-bind** — DFS traversal over all children, call `comp.bindEventHandler(screen)` (event registration).
- Auto-refresh** — DFS traversal over all children, call `await comp.refresh(screen)` (render dynamic data, e.g. `TListBox`, `TDataGrid`).
- Setup** — call `await onSetup(screen)` when set (extra binding).
- Event loop** — `_screen.loopUntilClose()` waits until the window is closed.

```
sequenceDiagram
    participant App as Aplikasi (main)
    participant Form as TForm.run()
    participant Comp as Komponen (TButton, TEdit, TDataGrid, ...)
    participant Screen as Screen (Emerald)

    App->>Form: await run()
    Form->>Form: loadCurrent() - theme
    Form->>Screen: mount(build())
    loop auto-bind (DFS)
        Form->>Comp: bindEventHandler(screen)
        Comp->>Screen: screen.on(id, "click"/"input", cb)
    end
```

```

loop auto-refresh (DFS)
  Form->>Comp: await refresh(screen)
  Comp->>Screen: setContent / setData
end
Form->>App: await onSetup(screen)
Form->>Screen: loopUntilClose()
loop event loop
  Screen->>Comp: event (click / input / timer)
  Comp->>Screen: screen.update(id, props)
end

```

Component Hierarchy

All components derive from the base class `TComponent` in `src/mirror/lib/cashew.ts`:

```

graph TD
  TC[TComponent] --> TForm
  TC --> TPanel
  TC --> TLabel
  TC --> TButton
  TC --> TEdit
  TC --> TMemo
  TC --> TCheckBox
  TC --> TListBox
  TC --> TRadioButton
  TC --> TComboBox
  TC --> TStatusBar
  TC --> TLineChart
  TC --> TRadialGauge
  TC --> TSevenSegment
  TC --> TIndicatorLamp
  TC --> TToggleSwitch
  TC --> TVerticalGauge
  TC --> TSensorCard
  TC --> TRelayCard
  TC --> TSlider
  TC --> TTimer
  TC --> TChart
  TC --> TDataGrid

```

[!NOTE] **Not classes.** `TDialogs` is a **static** utility (not a `TComponent`). `HStack`, `VStack`, `Spacer`, `TScrollBox`, `TFlowPanel`, `TGridPanel`, `TSplitHorizontal`, `TSplitVertical`, `TGroupBox` are **functions** that return `TComponent` — quick layout without new subclasses. Align constants (`alTop`, `alClient`, etc.) are available for `style.position`.

Why events are set at build time (mount-time)

`onClickId`/`onInputId` are written directly into `props` in the constructor or `build()` (example: `TButton` sets `this.props.onClickId = id`). When DOME mounts the node, the listener is attached once. Adding listeners via `app.on()` after mount risks colliding with DOME's `cloneNode` pattern, which **does not copy listeners** — see Module 19.

Basic components

Component	Function
<code>TForm</code>	Main window (<code>add()</code> , <code>alert()</code> , <code>confirm()</code> , <code>screen</code> , <code>onSetup</code> ; options <code>maximizable</code> / <code>resizable</code> / <code>fullscreen</code> / <code>frameless</code>)
<code>TPanel</code>	Container (with/without style)
<code>TLabel</code>	Text (<code>caption</code>)
<code>TButton</code>	Button (<code>onClick</code> , <code>caption</code> , <code>enabled</code>)
<code>TEdit</code>	Text input (<code>onInput</code> , <code>text</code> , <code>placeholder</code>)

Component	Function
<code>TMemo</code> , <code>TCheckBox</code> , <code>TListBox</code> , <code>TStatusBar</code> , <code>TRadioButton</code> , <code>TComboBox</code>	Others
<code>TScrollBar</code> , <code>TFlowPanel</code> , <code>TGridPanel</code> , <code>TSplitHorizontal/Vertical</code> , <code>TGroupBox</code>	Layout
<code>HStack</code> , <code>VStack</code> , <code>Spacer</code>	Quick layout

Complex & IoT widgets

Component	Function
<code>TDialogs</code>	Static dialog utility: <code>alert</code> , <code>confirm</code> , <code>input</code> , <code>openFile</code> , <code>saveFile</code>
<code>TTimer</code>	Delphi-style interval timer; auto-cleanup when the form closes (<code>onTimer</code>)
<code>TChart</code>	Real-time multi-series chart (Lightweight Charts via IPC DOME)
<code>TLineChart</code>	SVG line chart (spline, fill, scroll animation)
<code>TRadialGauge</code>	Circular speedometer-style gauge
<code>TSevenSegment</code>	LED 7-segment display (options <code>scale</code> , <code>height</code>)
<code>TIndicatorLamp</code>	ON/OFF indicator lamp with glow
<code>TToggleSwitch</code>	Clickable ON/OFF switch
<code>TVerticalGauge</code>	Vertical glass tube; value text dual-layer masking
<code>TSensorCard</code>	IoT sensor card with progress bar
<code>TRelayCard</code>	ON/OFF relay card
<code>TSlider</code>	Horizontal range slider

TDataGrid — one scroll container

`TDataGrid` wraps Emerald's `ConnectedDataGrid` (`src/mirror/lib/emerald.ts`). A 2026-08-03 change makes the grid use a **single scroll container**:

- **One scroll container** — header table + `<style>` + body table live inside one `body-scroll` (`overflow: auto`). The header is pinned via **th sticky** (`position: sticky; top: 0; z-index: 2`), the same pattern as the static `dataGrid()` .
- **No manual compensation** — the `thead-scroll` wrapper, relayed `scrollLeft`, `scrollbar-gutter`, and `calc(100% - scrollbarWidth)` were removed. Horizontal & vertical scrolling sync automatically; header width == body width == container width.
- **Column resize** — the resize scope is the grid wrapper `table.closest(".tsix-dgrid")` (not `table.parentElement`). The drag handle changes the width of `<col>` in **all colgroups** (header + body) at once.
- **col_resized** — when dragging finishes, the browser sends `col_resized` with `targetId = grid id` (wrapper `.tsix-dgrid`) → `ConnectedDataGrid` stores the width and re-applies it on every render (sort/refresh/setData).

[!IMPORTANT] **Mount-time event.** Set `onClickId/onInputId` at build time (mount-time) — not in `app.on()` — to avoid the `cloneNode` bug (see Module 19).

Flow / How It Works

1. Create a `TForm` object — two forms: sequential `new TForm(title, w, h, ...)` or object literal `new TForm({ title, ... })` .
2. Add components via `form.add(...)` — this forms a parent-child tree.
3. Set properties (`caption`, `text`, `value`) and event handlers (`onClick`, `onInput`, `onChange`, `onTimer`).

4. Call `await form.run()` — build → mount → auto-bind → auto-refresh → `onSetup` → loop.
5. Change properties at any time; bound components automatically call `screen.update(...)` to the screen.
6. Window closes → `loopUntilClose()` finishes; managed timers are cleaned up automatically.

Source Code

- `src/mirror/lib/cashew.ts` — the whole framework (TComponent, TForm, TDialogs, basic components, layout, IoT widgets, TDataGrid)
- `src/mirror/lib/emerald.ts` — `ConnectedDataGrid` (grid rendering & column width state)
- `src/mirror/opt/dome/dome-client-dom.js` — DOM build, `data-col-resize`, `col_resized` event
- `wiki/cashew-in-a-nutshell.md`, `wiki/emerald-in-a-nutshell.md` — summary

Snippet (code level)

Cashew quick start

```
import { Program } from "@tsix/Application";
import { TForm, TLabel, TButton, TStatusBar } from "@tsix/cashew";

export const main = Program(async () => {
  const form = new TForm("My App", 400, 300);
  let count = 0;

  const lblCounter = new TLabel("counter");
  lblCounter.caption = "Count: 0";
  form.add(lblCounter);

  const btnClick = new TButton("btn-click");
  btnClick.caption = "Klik";
  btnClick.onClick = () => { count++; lblCounter.caption = "Count: " + count; };
  form.add(btnClick);

  const status = new TStatusBar("status");
  status.text = "✅ Siap";
  form.add(status);

  // Tidak perlu bind manual — TForm.run() memanggil bindEventHandler()
  // untuk setiap komponen secara otomatis (auto-bind lifecycle).
  await form.run();
});
```

TForm.run() — auto-bind lifecycle (from cashew.ts)

```
async run(): Promise<void> {
  // 1. Theme — load dulu biar komponen pake warna yang benar
  try {
    const t = await ensureTheme();
    await t.loadCurrent();
    t.watch();
  } catch (_) { /* theme opsional — skip jika gagal */ }

  // 2. Build & mount ke Screen
  this._screen = new Screen(opts); // title, width, height, maximizable, ...
  await this._screen.mount(this.build());
  // (opsional) kirim WINDOW_THEME ke DOME untuk CSS variables

  // 3. Auto-bind: DFS — panggil bindEventHandler() untuk semua komponen
  const autoBind = (comp: TComponent) => {
    comp.bindEventHandler(this._screen);
    for (const child of comp.children) autoBind(child);
  };
  for (const child of this.children) autoBind(child);

  // 4. Auto-refresh: DFS — panggil refresh() (misal TListBox, TDataGrid)
```

```

const autoRefresh = async (comp: TComponent) => {
  await comp.refresh(this._screen);
  for (const child of comp.children) await autoRefresh(child);
};
for (const child of this.children) await autoRefresh(child);

// 5. Setup custom – setelah bind & refresh
if (this.onSetup) await this.onSetup(this._screen);

// 6. Event loop sampai window ditutup
await this._screen.loopUntilClose();
}

```

TComponent — bindEventHandler & refresh (base no-op)

```

export class TComponent {
  // Daftarkan event handler ke Screen. Otomatis dipanggil oleh TForm.run().
  bindEventHandler(screen: Screen): void {
    // Base: no-op – subclass override untuk register event
  }

  // Update tampilan setelah mount. Otomatis dipanggil oleh TForm.run().
  async refresh(screen: Screen): Promise<void> {
    // Base: no-op – subclass override untuk render dinamis
  }

  build(): IDOMNode {
    return {
      id: this.id,
      tag: this.tag,
      props: { ...this.props, style: { ...this.style } },
      children: this._children.map((c) => c.build()),
    };
  }
}

```

Widget example — TButton & TEdit (from cashew.ts)

```

// TButton – bind onClick saat auto-bind
// constructor: this.tag = "button"; this.props.onClickId = id;
export class TButton extends TComponent {
  public onClick: (() => void) | null = null;
  bindEventHandler(screen: Screen): void {
    this._screen = screen;
    if (this.onClick) screen.on(this.id, "click", this.onClick);
  }
}

// TEdit – bind onInput saat auto-bind
// constructor: this.tag = "input"; this.props.onInputId = id;
export class TEdit extends TComponent {
  public onInput: ((value: string) => void) | null = null;
  bindEventHandler(screen: Screen): void {
    this._screen = screen;
    if (this.onInput) {
      screen.on(this.id, "input", (ev: any) => {
        this.onInput!(ev?.value || "");
      });
    }
  }
}

```

Widget example — TDataGrid (from cashew.ts)

```

// Penggunaan di app
const grid = new TDataGrid(
  "sensor",
  [

```

```

    { key: "node_id", label: "Node", width: 140 },
    { key: "value", label: "Nilai", width: 80, align: "right" },
    { key: "timestamp", label: "Waktu", width: "40%" },
  ],
  [],
  { height: 300 },
);
form.add(grid);
grid.onSort = (key, dir) => { reloadSorted(key, dir); };
grid.onRowClick = (index, record) => { openDetail(record); }; // index = row-key stabil

// Data inkremental – hanya baris baru yang dikirim ke browser (hemat WS)
await grid.appendData(newRows);

```

```

// (dari cashew.ts – TDataGrid) auto-bind oleh TForm.run(): mount
// ConnectedDataGrid + daftarkan handler sort & klik row.
bindEventHandler(screen: Screen): void {
  this._screen = screen;
  void this.grid
    .mount(
      screen,
      (key, dir) => { if (this.onSort) this.onSort(key, dir); },
      (index, record) => { if (this.onRowClick) this.onRowClick(index, record); },
    )
    .catch(() => {});
}

// (dari cashew.ts – TDataGrid) auto-refresh: render data awal
async refresh(screen: Screen): Promise<void> {
  await this.grid.setData(this._data);
}

```

TTimer — Delphi-style interval

```

const timer = new TTimer("tmr-update", 1000);
timer.onTimer = () => { chart.pushData(Date.now(), sensor.value); };
timer.enabled = true;
form.add(timer); // bindEventHandler auto-start saat form.run()

```

Layout helpers

```

form.add(
  HStack(
    new TButton("btn-a"),
    new TButton("btn-b"),
    Spacer(), // flex: 1 – mendorong tombol berikutnya ke kanan
    new TButton("btn-c"),
  ),
);

// Dua panel bersebelahan dengan divider yang bisa di-drag
form.add(TSplitHorizontal(leftPanel, rightPanel, "lfr"));

```

Exercises / Practice

1. Read `wiki/cashew-in-a-nutshell.md` — follow the quick start and the layout components.
2. Build an app with `TForm` + `TPanel` + `TEdit` + `TListBox`: type in the `TEdit`, click the button, and the result goes into the `TListBox`.
3. Use `TDialogs.confirm()` for confirmation before deleting an item.
4. Read `src/mirror/lib/cashew.ts` — find the implementations of `TForm.run()`, `bindEventHandler()`, and `refresh()`.
5. Create a `TDataGrid` with `appendData()` for continuously growing data (log / telemetry).

References

- `wiki/cashew-in-a-nutshell.md`, `wiki/emerald-in-a-nutshell.md`
 - `wiki/course/00-overview.en.md` §10
 - `src/mirror/lib/cashew.ts`, `src/mirror/lib/emerald.ts`
 - `src/mirror/opt/dome/dome-client-dom.js`
 - Changelog: `wiki/changelogs/cashew.md`, `wiki/changelogs/dome.md`
-

Module 20 — done. Continue to [Module 21 — Asteracea & TDE](#).

RFC-TSIX-EDU-002 | Twenty-first module of the TSIX curriculum. Understand the TSIX Window Manager: an ordinary PixelSpace app (fullscreen frameless) that manages the lifecycle of other GUI apps.

Key point: **Asteracea is not part of the kernel/DOME** — it is an ordinary PixelSpace app (Ring 4, sandboxed worker) that runs fullscreen. It launches, controls, and manages the lifecycle of other GUI apps.

Learning Objectives

- ☐ Explain Asteracea's position in the architecture
- ☐ Explain the queue-based IPC philosophy (MessageBus)
- ☐ Explain the taskbar (pinned/running/foreign)
- ☐ Explain the launcher & fuzzy search
- ☐ Explain lifecycle events via `/opt/asteracea/wm-pid`

Core Concepts

Specification

Aspect	Description
Location	<code>src/mirror/opt/asteracea/asteracea.ts</code>
Mode	Fullscreen, frameless
Model	Queue-based IPC (MessageBus)
Protocol	PixelSpace Protocol via DOME
Privilege	Ring 4 (sandboxed worker)
Auto-start	<code>/etc/rc.local.ts</code> (after DOME is ready — poll <code>/var/run/dome.ready</code>)

Queue-based philosophy

All events enter a FIFO queue (MessageBus) and are processed one by one — preventing race conditions and starvation:

```
App A —(shell.send)→ Kernel —(ipc_message)→ MessageBus Queue → handlers
App B —(GUI_REQ)→ DOME —(ipc_message)→ MessageBus Queue → handlers
Browser —(click)→ DOME —(shell.send)→ MessageBus Queue → handlers
```

 Desktop notification flow: app → kernel → Asteracea → browser Source: wiki/diagram/ASTERACEA_WM-1.mmd

Components

- **Taskbar**: pinned / running / foreign windows
- **Launcher**: start menu + **fuzzy search** for apps
- **Login & wallpaper**
- **Desktop Context Menu (DCM)**
- **App State Manager**

IPC lifecycle events

The WM listens for `GUI_WINDOW_*` lifecycle events: `GUI_WINDOW_CREATED`, `GUI_WINDOW_MINIMIZED`, `GUI_WINDOW_RESTORED`, `GUI_WINDOW_MAXIMIZED`, `GUI_WINDOW_UNMAXIMIZED`, `GUI_WINDOW_CLOSED`, and `GUI_WINDOW_ERROR`. Other GUI apps communicate with the WM through `/opt/asteracea/wm-pid` (PID file) — Emerald broadcasts events to Asteracea.

Fixes & Latest Features

To stay in sync with the real code, the following behaviors were added:

- **Daemonize** — `main` starts with `shell.daemonize("Asteracea Window Manager")`. The process detaches from the `tty1` console; `stdio` is redirected to `/dev/null`. Logs still go to `/var/log/syslog` via VFS (`std.log/std.error`), and GUI rendering stays normal because it goes through DOME.
- **Taskbar icon & tooltip for foreign apps** — the `GUI_WINDOW_CREATED` handler forwards `payload.icon || "🖥️"` to `registerForeignApp()`. The icon is used for the taskbar button; `title` (the window title) becomes the tooltip via the `title` prop (translated to `data-tt` in DOME). Foreign apps can now show a custom icon.
- **GUI_WINDOW_MAXIMIZED / GUI_WINDOW_UNMAXIMIZED handlers** — both call `transitionTo(appId, "RUNNING")` and set the active taskbar style (same as `GUI_WINDOW_RESTORED`). WM state stays in sync after maximizing via the taskbar context menu; the next taskbar click becomes a minimize toggle.
- **Login without password prefill** — the password field is no longer filled with `value: "1"` (initialized as `loginPass = ""`). The old prefill would "stick" in front of the new password and make login always fail after `passwd` changes the password.
- **Wallpaper persistence** — before writing `/opt/asteracea/wallpaper/current-wp.b64`, `showWallpaperDialog` creates the `/opt/asteracea/wallpaper` folder (inside a try-catch). Previously the folder did not exist → `fs.writeFile` failed silently → blank wallpaper after reboot.

Source Code

File	Role
<code>src/mirror/opt/asteracea/asteracea.ts</code>	Window manager
<code>src/mirror/opt/asteracea/menu/*.menu</code>	Launcher menu configuration
<code>src/mirror/etc/rc.local.ts</code>	Auto-start DOME + Asteracea (poll <code>/var/run/dome.ready</code>)

Exercises / Practice

1. Log in to TSIX — observe the desktop: taskbar, launcher, wallpaper.
2. Open the launcher and type part of an app name — observe fuzzy search.
3. Read `wiki/ASTERACEA_WM.md` — learn about the MessageBus and the App State Manager.
4. Read `src/mirror/opt/asteracea/asteracea.ts` — find the `GUI_WINDOW_*` lifecycle handlers.

References

- `wiki/ASTERACEA_WM.md` — complete WM documentation
- `wiki/PIXELSPACE_DEVELOPER_GUIDE.md` §3, §9
- `wiki/course/00-overview.md` §10
- `src/mirror/opt/asteracea/asteracea.ts`, `src/mirror/opt/asteracea/menu/*.menu`

Module 21 — done. Continue to [Module 22 — State Replay & Persistence](#).

RFC-TSIX-EDU-002 | Module twenty-two of the TSIX curriculum. Understand how the UI is restored when the browser reconnects (F5): `windowStates`, `pruneWindowState`, orphan discard, `findElementById` across 3 layers, and navigation protection (`allowReload`).

F5 in the browser should **not clear the desktop**. Since navigation protection (DOME 2026-08-06), refresh requires confirmation first. DOME stores all `MOUNT_NODE` & `UPDATE_PROPS` in `windowStates` and resends them to the new browser after reload is confirmed. Pruning ensures no orphan state leaks into the replay.

Learning Objectives

- ☐ Explain the State Replay mechanism
- ☐ Explain `pruneWindowState` and why it is needed
- ☐ Explain browser-side orphan discard
- ☐ Explain `findElementById` across 3 layers
- ☐ Explain navigation protection (`allowReload`) and replay after reload
- ☐ Explain the `_savedRect` → `_origRect` fallback during restore
- ☐ Explain the bug scenarios that are prevented (dialog residue after F5)

Core Concepts

State Replay

`windowStates` = `Map<wid, state[]>` — stores all `MOUNT_NODE` & `UPDATE_PROPS` as "replay payloads".

Lifecycle (in `dome.ts`, server side):

- `CREATE_WINDOW` → `windowStates.set(wid, [])` — initializes the state history.
- `MOUNT_NODE` → push `{ type, wid, node, targetId }` to the array.
- `UPDATE_PROPS` → push `{ type, wid, targetId, props }` to the array.
- `UNMOUNT_NODE` → `pruneWindowState(wid, targetId)` then broadcast.
- `DESTROY_WINDOW` → `windowStates.delete(wid)` — the history is cleared.

Replay on reconnect (F5):

```
Browser F5 → WebSocket reconnect
→ DOME kirim CREATE_WINDOW (semua window aktif)
→ DOME kirim semua stored states (MOUNT_NODE + UPDATE_PROPS)
→ jika entry.isMaximized → kirim MAXIMIZE_WINDOW (status maximize dipulihkan)
→ DOME kirim tema terakhir (lastThemeColors)
→ Browser build ulang UI + terima event dari user
```

On the browser side, when the socket `onclose` fires, all windows from the old session are discarded (`state.windows.clear()`). Then `onopen` is ready to receive the replay from the server.

State Pruning (`pruneWindowState`)

When `UNMOUNT_NODE` is called, DOME cleans up the related state:

- Collect:** for each `MOUNT_NODE` state, gather ALL nodeIds from its tree (including children) via `collectNodeIds`.
- Filter:** remove the `MOUNT_NODE` if `state.targetId === targetId`, or if any nodeId in its tree equals `targetId`.
`UPDATE_PROPS` is only removed when `state.targetId === targetId` (exact match).

This prevents **child states** (from `setContent()`) from lingering after the parent is unmounted → prevents DOM residue on F5.

Browser-side orphan discard

If `handleMountNode` receives a `MOUNT_NODE` whose `targetId` is not found in the DOM, the node **is discarded** — not a fallback to `win.content`. This prevents:

- Wallpaper dialog child states from appearing on the desktop after F5
- Login overlay residue
- Orphan modal/dialog elements

Overlay layer search

`findElementById(wid, nodeId)` searches at 3 levels:

1. `win.el.querySelector([data-tsix-id=...])` — inside the window
2. `__global_start_menu__` — global start menu
3. `__tsix_overlay_layer__` — overlay layer (launcher, dialog)

This allows elements that were extracted to the overlay (e.g. `launcher-grid`) to still be found by mount/replay.

Navigation Protection (refresh confirmation)

Since DOME 2026-08-06, refresh no longer runs directly. The `protectNavigation()` IIFE in `dome-client-core.js` uses the `allowReload` flag:

- **F5 / Ctrl+R / Cmd+R** is intercepted in the capture phase → `preventDefault()` → `window.confirm("Yakin mau refresh TDE? ...")` dialog.
 - **OK** → sets `allowReload = true`, then `location.reload()`.
 - **Cancel** → the page stays intact, no reload.
- **back/forward, close tab, other navigation** → the `beforeunload` event shows the native browser dialog. It is skipped when `allowReload` is set (so an already-confirmed refresh does not prompt twice).
- `pointerdown` (once) marks interaction — prevents Chrome 91+ from disabling `beforeunload` on pages that have not been touched yet.

After a confirmed reload, **state replay still works**: DOME resends `CREATE_WINDOW` + stored states to the new WebSocket connection, so the desktop is fully restored.

Safe restore (`_savedRect` → `_origRect`)

When a window is restored (e.g. after minimize), `handleRestoreWindow` uses `_savedRect`. If `_savedRect` has zero dimensions (`width <= 0 || height <= 0`) — e.g. leftover from maximize while the window is still `display:none` — the code **falls back to** `win.el._origRect`. This prevents the window from being restored to `(0,0,0,0)` (top-left corner with odd dimensions).

Snippet (code level)

Pruning (`pruneWindowState`)

```
const collectNodeIds = (node: any, ids: Set<string>): void => {
  if (!node || typeof node !== "object") return;
  if (typeof node.id === "string" && node.id) ids.add(node.id);
  if (Array.isArray(node.children)) {
    for (const child of node.children) collectNodeIds(child, ids);
  }
};

const pruneWindowState = (wid: string, targetId: string): void => {
  const states = windowStates.get(wid) || [];
  const filtered = states.filter((state: any) => {
    if (state?.type === "MOUNT_NODE") {
      const nodeIds = new Set<string>();
      collectNodeIds(state.node, nodeIds);
      // hapus MOUNT_NODE jika targetId-nya = targetId, atau
```

```

    // nodeId di dalam tree-nya ada yang = targetId (termasuk child)
    if (state.targetId === targetId || nodeIds.has(targetId))
        return false;
    }
    if (state?.type === "UPDATE_PROPS") {
        if (state.targetId === targetId) return false; // exact match saja
    }
    return true;
});
windowStates.set(wid, filtered);
};

```

Browser orphan discard

```

if (targetId) {
    const parent = TSIX.findElementById(wid, targetId);
    if (parent) {
        parent.appendChild(domEl);
    } else {
        // Parent not found – discard (jangan fallback ke win.content)
    }
} else {
    win.content.appendChild(domEl);
}

```

Cross-layer search (`findElementById`)

```

function findElementById(wid, nodeId) {
    // 1) di dalam window milik app
    const win = state.windows.get(wid);
    if (win) {
        const el = win.el.querySelector(
            '[data-tsix-id="' + CSS.escape(nodeId) + '"]',
        );
        if (el) return el;
    }
    // 2) start menu global
    const gm = document.getElementById("__global_start_menu__");
    if (gm) {
        if (gm.getAttribute("data-tsix-id") === nodeId) return gm;
        const child = gm.querySelector(
            '[data-tsix-id="' + CSS.escape(nodeId) + '"]',
        );
        if (child) return child;
    }
    // 3) overlay layer (launcher, dialog)
    const overlay = document.getElementById("__tsix_overlay_layer__");
    if (overlay) {
        if (overlay.getAttribute("data-tsix-id") === nodeId) return overlay;
        const child = overlay.querySelector(
            '[data-tsix-id="' + CSS.escape(nodeId) + '"]',
        );
        if (child) return child;
    }
    return null;
}

```

Navigation protection (`protectNavigation`)

```

(function protectNavigation() {
    let allowReload = false; // true saat user sudah konfirmasi refresh

    // Tandai interaksi user – Chrome 91+ mematikan beforeunload
    // jika halaman belum pernah disentuh.
    document.addEventListener("pointerdown", function () { }, { once: true });

    const isRefreshKey = (e) =>
        e.key === "F5" ||

```

```

((e.key === "r" || e.key === "R") && (e.ctrlKey || e.metaKey));

document.addEventListener(
  "keydown",
  function (e) {
    if (!isRefreshKey(e) || e.repeat) return;
    e.preventDefault();
    const ok = window.confirm(
      "Yakin mau refresh TDE?\n\n" +
      "Semua window yang berjalan akan ditutup lalu di-restore " +
      "ulang dari server.\n\n" +
      "OK = refresh, Cancel = batal",
    );
    if (ok) {
      allowReload = true;
      window.location.reload();
    }
  },
  true, // fase capture
);

window.addEventListener("beforeunload", function (e) {
  if (allowReload) return; // refresh yang sudah dikonfirmasi
  e.preventDefault();
  e.returnValue = "";
});
})();

```

Restore fallback (`_savedRect` → `_origRect`)

```

let saved = win._savedRect;
// Safety: rect nol (mis. sisa maximize saat window masih hidden)
// → jangan restore ke (0,0,0,0), pakai posisi/ukuran asli.
if (saved && (saved.width <= 0 || saved.height <= 0)) {
  saved = win.el._origRect || null;
}

```

Exercises / Practice

1. Open several windows (including dialog/modal), then press F5 — a confirmation dialog appears. Choose **OK** — observe the UI restored via state replay. Choose **Cancel** — the desktop is unchanged.
2. Close a window that has a child dialog, then F5 (OK) — observe no dialog residue.
3. Read `src/mirror/opt/dome/dome.ts` — find `windowStates`, `pruneWindowState`, and the replay block on reconnect.
4. Read `src/mirror/opt/dome/dome-client-dom.js` — find the orphan discard in `handleMountNode`.
5. Read `src/mirror/opt/dome/dome-client-core.js` — find `findElementById` (3 layers) and `protectNavigation` (`allowReload`).
6. Read `src/mirror/opt/dome/dome-client-windows.js` — find the `_savedRect` → `_origRect` fallback in `handleRestoreWindow`.

References

- [wiki/PIXELSPACE_DEVELOPER_GUIDE.md §11](#)
- [wiki/course/00-overview.md §10](#)
- `src/mirror/opt/dome/dome.ts` (`windowStates`, `pruneWindowState`, `replay`)
- `src/mirror/opt/dome/dome-client-dom.js` (`orphan discard`)
- `src/mirror/opt/dome/dome-client-core.js` (`findElementById`, `protectNavigation`)
- `src/mirror/opt/dome/dome-client-windows.js` (`_savedRect` → `_origRect`)

Module 22 — complete. Part VII is done. Continue to [Module 23 — Development Workflow](#).

RFC-TSIX-EDU-002 | Twenty-third module of the TSIX curriculum. Understand the host↔VFS cycle: install, vfs-bootstrap, sync-vfs, vfs-pull, SYNC_TO_HOST, userlib-update, and SDK mirroring.

TSIX has two worlds: the **host** (`src/mirror` + `src/common` folders in the repo) and the **VFS** (file database, its path is read from `kernel.database` in `src/sysconfig.conf`, default `system.db`). Developers write on the host, then sync to the VFS. There is also the reverse path — applications inside the VFS write back to the host (root-only).

Learning Objectives

- ☐ Explain the host ↔ VFS cycle
- ☐ Distinguish `install`, `vfs-bootstrap`, `sync-vfs`, `vfs-pull`
- ☐ Explain `SYNC_TO_HOST` (syscall)
- ☐ Explain `userlib-update` (SDK mirroring)
- ☐ Explain the role of `tsconfig` paths (`@tsix/*`)

Core Concepts

Two worlds: host vs VFS

- Host** — the place to write code: `src/mirror/` (rootfs: `bin`, `lib`, `etc`, `sbin`, ...) and `src/common/` (framework: `SyscallCode`, `IPCTypes`, `GUITypes`).
- VFS** — the filesystem booted by the kernel: the file database `system.db` (BKFS/SQLite). The kernel reads its path from `kernel.database`.
- src/root/** — the result of `vfs-pull`: a copy of the VFS on the host for inspection / permanent changes.

All host scripts get the DB path via `scripts/lib/db-path.ts`, so its value is always in sync with the installed path.

[!IMPORTANT] Since `install.ts` exists, a fresh install **does not need** a separate `vfs:bootstrap` run — the installer already performs the whole rootfs sync itself.

Script table & sync direction

Command	File	Direction	Purpose
<code>npm run install</code>	<code>scripts/install.ts</code>	host → DB	Interactive fresh install: create new DB + write <code>sysconfig.conf</code> + full rootfs sync
<code>npm run vfs:bootstrap</code>	<code>scripts/vfs-bootstrap.ts</code>	host → DB	Bulk sync <code>src/mirror</code> → <code>/</code> and <code>src/common</code> → <code>/lib/common</code>
<code>npx ts-node scripts/sync-vfs.ts <path></code>	<code>scripts/sync-vfs.ts</code>	host → DB	Sync one file (wired to VS Code "run on save")
<code>npm run vfs:pull</code>	<code>scripts/vfs-pull.ts</code>	DB → host	Pull the whole VFS → <code>src/root</code> (except runtime folders)
syscall <code>SYNC_TO_HOST</code> (53)	<code>src/kernel/Syscalls.ts</code>	DB → host	Root-only: write one VFS file to the host (limited to project root)
<code>sudo userlib-update</code>	<code>src/mirror/sbin/userlib-update.ts</code>	DB → host	Inside TSIX: sync <code>/lib</code> → <code>src/.tsix_sdk/lib</code>
<code>npm run bkfs:create</code>	<code>scripts/create-bkfs.ts</code>	—	Create an empty DB (optional standard directory skeleton)

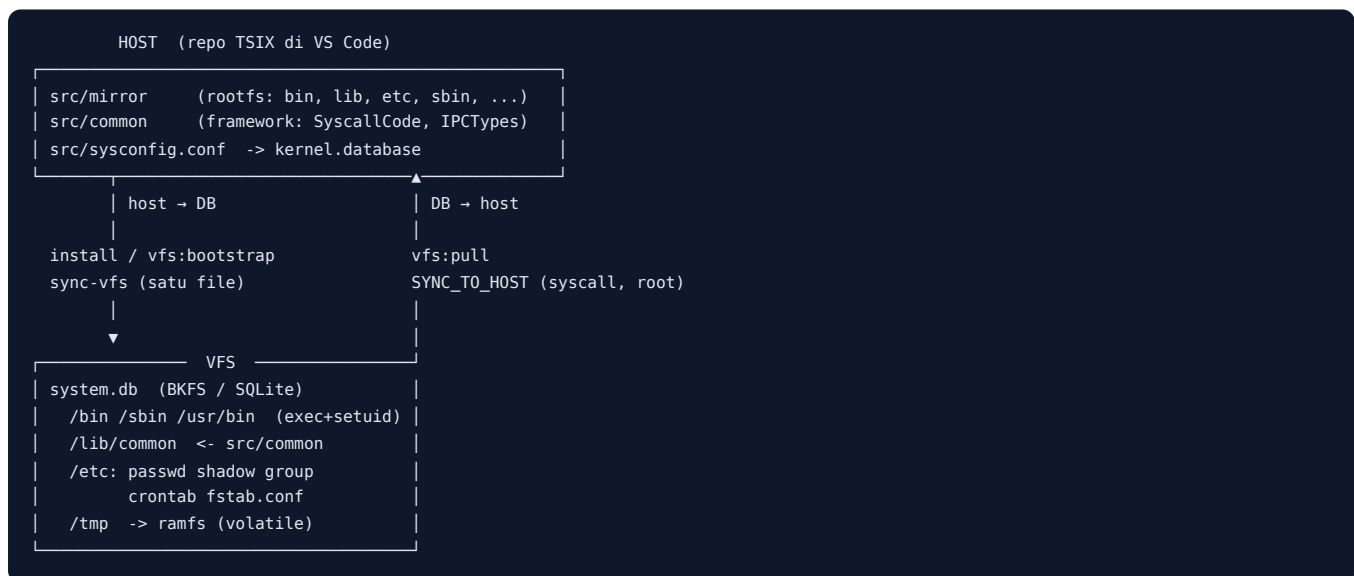
Shared configuration

- `src/sysconfig.conf` — `kernel.database` (DB path), `kernel.rootHostPath`, `scheduler.workerEntryPath`, `scheduler.bootEntry`, `network.interfaces`.
- `tsconfig.json` — `@tsix/*` → `src/.tsix_sdk/lib/*`, `src/root/lib/*`, `src/mirror/lib/*`; `@common/*` → `src/common/*`; `@bin/*` → `src/root/bin/*`, `src/.tsix_sdk/bin/*`.

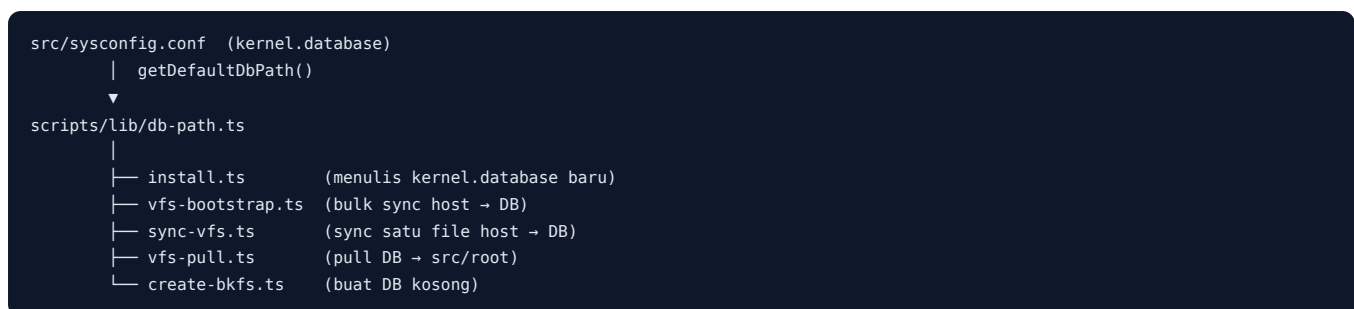
[!NOTE] There is no automatic host → VFS sync at boot. The host is the "writing surface"; the VFS is the runtime source of truth. Changes are moved via the scripts/syscalls above.

Flow / How It Works

Host ↔ VFS cycle diagram



sysconfig → db-path → scripts



1. Fresh install (npm run install)

1. Ask interactive configuration: hostname, user account (optional: username, password, confirm), MQTT broker, per-interface address, default MQTT port, kernel verbose, new DB path, (optional) root password.
2. Write the result to `src/sysconfig.conf` (stores the new `kernel.database`).
3. Create a new `.db` file. If the file already exists and without `--force` → stop; with `--force` → the old file is backed up to `*.bak-<timestamp>`.
4. Sync rootfs: `src/mirror` → `/`, then `src/common` → `/lib/common`. Each `.ts` is transpiled to a sidecar `.js`; executable directories get execute mode (`/sbin` = `0744`, others `0755`); `login`, `passwd`, `sudo` get SetUID (`4755` root).
5. Sync explicit `/etc` files without extension: `passwd`, `shadow` (mode `0640`), `group`, `crontab`, `profile`, `motd`, `fstab.md`, `pkg-demo.conf`.

6. Write a fresh `fstab.conf` — only `/tmp` as ramfs (mode `0o1777` sticky); device-specific mounts are not carried over.
7. Clear `/etc/crontab`.
8. If a user account was provided → add an entry to `/etc/passwd` + `/etc/shadow` (bcrypt) + `users` group, then create `/home/<username>` (mode `0700`, owned by the user).
9. If a root password is provided → it is bcrypt-hashed and written to `/etc/shadow`.
10. Verify: total nodes, number of `passwd` accounts, group list, `shadow` entries.

```
npm run install                # interaktif, path dari sysconfig
npm run install -- --path data/tsix.db    # path database tertentu
npm run install -- --path data/tsix.db --force  # timpa (auto-backup)
npm run install -- --defaults             # non-interaktif, semua default
npm run install -- --no-config            # skip tulis sysconfig.conf
```

2. Bulk sync (`npm run vfs:bootstrap`)

Mass sync of `src/mirror` → `/` and `src/common` → `/lib/common`. Useful for the initial device installation or after many edits at once.

```
npm run vfs:bootstrap          # DB default dari sysconfig
npm run vfs:bootstrap -- data/test.db # path lain (posisi argumen)
```

3. Sync one file (`sync-vfs`)

For one quick file (e.g. on save in the editor). Path mapping: `src/mirror/*` → `/*`, `src/common/*` → `/common/*`. `.ts` files are also transpiled to a `.js` sidecar.

```
npx ts-node scripts/sync-vfs.ts src/mirror/bin/hello.ts
```

4. Pull back (`npm run vfs:pull`)

Pulls the entire VFS → `src/root`, with mapping: `/etc/*` → `src/root/etc/*`, `/root/*` → `src/root/home/root/*`, others → `src/root/<path>`. Runtime folders are excluded: `dev`, `tmp`, `proc`, `logs`, `var`.

```
npm run vfs:pull
```

5. Syscall `SYNC_TO_HOST` (inside TSIX)

Applications inside the VFS (root) can write files to the host via the `SYNC_TO_HOST` syscall. `userlib-update` and `vfs-pull` (the `sbin` version) use it.

```
# dari shell TSIX, sebagai root
sudo userlib-update  # sync /lib → src/.tsix_sdk/lib
vfs-pull             # tarik seluruh VFS → host root
```

Source Code

- `scripts/install.ts` — fresh install agent (host → DB).
- `scripts/vfs-bootstrap.ts` — bulk sync host → DB.
- `scripts/sync-vfs.ts` — sync one file host → DB.
- `scripts/vfs-pull.ts` — pull DB → `src/root`.
- `scripts/create-bkfs.ts` — create an empty DB.
- `scripts/lib/db-path.ts` — shared DB path from `sysconfig`.
- `src/kernel/Syscalls.ts` — `SYNC_TO_HOST` / `SYNC_FROM_HOST` implementation.
- `src/mirror/lib/UserLib.ts` — `syncToHost()` / `syncFromHost()`.
- `src/mirror/sbin/userlib-update.ts`, `src/mirror/sbin/vfs-pull.ts` — commands inside TSIX.

Snippet (code level)

Shared DB path (scripts/lib/db-path.ts)

```
export function getDefaultDbPath(): string {
  // Satu parser dengan kernel: Config membaca src/sysconfig.conf (key-value,
  // gaya /etc/fstab.conf) + migrasi otomatis dari sysconfig.json lama.
  const database = Config.tryGet()?.kernel?.database;
  return typeof database === "string" && database.length > 0 ? database : "system.db";
}
```

Execute mode & SetUID (isSetuidBinary / applyBinaryMode)

```
const EXEC_DIRS = ["/bin", "/sbin", "/usr/bin", "/usr/local/bin"];

function isSetuidBinary(vfsPath: string): boolean {
  return /\bin\/(login|passwd|sudo)\.(ts|js)$/.test(vfsPath);
}

function isExecutableBinary(vfsPath: string): boolean {
  return EXEC_DIRS.some((d) => vfsPath.startsWith(d + "/"));
}

function applyBinaryMode(bkfs: BKFS, vfsPath: string, label = "INSTALL"): void {
  if (isSetuidBinary(vfsPath)) {
    bkfs.chmod(vfsPath, 0o4755); // setuid root
    bkfs.chown(vfsPath, 0, 0);
  } else if (isExecutableBinary(vfsPath)) {
    // /sbin = root-only (0o744), lainnya 0o755
    bkfs.chmod(vfsPath, vfsPath.startsWith("/sbin/") ? 0o744 : 0o755);
  }
}
```

Recursive syncDir (vfs-bootstrap.ts , identical in install.ts)

```
const syncDir = (hostDir: string, vfsDir: string) => {
  if (!bkfs.exists(vfsDir)) bkfs.mkdir(vfsDir, 0, 0, 0o755);
  const items = fs.readdirSync(hostDir);
  for (const item of items) {
    const fullHostPath = path.join(hostDir, item);
    const fullVfsPath = path.join(vfsDir, item).replace(/\\/g, "/");
    const stats = fs.statSync(fullHostPath);

    if (stats.isDirectory()) {
      syncDir(fullHostPath, fullVfsPath);
      continue;
    }

    const isTarget =
      item.endsWith(".ts") || item.endsWith(".js") ||
      item.endsWith(".json") || item.endsWith(".html") ||
      item.endsWith(".css") || item.endsWith(".menu") ||
      item.endsWith(".mp3") || item.endsWith(".wav");
    if (!isTarget) continue;

    // Binary (mp3/wav) disimpan sebagai latin1
    const isBinary = item.endsWith(".mp3") || item.endsWith(".wav");
    const content = isBinary
      ? fs.readFileSync(fullHostPath).toString("latin1")
      : fs.readFileSync(fullHostPath, "utf8");
    bkfs.touch(fullVfsPath, content);

    // Transpile TS → JS sidecar (yang dieksekusi runtime)
    if (fullVfsPath.endsWith(".ts")) {
      const result = esbuild.transformSync(content, {
        loader: "ts", format: "cjs", target: "node18", sourcemap: "inline",
      });
    }
  }
}
```

```

    if (result.code) {
      const jsPath = fullVfsPath.substring(0, fullVfsPath.length - 3) + ".js";
      bkfs.touch(jsPath, result.code);
      if (isSetuidBinary(jsPath) || isExecutableBinary(jsPath)) {
        applyBinaryMode(bkfs, jsPath);
      }
    }
  }
}

if ((isSetuidBinary(fullVfsPath) || isExecutableBinary(fullVfsPath)) &&
    (fullVfsPath.endsWith(".ts") || fullVfsPath.endsWith(".js"))) {
  applyBinaryMode(bkfs, fullVfsPath);
}
}
};

```

install.ts flow (summarized from source)

```

async function main() {
  const opts = parseArgs(process.argv.slice(2));
  const cfg = loadConfig();
  const dbRel = opts.dbPath || cfg.kernel.database || "system.db";
  let rootPassword = "";

  // 1. Konfigurasi interaktif → cfg (hostname, user, broker, port, verbose, db path, root password)
  // 2. Tulis src/sysconfig.conf
  if (opts.writeConfig) {
    cfg.kernel.database = path.relative(PROJECT_ROOT, dbPath).replace(/\\/g, "/");
    saveConfig(cfg);
  }
  // 3. Buat DB baru (file lama di-backup bila --force)
  const bkfs = new BKFS(dbPath);
  // 4. Sync rootfs
  syncDir(bkfs, MIRROR_ROOT, "/"); // src/mirror → /
  syncDir(bkfs, COMMON_ROOT, "/lib/common"); // src/common → /lib/common
  // 5. fstab fresh (hanya /tmp ramfs), crontab kosong, password root opsional
  // 6. Verifikasi & tutup DB
}

```

Syscall SYNC_TO_HOST (src/kernel/Syscalls.ts)

```

case SyscallCode.SYNC_TO_HOST: {
  if (!this.isRoot(pcb))
    throw new Error("Permission Denied: Only root or root group members can perform physical sync.");
  const { vfsPath, hostPath } = args;

  // Keamanan: hostPath tidak boleh keluar dari project root
  const projectRoot = process.cwd();
  const absoluteHostPath = path.resolve(projectRoot, hostPath);
  if (!absoluteHostPath.startsWith(projectRoot)) {
    throw new Error("Security Violation: Target path is outside project root.");
  }

  const { vfs, relativePath } = this.mountManager.resolve(vfsPath);
  const content = vfs.read(relativePath);
  if (content === null) throw new Error(`Source file not found in VFS: ${vfsPath}`);

  const parentDir = path.dirname(absoluteHostPath);
  if (!fs.existsSync(parentDir)) fs.mkdirSync(parentDir, { recursive: true });
  fs.writeFileSync(absoluteHostPath, content);
  this.logger.info(`[SYNC] Applied VFS:${vfsPath} -> HOST:${hostPath}`);
  return true;
}

```

UserLib.syncToHost (src/mirror/lib/UserLib.ts)

```

public async syncToHost(vfsPath: string, hostPath: string): Promise<boolean> {
  return await this.dispatch(SyscallCode.SYNC_TO_HOST, { vfsPath, hostPath });
}

```

```
}
```

Exercises / Practice

1. Run `npm run install -- --defaults` — check `src/sysconfig.conf` is written and a new DB is created, then `npm start`.
 2. Run `npm run install` (interactive) — fill in hostname & user login, set the root password, observe the per-file sync.
 3. Write a new app in `src/mirror/bin/`, then `npm run vfs:bootstrap` — run the app in the TSIX shell.
 4. Change one file then `npx ts-node scripts/sync-vfs.ts src/mirror/bin/<file>.ts` — compare its speed vs the full bootstrap.
 5. Change a file in the VFS (from the TSIX shell), then `npm run vfs:pull` — check the change appears in `src/root`.
 6. Read `wiki/Memulai.md` — follow the flow starting from scratch.
-

References

- `wiki/Memulai.md`, `wiki/Panduan-Developer.md`
 - `wiki/course/00-overview.en.md` §11 (Development Flow / VFS Sync)
 - `wiki/Virtual-File-System.md` — VFS ↔ Host Synchronization
 - `scripts/install.ts`, `scripts/vfs-bootstrap.ts`, `scripts/sync-vfs.ts`, `scripts/vfs-pull.ts`, `scripts/create-bkfs.ts`, `scripts/lib/db-path.ts`
 - `src/kernel/Syscalls.ts`, `src/mirror/lib/UserLib.ts`, `src/mirror/sbin/userlib-update.ts`, `src/mirror/sbin/vfs-pull.ts`
-

Module 23 — done. Continue to [Module 24 — Best Practices & App Writing](#).

RFC-TSIX-EDU-002 | Module twenty-four of the TSIX curriculum. Closes the curriculum with best practices: two app styles, error = window, themes, UPDATE_PROPS batching, and the Piagam Antigonon.

The curriculum is complete here. This module is the "practice book" — a collection of rules that keep TSIX applications consistent, secure, and easy to maintain.

Learning Objectives

- ☐ Distinguish the two app styles (`IProgram` vs `Program()`)
- ☐ Explain "error = window" (the 4-step `std.error`)
- ☐ Explain `UPDATE_PROPS` batching
- ☐ Explain the Piagam Antigonon (4 GUI rules)
- ☐ Apply best practices to new apps

Core Concepts

Two application styles

1. Class `main` implements `IProgram` (classic style, the majority in `/bin`):

The contract is in `src/mirror/lib/IProgram.ts`:

```
export interface OSContext {
  std: StdLib;
  fs: FsLib;
  shell: ShellLib;
  aux: AuxLib;
}

export interface IProgram {
  execute(os: OSContext, args: string[]): Promise<string | void>;
}
```

Minimal example (the `cat.ts` / `echo.ts` pattern):

```
import { IProgram, OSContext } from "../lib/IProgram";

export class main implements IProgram {
  async execute(os: OSContext, args: string[]): Promise<string | void> {
    const { std } = os;
    await std.print("Hello from TSIX!\n");
    // void → WorkerEntry memanggil shell.exit(0)
  }
}
```

2. `Program()` wrapper + proxy singletons (new, recommended for new apps & GUI):

```
import { Program, std } from "@tsix/Application";

export const main = Program(async (args: string[]) => {
  await std.print("Hello from TSIX!\n");
});
```

Alias advantage: **path independence** — the import stays the same no matter where the file is placed (/bin/ , /root/test/ , etc.). It eliminates "Module not found" errors when moving folders.

Comparison of the two styles:

Aspect	main implements IProgram	Program() wrapper
Import	<code>import { IProgram, OSContext } from "../lib/IProgram"</code>	<code>import { Program, std, fs, shell, net, db } from "@tsix/Application"</code>
Entry point	<code>async execute(os: OSContext, args)</code>	callback function <code>async (args) => ...</code>
Library access	<code>os.std, os.fs, os.shell, os.aux</code>	proxy singletons <code>std, fs, shell, net, db</code>
Path	relative import to the file location	absolute import <code>@tsix/*</code> — path independent
Error handling	manual (WorkerEntry catches other errors → <code>realExit(1)</code>)	automatic: the wrapper catches errors → <code>std.error()</code> → rethrow
Suitable for	CLI utilities in /bin	new apps, GUI (Emerald), daemons

How the wrapper works (`src/mirror/lib/Application.ts`): `Program(fn)` returns a class that implements `IProgram`. When `execute` is called, it stores the `OSContext` into `global._tsixOsc` and then calls `fn(args)`. If `fn` throws an error, the wrapper sends that error to the parent via `std.error()` (it becomes a desktop error window), then rethrows so that `WorkerEntry` also handles it. The `std/fs/shell/net/db` proxies read `UserLib` from `global._tsixLib` lazily — that is why the imports are path-independent.

Error = "Window"

Don't let an app crash silently. `std.error(message, context?, wid?)` displays the error as a **desktop window popup** (processed by `WM/Asteracea`). The 4-step flow (see `StdLib.error()` in `src/mirror/lib/UserLib.ts`):

1. **Log to syslog** — write a `[ERROR]` line + timestamp to `/var/log/syslog`.
2. **Extract fileHint** — read the stack trace, find the first source file that is not an internal library (`UserLib`, `Application`, `emerald`).
3. **Broadcast to the parent (WM)** — send an IPC `GUI_WINDOW_ERROR` containing `{ wid, pid, file, error, context, timestamp }` to the parent process; `Asteracea` displays it as a popup.
4. **Print red TTY** — print `\x1b[31m[ERROR]\x1b[0m <message>` to the terminal (stderr style).

Each step is non-fatal — if one fails (e.g., `syslog` not yet present), the other steps still run.

Batching UPDATE_PROPS

Don't send one `UPDATE_PROPS` per property change. **Batch** several changes and send them together — this reduces the IPC count and render latency. In `Emerald` (`src/mirror/lib/emerald.ts`, class `Window`) the mechanism is already automatic:

- `updateProps(targetId, props)` does not send immediately — the changes are merged into `dirtyProps` (a Map), then `scheduleFlush()` is scheduled.
- `scheduleFlush()` uses `setTimeout(..., 0)`: all changes within one async tick are collected first, then flushed at the end of the tick. If a flush is already scheduled, there is no double scheduling.
- `flushNow()` sends all pending `UPDATE_PROPS` at once.
- `Screen.update()` / `setText()` / `setVisible()` / `setStyle()` all use this batch path.
- `setContent()` is deliberately **not** batched (sent directly via `sendImmediate`) — one atomic `innerHTML=""` operation followed by `MOUNT_NODE` per child.
- `Screen.on()` calls `flush()` automatically after binding — ensuring the listener is attached in the browser.

Best practice: don't write a loop `for (...) await app.update(id, props)` expecting each item to be sent individually — `Emerald` already batches it for you.

Piagam Antigonon (4 GUI rules)

The Piagam Antigonon is a strict rule set for AI agents & developers who write TSIX GUIs (source: [wiki/PIXELSPACE_DEVELOPER_GUIDE.md](#)):

- 1. **NO DOM in Userland** — @tsix/emerald MUST NOT touch document.* or window.*. All rendering goes through the PixelSpace protocol (Worker → Kernel → DOME → Browser).
- 2. **State-Sync** — Don't send UPDATE_PROPS in a tight loop; use batching.
- 3. **Memory Cleanup** — Every unmounted node must have its event listeners cleaned up (prevent memory leaks).
- 4. **Type Safety** — All payloads must conform to IGUIPayload; malformed payload → SIGKILL.

[!TIP] Two related practices are equally important: attach **mount-time** listeners (onClickId / onInputId in props at build time, not app.on() after mount) to avoid the cloneNode bug (Modules 19 & 20), and keep the UI safe for **state replay** after F5 (Module 22).

Flow / How It Works

Steps to write a correct TSIX app (the Program() style):

- 1. **Choose the style** — short CLI utilities: main implements IProgram. New apps & GUI: Program().
- 2. **Write the entry point** — export const main = Program(async (args) => {...}) (or export class main implements IProgram).
- 3. **Declare GUI mode** (if the app shows a window) — export const appMode = "gui"; so WorkerEntry handles the GUI.
- 4. **Access libraries via proxy** — std, fs, shell, net, db — import once from @tsix/Application.
- 5. **GUI: use Emerald** — new Screen({...}), app.mount(...), app.on(...), app.loopUntilClose().
- 6. **Error = window** — call await std.error(msg, context) on failure; don't let it crash silently.
- 7. **Theme** — await theme.loadCurrent() + theme.watch() so colors follow Asteracea preferences.
- 8. **Batch updates** — leave batching to Emerald; don't send UPDATE_PROPS per item.
- 9. **Follow the Piagam Antigonon** — no direct DOM, no tight loops, clean up listeners, payloads conform to IGUIPayload.

Source Code

File	Contents
src/mirror/lib/IProgram.ts	IProgram & OSContext contract
src/mirror/lib/Application.ts	Program() wrapper + proxy singletons (std, fs, shell, net, db)
src/mirror/lib/UserLib.ts	StdLib (print, log, error), FsLib
src/mirror/lib/emerald.ts	Window/Screen, UPDATE_PROPS batching, bindHandler
src/mirror/lib/theme.ts	ThemeProvider — loadCurrent, watch, switchTo, applyToDome
src/mirror/bin/cat.ts, echo.ts	IProgram style examples
src/mirror/root/ps-sample2.ts, ps-sample3.ts	Program() + Emerald examples
src/mirror/opt/set-theme/set-theme.ts	Theme example (dark/light switch)
wiki/PIXELSPACE_DEVELOPER_GUIDE.md	Piagam Antigonon

Snippet (code level)

App CLI — IProgram style (readfile)

execute(os, args) uses os.std / os.fs (the OSContext contract). The returned string is printed by WorkerEntry; void → shell.exit(0):

```
import { IProgram, OSContext } from "../lib/IProgram";

export class main implements IProgram {
  async execute(os: OSContext, args: string[]): Promise<string | void> {
    const { std, fs } = os;
    if (args.length === 0) {
      return "Usage: readfile <path>"; // string return → dicetak WorkerEntry
    }
    const fd = await fs.open(args[0], "r"); // fd < 0 = gagal buka
    if (fd < 0) {
      await std.print(`Error: Cannot open ${args[0]}\n`);
      return;
    }
    const content = await fs.read(fd);
    await std.print(content ?? "");
    await fs.close(fd);
  }
}
```

[!TIP] To read a file in one shot, FsLib already provides `readFile(path)`, which wraps `open → read → close`.

App CLI — `Program()` style (hello)

```
import { Program, std } from "@tsix/Application";

export const main = Program(async (args: string[]) => {
  await std.print(`Hello, ${args[0]} ?? "world"!\\n`);
});
```

`std.error` — error becomes a window

```
// Log syslog → ekstrak fileHint → broadcast GUI_WINDOW_ERROR → TTY merah
await std.error("Disk full", "myapp");
await std.error("Connection timeout", "net", app.wid); // wid opsional
```

When the app calls this, WM/Asteracea shows an error popup on the desktop — not a console crash.

Theme — import & apply

```
import { theme } from "@tsix/theme";
import { Screen, div } from "@tsix/emerald";

// Muat tema aktif (terang/gelap) sesuai prefs Asteracea
await theme.loadCurrent();
theme.watch(); // ikut perubahan tema saat app berjalan

// Pakai warna terpusat — tanpa hardcode hex
const app = new Screen({ title: "App", width: 400, height: 300 });
await app.mount(
  div({
    id: "root",
    style: { background: theme.colors.bg, color: theme.colors.text },
  }),
);

// Ganti tema global + broadcast THEME_CHANGED ke DOME
await theme.switchTo("theme-light.json");
// Terapkan ke window ini (titlebar, border, shadow)
await theme.applyToDome(domePid, app.wid);
```

Theme files are located in `/opt/asteracea/theme-*.json` (`theme-dark.json`, `theme-light.json`). Full example: `src/mirror/opt/set-theme/set-theme.ts`.

Exercises / Practice

1. Write a "Hello" app in both styles — compare the structure & imports.
 2. Add `std.error()` to an app that deliberately crashes — observe the error window on the desktop.
 3. Refactor a GUI app to batch `UPDATE_PROPS` — measure the difference in responsiveness.
 4. Read `wiki/DEVELOPER_GUIDE_SCRIPTING-V2.md` — work through the v2.1 script examples.
-

References

- `wiki/PIXELSPACE_DEVELOPER_GUIDE.md` — Piagam Antigonon & PixelSpace guide
 - `wiki/DEVELOPER_GUIDE_SCRIPTING-V2.md`, `wiki/Panduan-Developer.md`
 - `wiki/course/00-overview.md` §9
 - `src/mirror/lib/IProgram.ts`, `src/mirror/lib/Application.ts`, `src/mirror/lib/UserLib.ts`
 - `src/mirror/lib/emerald.ts`, `src/mirror/lib/theme.ts`
 - `src/mirror/root/ps-sample2.ts`, `src/mirror/root/ps-sample3.ts`
 - `src/mirror/opt/set-theme/set-theme.ts`, `src/mirror/opt/test/gui-demo.ts`, `src/mirror/opt/file-cruiser/file-cruiser.ts`
 - `src/mirror/bin/cat.ts`, `src/mirror/bin/echo.ts`
-

Module 24 — complete. The TSIX curriculum is complete! 🎉 Keep writing? Update `toc.md` and create the `.en.md` translation (FORMAT §5).